

BDCC 200 X 300					TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																																
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																																		
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,15 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,38 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,44 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa																																		
LISTA DE FERROS (POR METRO LINEAR)																																																																					
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																																		
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																														
1	8,0	15,4	205	c.13	1	10,0	16,7	205	c.12	1	10,0	18,2	205	c.11	1	12,5	16,7	205	c.12	1	12,5	16,7	209	c.12	1	16,0	14,3	209	c.14	1	16,0	15,4	209	c.13	1	16,0	16,7	212	c.12																														
2	6,3	40,0	212	c.10	2	8,0	36,4	212	c.11	2	8,0	40,0	212	c.10	2	10,0	33,3	212	c.12	2	10,0	36,4	217	c.11	2	10,0	40,0	217	c.10	2	10,0	40,0	217	c.10	2	12,5	33,3	222	c.12																														
3	6,3	11,8	290	c.34	3	8,0	16,7	290	c.24	3	6,3	20,0	290	c.20	3	8,0	20,0	290	c.20	3	6,3	20,0	295	c.20	3	6,3	18,2	295	c.22	3	10,0	15,4	295	c.26	3	8,0	16,7	300	c.24																														
4	6,3	11,8	245	c.34	4	8,0	16,7	245	c.24	4	6,3	20,0	245	c.20	4	8,0	20,0	245	c.20	4	6,3	20,0	249	c.20	4	6,3	18,2	249	c.22	4	10,0	15,4	249	c.26	4	8,0	16,7	252	c.24																														
5	6,3	8,0	332	c.25	5	6,3	8,0	332	c.25	5	6,3	8,0	332	c.25	5	6,3	8,0	332	c.25	5	6,3	9,5	342	c.21	5	6,3	9,5	342	c.21	5	6,3	9,5	342	c.21	5	6,3	11,8	352	c.17																														
6	6,3	8,0	332	c.25	6	6,3	8,0	332	c.25	6	6,3	8,0	332	c.25	6	6,3	8,0	332	c.25	6	6,3	9,5	342	c.21	6	6,3	9,5	342	c.21	6	6,3	9,5	342	c.21	6	6,3	11,8	352	c.17																														
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	162	c.20	7	6,3	20,0	162	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20																														
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	232	c.20	8	6,3	10,0	232	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20																														
9	6,3	40,0	Corr.	c.20	9	6,3	40,0	Corr.	c.20	9	6,3	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20	9	6,3	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20																														
10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	8,0	40,0	Corr.	c.20	10	8,0	40,0	Corr.	c.20	10	8,0	40,0	Corr.	c.20	10	8,0	40,0	Corr.	c.20																														
11	6,3	28,0	Corr.	c.20	11	6,3	30,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20																														
12	6,3	56,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20																														
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----																														
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	100,0	Var.	c.16	14	6,3	145,5	Var.	c.11	14	8,0	123,1	Var.	c.13																														
15	----	----	----	----	15	----	----	----	----	15	----	----	----	----	15	6,3	114,3	Var.	c.14	15	6,3	114,3	Var.	c.14	15	8,0	106,7	Var.	c.15	15	8,0	145,5	Var.	c.11	15	10,0	114,3	Var.	c.14																														
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																																					
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																																
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																															
6,3	0,252	106,0	6,3	0,252	56,2	6,3	0,252	95,7	6,3	0,252	72,8	6,3	0,252	106,6	6,3	0,252	67,2	6,3	0,252	73,4	6,3	0,252	60,1	6,3	0,252	73,4	6,3	0,252	62,3	8,0	0,393	96,2	8,0	0,393	12,4	8,0	0,393	65,4	8,0	0,393	33,3																												
8,0	0,393	12,4	8,0	0,393	65,4	8,0	0,393	33,3	8,0	0,393	57,8	8,0	0,393	15,7	8,0	0,393	93,0	8,0	0,393	62,3	8,0	0,393	96,2	8,0	0,393	62,3	8,0	0,393	62,3	10,0	0,624	42,1	10,0	0,624	21,4	10,0	0,624	23,3																															
10,0	0,624	----	10,0	0,624	21,4	10,0	0,624	23,3	10,0	0,624	44,1	10,0	0,624	49,3	10,0	0,624	54,2	10,0	0,624	106,4	10,0	0,624	42,1	10,0	0,624	106,4	10,0	0,624	106,4	12,5	0,988	73,0	12,5	0,988	----	12,5	0,988	----	12,5	0,988	33,8																												
12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	33,8	12,5	0,988	34,5	12,5	0,988	----	12,5	0,988	----	12,5	0,988	73,0	12,5	0,988	----	12,5	0,988	----	16,0	1,570	83,8	16,0	1,570	28,3	16,0	1,570	28,3																															
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	75,2	16,0	1,570	78,8	16,0	1,570	83,8	16,0	1,570	78,8	16,0	1,570	78,8	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----																															
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----																												
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	TOTAL (Kg/m)	146,7	TOTAL (Kg/m)	171,3	TOTAL (Kg/m)	180,6	TOTAL (Kg/m)	236,8	TOTAL (Kg/m)	234,4	TOTAL (Kg/m)	289,6	TOTAL (Kg/m)	320,9	TOTAL (Kg/m)	355,2																					
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																																					
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																															
1	80	45	1	80	45	1	80	45	1	80	45	1	80	49	1	80	49	1	80	49	1	80	49	1	80	49	1	80	49	1	80	52	1	80	52	1	80	52																															
2	200	12	2	200	12	2	200	12	2	200	12	2	200	17	2	200	17	2	200	17	2	200	17	2	200	17	2	200	17	2	200	22	2	200	22	2	200	22																															
3	80	210	3	80	210	3	80	210	3	80	210	3	80	215	3	80	215	3	80	215	3	80	215	3	80	215	3	80	215	3	80	220	3	80	220	3	80	220																															
4	125	120	4	125	120	4	125	120	4	125	120	4	125	120	4	129	120	4	129	120	4	129	120	4	129	120	4	129	120	4	132	120	4	132	120	4	132	120																															
5	332	----	5	332	----	5	332	----	5	332	----	5	342	----	5	342	----	5	342	----	5	342	----	5	342	----	5	342	----	5	352	----	5	352	----	5	352	----																															
6	332	----	6	332	----	6	332	----	6	332	----	6	342	----	6	342	----	6	342	----	6	342	----	6	342	----	6	342	----	6	352	----	6	352	----	6	352	----																															
7	122	----	7	----	----	7	122	----	7	122	----	7	143	----	7	143	----	7	143	----	7	143	----	7	143	----	7	143	----	7	164	----	7	164	----	7	164	----																															
8	116	----	8	----	----	8	116	----	8	116	----	8	135	----	8	135	----	8	135	----	8	135	----	8	135	----	8	135	----	8	153	----	8	153	----	8	153	----																															
															ASSINATURA DAS AUTORIDADES																				DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																		
															Eng ^o Selma Schwab Coordenadora da DP/GNT															Eng ^o Mário S.Bortone Gerente da DP/GPE										Eng ^o Roger Gama Veloso Diretor de Projetos										Eng ^o Nelson A. Reis Vice - Diretor Geral										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)									
																																			DES. 61																																		

BDCC 200 X 350	TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	15,4	205	c.13	1	10,0	16,7	205	c.12	1	10,0	16,7	205	c.12	1	12,5	15,4	205	c.13	1	12,5	18,2	209	c.11
2	6,3	40,0	212	c.10	2	8,0	40,0	212	c.10	2	8,0	36,4	212	c.11	2	10,0	30,8	217	c.13	2	10,0	36,4	222	c.11
3	6,3	20,0	315	c.20	3	10,0	16,7	315	c.24	3	10,0	15,4	315	c.26	3	12,5	13,3	315	c.30	3	10,0	20,0	320	c.20
4	6,3	20,0	265	c.20	4	10,0	16,7	265	c.24	4	10,0	15,4	265	c.26	4	12,5	13,3	265	c.30	4	10,0	18,2	269	c.22
5	6,3	8,0	382	c.25	5	6,3	8,0	382	c.25	5	6,3	8,0	382	c.25	5	6,3	9,5	392	c.21	5	6,3	9,5	392	c.21
6	6,3	8,0	382	c.25	6	6,3	8,0	382	c.25	6	6,3	8,0	382	c.25	6	6,3	9,5	392	c.21	6	6,3	11,8	402	c.17
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	162	c.20	7	6,3	20,0	162	c.20	7	6,3	20,0	183	c.20
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	232	c.20	8	6,3	10,0	232	c.20	8	6,3	10,0	270	c.20
9	6,3	40,0	Corr.	c.20	9	6,3	40,0	Corr.	c.20	9	6,3	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20	9	8,0	40,0	Corr.	c.20
10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	6,3	40,0	Corr.	c.20	10	8,0	40,0	Corr.	c.20
11	6,3	32,0	Corr.	c.20	11	6,3	36,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	8,0	32,0	Corr.	c.20	11	8,0	32,0	Corr.	c.20
12	6,3	64,0	Corr.	c.20	12	6,3	72,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	114,3	Var.	c.14
15	----	----	----	----	15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	6,3	106,7	Var.	c.15	15	6,3	160,0	Var.	c.10

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																				
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	124,4	6,3	0,252	62,8	6,3	0,252	73,8	6,3	0,252	68,0	6,3	0,252	75,5	6,3	0,252	88,3	6,3	0,252	77,8
8,0	0,393	12,4	8,0	0,393	33,3	8,0	0,393	30,3	8,0	0,393	28,3	8,0	0,393	28,3	8,0	0,393	44,0	8,0	0,393	68,8
10,0	0,624	----	10,0	0,624	81,8	10,0	0,624	77,1	10,0	0,624	40,7	10,0	0,624	108,6	10,0	0,624	122,8	10,0	0,624	118,2
12,5	0,988	----	12,5	0,988	----	12,5	0,988	107,4	12,5	0,988	107,4	12,5	0,988	31,8	12,5	0,988	37,6	12,5	0,988	35,0
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		165,1	TOTAL (Kg/m)		206,3	TOTAL (Kg/m)		209,5	TOTAL (Kg/m)		272,7	TOTAL (Kg/m)		272,5	TOTAL (Kg/m)		321,0	TOTAL (Kg/m)		328,1

DIMENSÕES DAS DOBRAS DAS ARMADURAS																				
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)
1	80	45	1	80	45	1	80	45	1	80	45	1	80	49	1	80	49	1	80	52
2	200	12	2	200	12	2	200	12	2	200	12	2	200	17	2	200	17	2	200	22
3	80	235	3	80	235	3	80	235	3	80	235	3	80	240	3	80	240	3	80	245
4	125	140	4	125	140	4	125	140	4	125	140	4	129	140	4	129	140	4	132	140
5	382	----	5	382	----	5	382	----	5	382	----	5	392	----	5	392	----	5	402	----
6	382	----	6	382	----	6	382	----	6	382	----	6	392	----	6	392	----	6	402	----
7	122	----	7	----	----	7	122	----	7	122	----	7	143	----	7	143	----	7	164	----
8	116	----	8	----	----	8	116	----	8	116	----	8	135	----	8	135	----	8	153	----

ASSINATURA DAS AUTORIDADES					DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS			
	CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)				DES. 62			

BDCC 250 X 150					TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																						
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,30 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,36 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,54 MPa																								
LISTA DE FERROS (POR METRO LINEAR)																																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																				
1	8,0	18,2	250	c.11	1	12,5	16,7	250	c.12	1	16,0	14,3	250	c.14	1	16,0	16,7	254	c.12	1	16,0	18,2	258	c.11	1	16,0	20,0	258	c.10	1	20,0	15,4	258	c.13	1	20,0	18,2	261	c.11																				
2	8,0	28,6	262	c.14	2	10,0	36,4	262	c.11	2	12,5	28,6	262	c.14	2	12,5	33,3	267	c.12	2	12,5	36,4	272	c.11	2	12,5	40,0	272	c.10	2	16,0	30,8	272	c.13	2	16,0	36,4	277	c.11																				
3	6,3	8,0	235	c.50	3	6,3	11,1	235	c.36	3	6,3	8,0	235	c.50	3	6,3	8,0	240	c.50	3	6,3	8,0	245	c.50	3	6,3	8,0	245	c.50	3	6,3	8,0	245	c.50	3	6,3	8,0	250	c.50																				
4	6,3	8,0	210	c.50	4	6,3	11,1	210	c.36	4	6,3	8,0	210	c.50	4	6,3	8,0	214	c.50	4	6,3	8,0	218	c.50	4	6,3	8,0	218	c.50	4	6,3	8,0	218	c.50	4	6,3	8,0	221	c.50																				
5	6,3	8,0	182	c.25	5	6,3	8,0	182	c.25	5	6,3	8,0	182	c.25	5	6,3	9,5	192	c.21	5	6,3	11,8	202	c.17	5	6,3	11,8	202	c.17	5	6,3	11,8	202	c.17	5	8,0	9,1	212	c.22																				
6	6,3	8,0	182	c.25	6	6,3	8,0	182	c.25	6	6,3	8,0	182	c.25	6	6,3	9,5	192	c.21	6	6,3	11,8	202	c.17	6	6,3	11,8	202	c.17	6	6,3	11,8	202	c.17	6	8,0	9,1	212	c.22																				
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	162	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	225	c.20																				
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	232	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	344	c.20																				
9	6,3	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	10,0	52,0	Corr.	c.20	9	10,0	52,0	Corr.	c.20																				
10	6,3	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	12,5	52,0	Corr.	c.20																				
11	6,3	12,0	Corr.	c.20	11	6,3	16,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20	11	6,3	12,0	Corr.	c.20																				
12	6,3	24,0	Corr.	c.20	12	6,3	32,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20	12	6,3	24,0	Corr.	c.20																				
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----																				
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	88,9	Var.	c.18	14	6,3	133,3	Var.	c.12	14	8,0	114,3	Var.	c.14																				
15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	6,3	145,5	Var.	c.11	15	6,3	160,0	Var.	c.10	15	8,0	106,7	Var.	c.15	15	8,0	145,5	Var.	c.11	15	10,0	123,1	Var.	c.13	15	10,0	145,5	Var.	c.11																				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																											
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																				
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																		
6,3	0,252	65,6	6,3	0,252	55,3	6,3	0,252	57,4	6,3	0,252	65,2	6,3	0,252	48,4	6,3	0,252	48,4	6,3	0,252	43,6	6,3	0,252	50,2	6,3	0,252	50,2	6,3	0,252	43,6	6,3	0,252	50,2	6,3	0,252	18,6	6,3	0,252	18,6																					
8,0	0,393	47,3	8,0	0,393	20,4	8,0	0,393	40,9	8,0	0,393	40,9	8,0	0,393	65,6	8,0	0,393	65,6	8,0	0,393	82,2	8,0	0,393	28,1	8,0	0,393	28,1	8,0	0,393	82,2	8,0	0,393	28,1	8,0	0,393	75,1	8,0	0,393	75,1																					
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	----	10,0	0,624	----	10,0	0,624	----	10,0	0,624	----	10,0	0,624	32,4	10,0	0,624	110,2	10,0	0,624	110,2	10,0	0,624	32,4	10,0	0,624	110,2	10,0	0,624	90,6	10,0	0,624	90,6																					
12,5	0,988	----	12,5	0,988	41,2	12,5	0,988	74,0	12,5	0,988	87,8	12,5	0,988	97,8	12,5	0,988	97,8	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	----	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	51,4	12,5	0,988	51,4																					
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	84,4	16,0	1,570	94,9	16,0	1,570	102,0	16,0	1,570	102,0	16,0	1,570	109,3	16,0	1,570	159,8	16,0	1,570	159,8	16,0	1,570	109,3	16,0	1,570	159,8	16,0	1,570	186,6	16,0	1,570	186,6																					
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	98,5	20,0	2,480	117,8	20,0	2,480	117,8																		
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----																		
TOTAL (Kg/m)		141,2	TOTAL (Kg/m)		204,7	TOTAL (Kg/m)		256,7	TOTAL (Kg/m)		288,8	TOTAL (Kg/m)		313,8	TOTAL (Kg/m)		375,0	TOTAL (Kg/m)		446,8	TOTAL (Kg/m)		540,1																																				
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																											
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																					
1	100	50	1	100	50	1	100	50	1	100	54	1	100	58	1	100	58	1	100	58	1	100	58	1	100	58	1	100	58	1	100	61	1	100	61																								
2	250	12	2	250	12	2	250	12	2	250	17	2	250	22	2	250	22	2	250	22	2	250	22	2	250	22	2	250	22	2	250	27	2	250	27																								
3	100	135	3	100	135	3	100	135	3	100	140	3	100	145	3	100	145	3	100	145	3	100	145	3	100	145	3	100	145	3	100	150	3	100	150																								
4	150	60	4	150	60	4	150	60	4	154	60	4	158	60	4	158	60	4	158	60	4	158	60	4	158	60	4	158	60	4	158	60	4	161	60																								
5	182	----	5	182	----	5	182	----	5	192	----	5	202	----	5	202	----	5	202	----	5	202	----	5	202	----	5	202	----	5	202	----	5	212	----																								
6	182	----	6	182	----	6	182	----	6	192	----	6	202	----	6	202	----	6	202	----	6	202	----	6	202	----	6	202	----	6	202	----	6	212	----																								
7	122	----	7	----	----	7	122	----	7	143	----	7	164	----	7	164	----	7	164	----	7	164	----	7	164	----	7	164	----	7	164	----	7	185	----																								
8	116	----	8	----	----	8	116	----	8	135	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	172	----																								
										ASSINATURA DAS AUTORIDADES										 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																							
										Engº Selma Schwab Coordenadora da DP/GNT										Engº Mário S.Bortone Gerente da DP/GPE										Engº Roger Gama Veloso Diretor de Projetos										Engº Nelson A. Reis Vice - Diretor Geral										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular - Fundação Direta (Por Metro Linear)									
																																																		DES. 63									

BDCC 250 X 200	TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,24 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,31 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,36 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,43 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,55 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	18,2	250	c.11	1	12,5	16,7	250	c.12	1	12,5	20,0	250	c.10	1	16,0	16,7	254	c.12	1	16,0	18,2	258	c.11
2	8,0	30,8	262	c.13	2	10,0	36,4	262	c.11	2	12,5	28,6	262	c.14	2	12,5	33,3	267	c.12	2	12,5	36,4	272	c.11
3	6,3	8,0	260	c.50	3	6,3	13,3	260	c.30	3	6,3	9,1	260	c.44	3	6,3	8,0	265	c.50	3	6,3	8,0	270	c.50
4	6,3	8,0	230	c.50	4	6,3	13,3	230	c.30	4	6,3	9,1	230	c.44	4	6,3	8,0	234	c.50	4	6,3	8,0	238	c.50
5	6,3	8,0	232	c.25	5	6,3	8,0	232	c.25	5	6,3	8,0	232	c.25	5	6,3	9,5	242	c.21	5	6,3	11,8	252	c.17
6	6,3	8,0	232	c.25	6	6,3	8,0	232	c.25	6	6,3	8,0	232	c.25	6	6,3	9,5	242	c.21	6	6,3	11,8	252	c.17
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	162	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	232	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20
9	6,3	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20
10	6,3	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20
11	6,3	18,0	Corr.	c.20	11	6,3	20,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20
12	6,3	36,0	Corr.	c.20	12	6,3	40,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	94,1	Var.	c.17
15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	6,3	133,3	Var.	c.12	15	6,3	160,0	Var.	c.10	15	8,0	100,0	Var.	c.16

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	73,1	6,3	0,252	64,3	6,3	0,252	77,8	6,3	0,252	73,1	6,3	0,252	56,8	6,3	0,252	52,8	6,3	0,252	42,4	6,3	0,252	24,2
8,0	0,393	49,6	8,0	0,393	20,4	8,0	0,393	20,4	8,0	0,393	40,9	8,0	0,393	64,1	8,0	0,393	82,2	8,0	0,393	90,2	8,0	0,393	85,7
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	----	10,0	0,624	----	10,0	0,624	----	10,0	0,624	32,4	10,0	0,624	64,9	10,0	0,624	89,8
12,5	0,988	----	12,5	0,988	41,2	12,5	0,988	123,4	12,5	0,988	87,8	12,5	0,988	97,8	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	51,4
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	94,9	16,0	1,570	102,0	16,0	1,570	109,3	16,0	1,570	162,2	16,0	1,570	189,4
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	99,7	20,0	2,480	109,8
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		151,0	TOTAL (Kg/m)		213,7	TOTAL (Kg/m)		249,9	TOTAL (Kg/m)		296,7	TOTAL (Kg/m)		320,7	TOTAL (Kg/m)		384,2	TOTAL (Kg/m)		459,4	TOTAL (Kg/m)		550,3

DIMENSÕES DAS DOBRAS DAS ARMADURAS																			
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)
1	100	50	1	100	50	1	100	50	1	100	54	1	100	58	1	100	58	1	100
2	250	12	2	250	12	2	250	12	2	250	17	2	250	22	2	250	22	2	250
3	100	160	3	100	160	3	100	160	3	100	165	3	100	170	3	100	170	3	100
4	150	80	4	150	80	4	150	80	4	154	80	4	158	80	4	158	80	4	161
5	232	----	5	232	----	5	232	----	5	242	----	5	252	----	5	252	----	5	262
6	232	----	6	232	----	6	232	----	6	242	----	6	252	----	6	252	----	6	262
7	122	----	7	----	----	7	122	----	7	143	----	7	164	----	7	164	----	7	185
8	116	----	8	----	----	8	116	----	8	135	----	8	153	----	8	153	----	8	172

BDCC 250 X 250					TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																										
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000												
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,19 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,25 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,37 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,44 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa												
LISTA DE FERROS (POR METRO LINEAR)																																															
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS												
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)								
1	8,0	18,2	250	c.11	1	12,5	18,2	250	c.11	1	12,5	18,2	254	c.11	1	16,0	16,7	254	c.12	1	16,0	18,2	258	c.11	1	16,0	20,0	258	c.10	1	16,0	20,0	261	c.10	1	20,0	16,7	265	c.12								
2	8,0	33,3	262	c.12	2	10,0	36,4	262	c.11	2	10,0	40,0	267	c.10	2	12,5	33,3	267	c.12	2	12,5	36,4	272	c.11	2	12,5	40,0	272	c.10	2	16,0	30,8	277	c.13	2	16,0	33,3	282	c.12								
3	6,3	8,0	285	c.50	3	6,3	18,2	285	c.22	3	6,3	8,0	290	c.50	3	6,3	10,0	290	c.40	3	6,3	8,0	295	c.50	3	6,3	8,0	295	c.50	3	6,3	8,0	300	c.50	3	6,3	8,0	305	c.50								
4	6,3	8,0	250	c.50	4	6,3	18,2	250	c.22	4	6,3	8,0	254	c.50	4	6,3	10,0	254	c.40	4	6,3	8,0	258	c.50	4	6,3	8,0	258	c.50	4	6,3	8,0	261	c.50	4	6,3	8,0	265	c.50								
5	6,3	8,0	282	c.25	5	6,3	8,0	282	c.25	5	6,3	9,5	292	c.21	5	6,3	9,5	292	c.21	5	6,3	11,8	302	c.17	5	6,3	11,8	302	c.17	5	8,0	9,1	312	c.22	5	8,0	10,5	322	c.19								
6	6,3	8,0	282	c.25	6	6,3	8,0	282	c.25	6	6,3	9,5	292	c.21	6	6,3	9,5	292	c.21	6	6,3	11,8	302	c.17	6	6,3	11,8	302	c.17	6	8,0	9,1	312	c.22	6	8,0	10,5	322	c.19								
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	183	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	246	c.20								
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	270	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	380	c.20								
9	6,3	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	10,0	52,0	Corr.	c.20								
10	6,3	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20								
11	6,3	22,0	Corr.	c.20	11	6,3	26,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20	11	6,3	22,0	Corr.	c.20								
12	6,3	44,0	Corr.	c.20	12	6,3	52,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20	12	6,3	44,0	Corr.	c.20								
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----								
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	100,0	Var.	c.16	14	6,3	123,1	Var.	c.13	14	8,0	106,7	Var.	c.15								
15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	----	----	----	----	15	----	----	----	----	15	6,3	145,5	Var.	c.11	15	8,0	100,0	Var.	c.16	15	8,0	145,5	Var.	c.11	15	10,0	123,1	Var.	c.13								
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																															
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																										
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)									
6,3	0,252	79,0	6,3	0,252	79,0	6,3	0,252	70,7	6,3	0,252	80,1	6,3	0,252	63,7	6,3	0,252	60,6	6,3	0,252	47,8	6,3	0,252	28,1	8,0	0,393	52,2	8,0	0,393	20,4	8,0	0,393	20,4	8,0	0,393	40,9	8,0	0,393	64,1	8,0	0,393	82,2	8,0	0,393	110,6	8,0	0,393	89,8
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	66,6	10,0	0,624	----	10,0	0,624	----	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	117,9	12,5	0,988	----	12,5	0,988	45,0	12,5	0,988	45,7	12,5	0,988	87,8	12,5	0,988	97,8	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	----
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	94,9	16,0	1,570	102,0	16,0	1,570	109,3	16,0	1,570	244,2	16,0	1,570	175,7	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	109,8			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----			
TOTAL (Kg/m)		159,5	TOTAL (Kg/m)		232,2	TOTAL (Kg/m)		231,7	TOTAL (Kg/m)		303,7	TOTAL (Kg/m)		327,6	TOTAL (Kg/m)		392,0	TOTAL (Kg/m)		435,0	TOTAL (Kg/m)		521,3																								
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																															
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)									
1	100	50	1	100	50	1	100	54	1	100	54	1	100	58	1	100	58	1	100	61	1	100	65	2	250	12	2	250	12	2	250	17	2	250	22	2	250	22	2	250	27	2	250	32			
3	100	185	3	100	185	3	100	190	3	100	190	3	100	195	3	100	195	3	100	200	3	100	205	4	150	100	4	150	100	4	154	100	4	154	100	4	158	100	4	158	100	4	161	100	4	165	100
5	282	----	5	282	----	5	292	----	5	292	----	5	302	----	5	302	----	5	312	----	5	322	----	6	282	----	6	282	----	6	292	----	6	292	----	6	302	----	6	302	----	6	312	----	6	312	----
7	122	----	7	----	----	7	143	----	7	143	----	7	164	----	7	164	----	7	185	----	7	206	----	8	116	----	8	----	----	8	135	----	8	135	----	8	153	----	8	153	----	8	172	----	8	190	----

				ASSINATURA DAS AUTORIDADES				 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																			
				Engª Selma Schwab Coordenadora da DP/GNT				Engº Mário S.Bortone Gerente da DP/GPE				Engº Roger Gama Veloso Diretor de Projetos				Engº Nelson A. Reis Vice - Diretor Geral				CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)				DES. 65			

BDCC 250 X 300					TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																						
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,15 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,38 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa																								
LISTA DE FERROS (POR METRO LINEAR)																																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																				
1	8,0	18,2	250	c.11	1	12,5	18,2	250	c.11	1	12,5	18,2	254	c.11	1	16,0	16,7	254	c.12	1	16,0	18,2	258	c.11	1	16,0	20,0	258	c.10	1	16,0	20,0	261	c.10	1	20,0	16,7	265	c.12																				
2	8,0	36,4	262	c.11	2	10,0	36,4	262	c.11	2	10,0	40,0	267	c.10	2	12,5	30,8	267	c.13	2	12,5	36,4	272	c.11	2	12,5	40,0	272	c.10	2	12,5	40,0	277	c.10	2	16,0	33,3	282	c.12																				
3	6,3	11,8	310	c.34	3	8,0	20,0	310	c.20	3	6,3	13,3	315	c.30	3	6,3	18,2	315	c.22	3	6,3	13,3	320	c.30	3	6,3	16,7	320	c.24	3	6,3	9,1	325	c.44	3	6,3	8,0	330	c.50																				
4	6,3	11,8	270	c.34	4	8,0	20,0	270	c.20	4	6,3	13,3	274	c.30	4	6,3	18,2	274	c.22	4	6,3	13,3	278	c.30	4	6,3	16,7	278	c.24	4	6,3	9,1	281	c.44	4	6,3	8,0	285	c.50																				
5	6,3	8,0	332	c.25	5	6,3	8,0	332	c.25	5	6,3	9,5	342	c.21	5	6,3	9,5	342	c.21	5	6,3	11,8	352	c.17	5	6,3	11,8	352	c.17	5	8,0	9,1	362	c.22	5	8,0	10,5	372	c.19																				
6	6,3	8,0	332	c.25	6	6,3	8,0	332	c.25	6	6,3	9,5	342	c.21	6	6,3	9,5	342	c.21	6	6,3	11,8	352	c.17	6	6,3	11,8	352	c.17	6	8,0	9,1	362	c.22	6	8,0	10,5	372	c.19																				
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	183	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	246	c.20																				
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	270	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	380	c.20																				
9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	10,0	52,0	Corr.	c.20																				
10	6,3	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20																				
11	6,3	28,0	Corr.	c.20	11	6,3	30,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20	11	6,3	28,0	Corr.	c.20																				
12	6,3	56,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20	12	6,3	56,0	Corr.	c.20																				
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----																				
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	106,7	Var.	c.15	14	6,3	123,1	Var.	c.13																				
15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	----	----	----	----	15	6,3	145,5	Var.	c.11	15	6,3	160,0	Var.	c.10	15	8,0	133,3	Var.	c.12	15	8,0	145,5	Var.	c.11	15	10,0	123,1	Var.	c.13																				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																											
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																			
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																					
6,3	0,252	92,0	6,3	0,252	46,4	6,3	0,252	86,4	6,3	0,252	100,4	6,3	0,252	103,9	6,3	0,252	83,1	6,3	0,252	54,9	6,3	0,252	33,6	6,3	0,252	54,9	6,3	0,252	33,6	6,3	0,252	33,6	6,3	0,252	33,6	6,3	0,252	33,6																					
8,0	0,393	55,4	8,0	0,393	86,5	8,0	0,393	20,4	8,0	0,393	40,9	8,0	0,393	40,9	8,0	0,393	79,4	8,0	0,393	32,4	8,0	0,393	114,1	8,0	0,393	32,4	8,0	0,393	114,1	8,0	0,393	32,4	8,0	0,393	114,1	8,0	0,393	114,1																					
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	66,6	10,0	0,624	----	10,0	0,624	----	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	32,4	10,0	0,624	117,9																					
12,5	0,988	----	12,5	0,988	45,0	12,5	0,988	45,7	12,5	0,988	81,2	12,5	0,988	97,8	12,5	0,988	107,5	12,5	0,988	109,5	12,5	0,988	109,5	12,5	0,988	109,5	12,5	0,988	109,5	12,5	0,988	109,5	12,5	0,988	109,5	12,5	0,988	----																					
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	94,9	16,0	1,570	102,0	16,0	1,570	109,3	16,0	1,570	110,2	16,0	1,570	110,2	16,0	1,570	110,2	16,0	1,570	110,2	16,0	1,570	110,2	16,0	1,570	110,2	16,0	1,570	175,7																					
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	109,8																					
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----																					
TOTAL (Kg/m)		175,7	TOTAL (Kg/m)		265,7	TOTAL (Kg/m)		247,4	TOTAL (Kg/m)		317,4	TOTAL (Kg/m)		344,6	TOTAL (Kg/m)		411,7	TOTAL (Kg/m)		421,1	TOTAL (Kg/m)		533,0																																				
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																											
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																					
1	100	50	1	100	50	1	100	54	1	100	54	1	100	58	1	100	58	1	100	58	1	100	61	1	100	61	1	100	65	1	100	65	1	100	65	1	100	65																					
2	250	12	2	250	12	2	250	17	2	250	17	2	250	22	2	250	22	2	250	22	2	250	27	2	250	27	2	250	32	2	250	32	2	250	32	2	250	32																					
3	100	210	3	100	210	3	100	215	3	100	215	3	100	220	3	100	220	3	100	220	3	100	225	3	100	225	3	100	230	3	100	230	3	100	230	3	100	230																					
4	150	120	4	150	120	4	154	120	4	154	120	4	154	120	4	158	120	4	158	120	4	158	120	4	161	120	4	161	120	4	165	120	4	165	120	4	165	120																					
5	332	----	5	332	----	5	342	----	5	342	----	5	352	----	5	352	----	5	352	----	5	362	----	5	362	----	5	362	----	5	372	----	5	372	----	5	372	----																					
6	332	----	6	332	----	6	342	----	6	342	----	6	342	----	6	352	----	6	352	----	6	362	----	6	362	----	6	362	----	6	372	----	6	372	----	6	372	----																					
7	122	----	7	----	----	7	143	----	7	143	----	7	143	----	7	164	----	7	164	----	7	164	----	7	185	----	7	185	----	7	206	----	7	206	----	7	206	----																					
8	116	----	8	----	----	8	135	----	8	135	----	8	135	----	8	153	----	8	153	----	8	153	----	8	172	----	8	172	----	8	190	----	8	190	----	8	190	----																					
										ASSINATURA DAS AUTORIDADES										 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																							
										Eng° Selma Schwab Coordenadora da DP/GNT										Eng° Mário S.Bortone Gerente da DP/GPE										Eng° Roger Gama Veloso Diretor de Projetos										Eng° Nelson A. Reis Vice - Diretor Geral										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)									
																														DES. 66																													

BDCC 250 X 350	TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,27 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,46 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	20,0	250	c.10	1	12,5	18,2	250	c.11	1	12,5	16,7	254	c.12	1	16,0	15,4	254	c.13	1	16,0	18,2	258	c.11
2	8,0	40,0	262	c.10	2	10,0	36,4	262	c.11	2	10,0	40,0	267	c.10	2	12,5	30,8	267	c.13	2	12,5	36,4	272	c.11
3	6,3	20,0	335	c.20	3	10,0	16,7	335	c.24	3	8,0	16,7	340	c.24	3	10,0	16,7	340	c.24	3	8,0	16,7	345	c.24
4	6,3	20,0	290	c.20	4	10,0	16,7	290	c.24	4	8,0	16,7	294	c.24	4	10,0	16,7	294	c.24	4	8,0	18,2	298	c.22
5	6,3	8,0	382	c.25	5	6,3	8,0	382	c.25	5	6,3	9,5	392	c.21	5	6,3	9,5	392	c.21	5	6,3	11,8	402	c.17
6	6,3	8,0	382	c.25	6	6,3	8,0	382	c.25	6	6,3	9,5	392	c.21	6	6,3	11,8	402	c.17	6	6,3	11,8	402	c.17
7	6,3	20,0	162	c.20	7	----	----	----	----	7	6,3	20,0	183	c.20	7	6,3	20,0	183	c.20	7	6,3	20,0	204	c.20
8	6,3	10,0	232	c.20	8	----	----	----	----	8	6,3	10,0	270	c.20	8	6,3	10,0	270	c.20	8	6,3	10,0	306	c.20
9	6,3	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	8,0	52,0	Corr.	c.20	9	10,0	52,0	Corr.	c.20
10	6,3	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	8,0	52,0	Corr.	c.20	10	10,0	52,0	Corr.	c.20
11	6,3	32,0	Corr.	c.20	11	6,3	36,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20
12	6,3	64,0	Corr.	c.20	12	6,3	72,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	114,3	Var.	c.14
15	----	----	----	----	15	6,3	100,0	Var.	c.16	15	----	----	----	----	15	6,3	133,3	Var.	c.12	15	6,3	145,5	Var.	c.11

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50					
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	111,3	6,3	0,252	53,0	6,3	0,252	59,0	6,3	0,252	77,1	6,3	0,252	87,7	6,3	0,252	65,1	6,3	0,252	45,7	6,3	0,252	16,1
8,0	0,393	60,8	8,0	0,393	40,9	8,0	0,393	82,5	8,0	0,393	40,9	8,0	0,393	86,9	8,0	0,393	77,0	8,0	0,393	140,0	8,0	0,393	109,8
10,0	0,624	----	10,0	0,624	124,6	10,0	0,624	66,6	10,0	0,624	66,1	10,0	0,624	----	10,0	0,624	99,5	10,0	0,624	64,9	10,0	0,624	192,1
12,5	0,988	----	12,5	0,988	45,0	12,5	0,988	41,9	12,5	0,988	81,2	12,5	0,988	89,5	12,5	0,988	97,8	12,5	0,988	109,5	12,5	0,988	----
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	89,7	16,0	1,570	95,9	16,0	1,570	102,0	16,0	1,570	110,2	16,0	1,570	162,2
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	99,7
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		200,4	TOTAL (Kg/m)		291,8	TOTAL (Kg/m)		278,3	TOTAL (Kg/m)		355,0	TOTAL (Kg/m)		360,0	TOTAL (Kg/m)		441,4	TOTAL (Kg/m)		470,3	TOTAL (Kg/m)		579,9

DIMENSÕES DAS DOBRAS DAS ARMADURAS																							
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)
1	100	50	1	100	50	1	100	54	1	100	54	1	100	58	1	100	58	1	100	61	1	100	61
2	250	12	2	250	12	2	250	17	2	250	17	2	250	22	2	250	22	2	250	27	2	250	27
3	100	235	3	100	235	3	100	240	3	100	240	3	100	245	3	100	245	3	100	250	3	100	250
4	150	140	4	150	140	4	154	140	4	154	140	4	158	140	4	158	140	4	161	140	4	161	140
5	382	----	5	382	----	5	392	----	5	392	----	5	402	----	5	402	----	5	412	----	5	412	----
6	382	----	6	382	----	6	392	----	6	392	----	6	402	----	6	402	----	6	412	----	6	412	----
7	122	----	7	----	----	7	143	----	7	143	----	7	164	----	7	164	----	7	185	----	7	185	----
8	116	----	8	----	----	8	135	----	8	135	----	8	153	----	8	153	----	8	172	----	8	172	----

ASSINATURA DAS AUTORIDADES												 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS									
												CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear) DES. 67									
Eng ^o Selma Schwab Coordenadora da DP/GNT				Eng ^o Mário S.Bortone Gerente da DP/GPE				Eng ^o Roger Gama Veloso Diretor de Projetos				Eng ^o Nelson A. Reis Vice - Diretor Geral									

BDCC 300 X 200		TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																									
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,19 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,25 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,31 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,37 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,43 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa																								
LISTA DE FERROS (POR METRO LINEAR)																																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																				
1	8,0	20,0	299	c.10	1	12,5	18,2	299	c.11	1	16,0	16,7	299	c.12	1	16,0	20,0	302	c.10	1	20,0	15,4	302	c.13	1	20,0	16,7	306	c.12	1	20,0	18,2	310	c.11	1	20,0	20,0	314	c.10																				
2	8,0	30,8	317	c.13	2	10,0	40,0	317	c.10	2	12,5	33,3	317	c.12	2	12,5	40,0	322	c.10	2	16,0	30,8	322	c.13	2	16,0	33,3	327	c.12	2	16,0	36,4	332	c.11	2	20,0	30,8	337	c.13																				
3	6,3	8,0	285	c.50	3	6,3	12,5	285	c.32	3	6,3	10,0	285	c.40	3	6,3	8,0	290	c.50	3	6,3	10,5	290	c.38	3	6,3	8,0	295	c.50	3	6,3	8,0	300	c.50	3	6,3	8,0	305	c.50																				
4	6,3	8,0	259	c.50	4	6,3	12,5	259	c.32	4	6,3	10,0	259	c.40	4	6,3	8,0	262	c.50	4	6,3	10,5	262	c.38	4	6,3	8,0	266	c.50	4	6,3	8,0	270	c.50	4	6,3	8,0	274	c.50																				
5	6,3	9,5	342	c.21	5	6,3	9,5	242	c.21	5	6,3	9,5	242	c.21	5	6,3	11,8	252	c.17	5	6,3	11,8	252	c.17	5	8,0	9,1	262	c.22	5	8,0	10,5	272	c.19	5	8,0	11,8	282	c.17																				
6	6,3	9,5	342	c.21	6	6,3	9,5	242	c.21	6	6,3	9,5	242	c.21	6	6,3	11,8	252	c.17	6	6,3	11,8	252	c.17	6	8,0	9,1	262	c.22	6	8,0	10,5	272	c.19	6	8,0	11,8	282	c.17																				
7	6,3	20,0	183	c.20	7	----	----	----	----	7	6,3	20,0	183	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	204	c.20	7	8,0	20,0	225	c.20	7	10,0	20,0	246	c.20	7	10,0	20,0	267	c.20																				
8	6,3	10,0	270	c.20	8	----	----	----	----	8	6,3	10,0	270	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	306	c.20	8	8,0	10,0	344	c.20	8	10,0	10,0	380	c.20	8	10,0	10,0	416	c.20																				
9	6,3	60,0	Corr.	c.20	9	6,3	60,0	Corr.	c.20	9	8,0	60,0	Corr.	c.20	9	8,0	60,0	Corr.	c.20	9	10,0	60,0	Corr.	c.20	9	10,0	60,0	Corr.	c.20	9	10,0	60,0	Corr.	c.20	9	10,0	60,0	Corr.	c.20																				
10	6,3	60,0	Corr.	c.20	10	8,0	60,0	Corr.	c.20	10	8,0	60,0	Corr.	c.20	10	10,0	60,0	Corr.	c.20	10	10,0	60,0	Corr.	c.20	10	10,0	60,0	Corr.	c.20	10	12,5	60,0	Corr.	c.20	10	12,5	60,0	Corr.	c.20																				
11	6,3	18,0	Corr.	c.20	11	6,3	20,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20	11	6,3	18,0	Corr.	c.20																				
12	6,3	36,0	Corr.	c.20	12	6,3	40,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20	12	6,3	36,0	Corr.	c.20																				
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----																				
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	80,0	Var.	c.20	14	6,3	106,7	Var.	c.15	14	6,3	133,3	Var.	c.12	14	8,0	123,1	Var.	c.12	14	8,0	123,1	Var.	c.13																				
15	----	----	----	----	15	6,3	84,2	Var.	c.19	15	6,3	133,3	Var.	c.12	15	8,0	100,0	Var.	c.16	15	8,0	160,0	Var.	c.10	15	10,0	100,0	Var.	c.16	15	10,0	106,7	Var.	c.15	15	10,0	145,5	Var.	c.11																				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																											
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																			
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																					
6,3	0,252	82,4	6,3	0,252	68,7	6,3	0,252	73,1	6,3	0,252	39,7	6,3	0,252	55,1	6,3	0,252	42,1	6,3	0,252	48,3	6,3	0,252	25,3	8,0	0,393	61,9	8,0	0,393	23,6	8,0	0,393	47,2	8,0	0,393	74,8	8,0	0,393	65,2	8,0	0,393	49,9	8,0	0,393	22,4	8,0	0,393	62,0												
10,0	0,624	----	10,0	0,624	79,1	10,0	0,624	----	10,0	0,624	37,4	10,0	0,624	74,9	10,0	0,624	114,8	10,0	0,624	137,8	10,0	0,624	163,9	12,5	0,988	----	12,5	0,988	53,8	12,5	0,988	104,3	12,5	0,988	127,3	12,5	0,988	----	12,5	0,988	59,3	12,5	0,988	59,3															
16,0	1,570	28,3	16,0	1,570	28,3	16,0	1,570	106,7	16,0	1,570	123,1	16,0	1,570	184,0	16,0	1,570	199,2	16,0	1,570	218,0	16,0	1,570	28,3	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	115,3	20,0	2,480	126,7	20,0	2,480	139,9	20,0	2,480	413,2															
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----															
TOTAL (Kg/m)		172,6	TOTAL (Kg/m)		253,5	TOTAL (Kg/m)		331,3	TOTAL (Kg/m)		402,3	TOTAL (Kg/m)		494,5	TOTAL (Kg/m)		532,7	TOTAL (Kg/m)		625,7	TOTAL (Kg/m)		752,0																																				
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																											
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																					
1	120	59	1	120	59	1	120	59	1	120	62	1	120	62	1	120	66	1	120	70	1	120	74	2	300	17	2	300	17	2	300	32	2	300	37	3	120	165	3	120	165	3	120	170	3	120	170	3	120	170	3	120	185						
4	179	80	4	179	80	4	179	80	4	182	80	4	182	80	4	186	80	4	190	80	4	194	80	5	242	----	5	242	----	5	252	----	5	262	----	5	272	----	5	282	----	6	242	----	6	242	----	6	252	----	6	262	----	6	272	----	6	282	----
7	143	----	7	----	----	7	143	----	7	164	----	7	164	----	7	185	----	7	206	----	7	227	----	8	135	----	8	----	----	8	135	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----	8	153	----			

				ASSINATURA DAS AUTORIDADES				 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																			
				Engª Selma Schwab Coordenadora da DP/GNT				Engº Mário S.Bortone Gerente da DP/GPE				Engº Roger Gama Veloso Diretor de Projetos				Engº Nelson A. Reis Vice - Diretor Geral				CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)				DES. 68			

BDCC 350 X 350					TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																		
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000				
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,22 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,28 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,34 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,47 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,59 MPa				
LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	10,0	18,2	344	c.11	1	16,0	16,7	344	c.11	1	16,0	20,0	348	c.10	1	20,0	16,7	351	c.12	1	20,0	18,2	351	c.11	1	20,0	20,0	355	c.10	1	20,0	20,0	359	c.10	1	25,0	16,7	362	c.12
2	10,0	33,3	367	c.12	2	12,5	36,4	367	c.12	2	12,5	40,0	372	c.10	2	16,0	33,3	377	c.12	2	16,0	36,4	377	c.11	2	16,0	40,0	382	c.10	2	20,0	30,8	387	c.13	2	20,0	36,4	392	c.11
3	6,3	14,3	380	c.28	3	10,0	16,7	380	c.24	3	8,0	15,4	385	c.26	3	6,3	20,0	390	c.20	3	8,0	18,2	390	c.22	3	8,0	15,4	395	c.26	3	6,3	15,4	400	c.26	3	6,3	11,8	405	c.34
4	6,3	14,3	344	c.28	4	10,0	16,7	344	c.24	4	8,0	15,4	348	c.26	4	6,3	20,0	351	c.20	4	8,0	18,2	351	c.22	4	8,0	15,4	355	c.26	4	6,3	15,4	359	c.26	4	6,3	11,8	362	c.34
5	6,3	9,5	392	c.21	5	6,3	9,5	392	c.21	5	6,3	11,8	402	c.17	5	8,0	9,1	412	c.22	5	8,0	9,1	412	c.22	5	8,0	10,5	422	c.19	5	8,0	11,8	432	c.17	5	10,0	8,7	442	c.23
6	6,3	9,5	392	c.21	6	6,3	9,5	392	c.21	6	6,3	11,8	402	c.17	6	8,0	9,1	412	c.22	6	8,0	9,1	412	c.22	6	8,0	10,5	422	c.19	6	8,0	11,8	432	c.17	6	10,0	8,7	442	c.23
7	6,3	20,0	183	c.20	7	----	----	----	----	7	6,3	20,0	204	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	246	c.20	7	10,0	20,0	267	c.20	7	10,0	20,0	288	c.20
8	6,3	10,0	270	c.20	8	----	----	----	----	8	6,3	10,0	306	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	380	c.20	8	10,0	10,0	416	c.20	8	10,0	10,0	454	c.20
9	6,3	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	12,5	72,0	Corr.	c.20
10	6,3	72,0	Corr.	c.20	10	8,0	72,0	Corr.	c.20	10	10,0	72,0	Corr.	c.20	10	10,0	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	16,0	72,0	Corr.	c.20
11	6,3	32,0	Corr.	c.20	11	6,3	36,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20	11	6,3	32,0	Corr.	c.20
12	6,3	64,0	Corr.	c.20	12	6,3	72,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20	12	6,3	64,0	Corr.	c.20
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	100,0	Var.	c.16	14	6,3	133,3	Var.	c.12	14	6,3	100,0	Var.	c.16	14	8,0	100,0	Var.	c.16	14	8,0	145,5	Var.	c.11
15	----	----	----	----	15	6,3	123,1	Var.	c.13	15	6,3	123,1	Var.	c.13	15	8,0	100,0	Var.	c.16	15	8,0	160,0	Var.	c.10	15	10,0	106,7	Var.	c.15	15	10,0	114,3	Var.	c.14	15	10,0	160,0	Var.	c.10

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	121,4	6,3	0,252	60,3	6,3	0,252	84,4	6,3	0,252	61,5	6,3	0,252	40,3	6,3	0,252	47,4	6,3	0,252	53,6	6,3	0,252	47,0
8,0	0,393	---	8,0	0,393	28,3	8,0	0,393	44,4	8,0	0,393	85,8	8,0	0,393	153,9	8,0	0,393	114,5	8,0	0,393	69,1	8,0	0,393	45,2
10,0	0,624	115,3	10,0	0,624	120,4	10,0	0,624	89,9	10,0	0,624	89,9	10,0	0,624	44,9	10,0	0,624	90,9	10,0	0,624	157,0	10,0	0,624	191,1
12,5	0,988	---	12,5	0,988	132,0	12,5	0,988	147,0	12,5	0,988	---	12,5	0,988	71,1	12,5	0,988	71,1	12,5	0,988	71,1	12,5	0,988	71,1
16,0	1,570	28,3	16,0	1,570	118,5	16,0	1,570	137,5	16,0	1,570	225,4	16,0	1,570	243,7	16,0	1,570	268,2	16,0	1,570	28,3	16,0	1,570	141,3
20,0	2,480	---	20,0	2,480	---	20,0	2,480	---	20,0	2,480	145,4	20,0	2,480	158,4	20,0	2,480	176,1	20,0	2,480	473,7	20,0	2,480	353,9
25,0	3,980	---	25,0	3,980	---	25,0	3,980	---	25,0	3,980	---	25,0	3,980	---	25,0	3,980	---	25,0	3,980	---	25,0	3,980	240,6
TOTAL (Kg/m)		265,0	TOTAL (Kg/m)		459,5	TOTAL (Kg/m)		503,2	TOTAL (Kg/m)		608,0	TOTAL (Kg/m)		712,3	TOTAL (Kg/m)		768,2	TOTAL (Kg/m)		852,8	TOTAL (Kg/m)		1090,2

DIMENSÕES DAS DOBRAS DAS ARMADURAS																					
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	
1	140	64	1	140	64	1	140	68	1	140	71	1	140	71	1	140	75	1	140	79	
2	350	17	2	350	17	2	350	22	2	350	27	2	350	27	2	350	32	2	350	37	
3	140	240	3	140	240	3	140	245	3	140	250	3	140	250	3	140	255	3	140	260	
4	204	140	4	204	140	4	208	140	4	211	140	4	211	140	4	215	140	4	219	140	
5	392	----	5	392	----	5	402	----	5	412	----	5	412	----	5	422	----	5	432	----	
6	392	----	6	392	----	6	402	----	6	412	----	6	412	----	6	422	----	6	432	----	
7	143	----	7	----	----	7	164	----	7	185	----	7	185	----	7	206	----	7	227	----	
8	135	----	8	----	----	8	153	----	8	172	----	8	172	----	8	190	----	8	208	----	

[illegible]

BDCC 350 X 400	TABELA DE ARMADURA PARA BUEIRO DUPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,29 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,42 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,48 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,60 MPa

LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	10,0	20,0	344	c.10	1	16,0	18,2	344	c.11	1	16,0	20,0	348	c.10	1	20,0	15,4	351	c.13	1	20,0	18,2	351	c.11	1	20,0	20,0	355	c.10	1	20,0	20,0	359	c.10	1	25,0	16,7	362	c.12
2	10,0	36,4	367	c.11	2	12,5	36,4	367	c.11	2	12,5	40,0	372	c.10	2	16,0	33,3	377	c.12	2	16,0	36,4	377	c.11	2	16,0	40,0	382	c.10	2	16,0	40,0	387	c.10	2	20,0	33,3	392	c.12
3	6,3	20,0	405	c.20	3	10,0	20,0	405	c.20	3	10,0	15,4	410	c.26	3	8,0	20,0	415	c.20	3	10,0	20,0	415	c.20	3	10,0	16,7	420	c.24	3	8,0	18,2	425	c.22	3	8,0	16,7	430	c.24
4	6,3	20,0	364	c.20	4	10,0	20,0	364	c.20	4	10,0	15,4	368	c.26	4	8,0	20,0	371	c.20	4	10,0	20,0	371	c.20	4	10,0	16,7	375	c.24	4	8,0	18,2	379	c.22	4	8,0	16,7	382	c.24
5	6,3	9,5	442	c.21	5	6,3	9,5	442	c.21	5	6,3	11,8	452	c.17	5	8,0	9,1	462	c.22	5	8,0	9,1	462	c.22	5	8,0	10,5	472	c.19	5	8,0	11,8	482	c.17	5	10,0	8,7	492	c.23
6	6,3	9,5	442	c.21	6	6,3	9,5	442	c.21	6	6,3	11,8	452	c.17	6	8,0	9,1	462	c.22	6	8,0	9,1	462	c.22	6	8,0	10,5	472	c.19	6	8,0	11,8	482	c.17	6	10,0	8,7	492	c.23
7	6,3	20,0	183	c.20	7	----	----	----	----	7	6,3	20,0	204	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	225	c.20	7	8,0	20,0	246	c.20	7	10,0	20,0	267	c.20	7	10,0	20,0	288	c.20
8	6,3	10,0	270	c.20	8	----	----	----	----	8	6,3	10,0	306	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	344	c.20	8	8,0	10,0	380	c.20	8	10,0	10,0	416	c.20	8	10,0	10,0	454	c.20
9	6,3	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	10,0	72,0	Corr.	c.20	9	12,5	72,0	Corr.	c.20	9	12,5	72,0	Corr.	c.20	9	12,5	72,0	Corr.	c.20	9	12,5	72,0	Corr.	c.20
10	8,0	72,0	Corr.	c.20	10	8,0	72,0	Corr.	c.20	10	10,0	72,0	Corr.	c.20	10	10,0	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20	10	12,5	72,0	Corr.	c.20
11	6,3	38,0	Corr.	c.20	11	8,0	40,0	Corr.	c.20	11	6,3	38,0	Corr.	c.20	11	8,0	38,0	Corr.	c.20	11	8,0	38,0	Corr.	c.20	11	6,3	38,0	Corr.	c.20	11	6,3	38,0	Corr.	c.20	11	6,3	38,0	Corr.	c.20
12	6,3	76,0	Corr.	c.20	12	6,3	80,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20	12	6,3	76,0	Corr.	c.20
13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----	13	16,0	18,0	Corr.	----
14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	----	----	----	----	14	6,3	106,7	Var.	c.15	14	6,3	145,5	Var.	c.11	14	8,0	106,7	Var.	c.15	14	8,0	145,5	Var.	c.11
15	----	----	----	----	15	6,3	123,1	Var.	c.13	15	6,3	123,1	Var.	c.13	15	8,0	100,0	Var.	c.16	15	8,0	145,5	Var.	c.11	15	10,0	106,7	Var.	c.15	15	10,0	114,3	Var.	c.14	15	10,0	145,5	Var.	c.11

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	122,8	6,3	0,252	55,6	6,3	0,252	91,9	6,3	0,252	28,7	6,3	0,252	36,4	6,3	0,252	54,0	6,3	0,252	28,7	6,3	0,252	28,7
8,0	0,393	28,3	8,0	0,393	44,0	8,0	0,393	----	8,0	0,393	151,2	8,0	0,393	115,8	8,0	0,393	73,2	8,0	0,393	133,2	8,0	0,393	98,5
10,0	0,624	126,3	10,0	0,624	140,9	10,0	0,624	164,6	10,0	0,624	89,9	10,0	0,624	98,1	10,0	0,624	128,8	10,0	0,624	112,1	10,0	0,624	189,4
12,5	0,988	----	12,5	0,988	132,0	12,5	0,988	147,0	12,5	0,988	----	12,5	0,988	142,3	12,5	0,988	142,3	12,5	0,988	142,3	12,5	0,988	142,3
16,0	1,570	28,3	16,0	1,570	126,6	16,0	1,570	137,5	16,0	1,570	225,4	16,0	1,570	243,7	16,0	1,570	268,2	16,0	1,570	271,3	16,0	1,570	28,3
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	134,1	20,0	2,480	158,4	20,0	2,480	176,1	20,0	2,480	178,1	20,0	2,480	323,7
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	240,6
TOTAL (Kg/m)		305,7	TOTAL (Kg/m)		499,1	TOTAL (Kg/m)		541,0	TOTAL (Kg/m)		629,3	TOTAL (Kg/m)		794,7	TOTAL (Kg/m)		842,6	TOTAL (Kg/m)		865,7	TOTAL (Kg/m)		1051,5

DIMENSÕES DAS DOBRAS DAS ARMADURAS																							
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)
1	140	64	1	140	64	1	140	68	1	140	71	1	140	71	1	140	75	1	140	79	1	140	82
2	350	17	2	350	17	2	350	22	2	350	27	2	350	27	2	350	32	2	350	37	2	350	42
3	140	265	3	140	265	3	140	270	3	140	275	3	140	275	3	140	280	3	140	285	3	140	290
4	204	160	4	204	160	4	208	160	4	211	160	4	211	160	4	215	160	4	219	160	4	222	160
5	442	----	5	442	----	5	452	----	5	462	----	5	462	----	5	472	----	5	482	----	5	492	----
6	442	----	6	442	----	6	452	----	6	462	----	6	462	----	6	472	----	6	482	----	6	492	----
7	143	----	7	----	----	7	164	----	7	185	----	7	185	----	7	206	----	7	227	----	7	248	----
8	135	----	8	----	----	8	153	----	8	172	----	8	172	----	8	190	----	8	208	----	8	227	----

ASSINATURA DAS AUTORIDADES												 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS
												CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Duplo Celular Fundação Direta (Por Metro Linear)

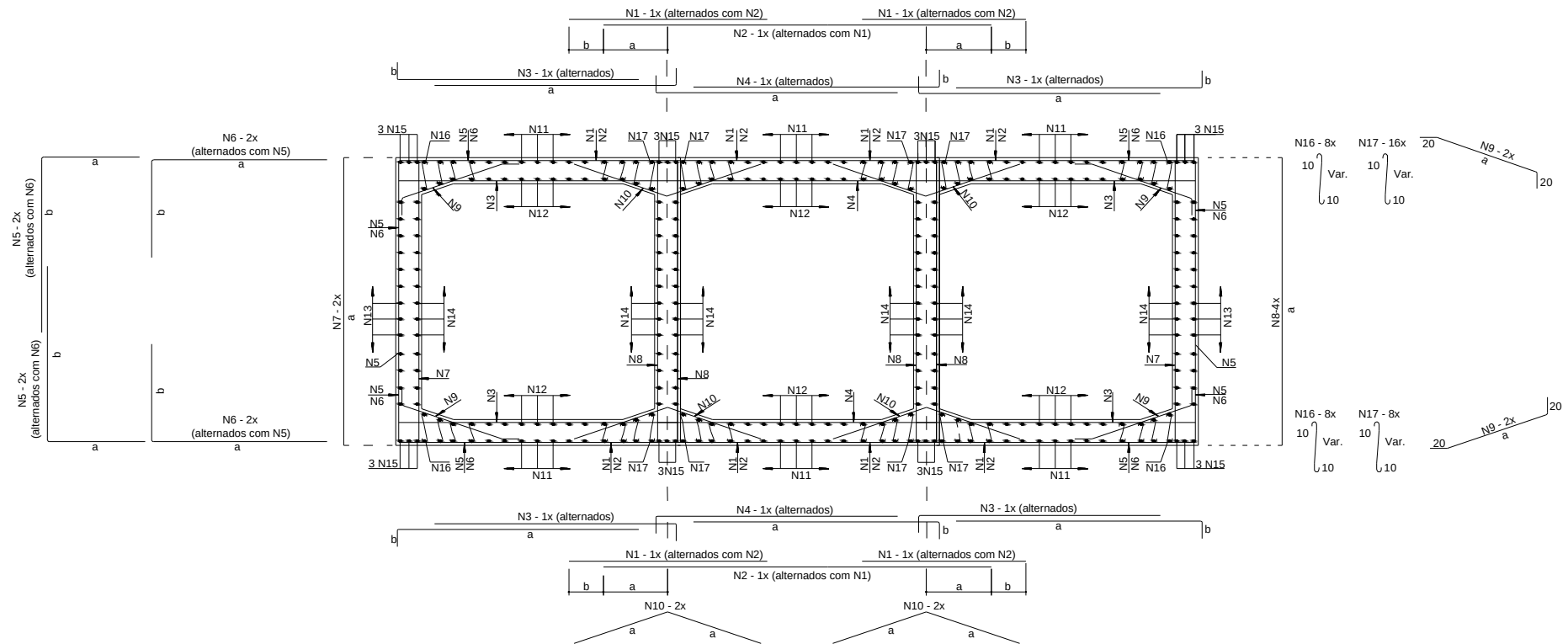
Eng^o Selma Schwab
Coordenadora da DP/GNT

Eng^o Mário S.Bortone
Gerente da DP/GPE

Eng^o Roger Gama Veloso
Diretor de Projetos

Eng^o Nelson A. Reis
Vice - Diretor Geral

BUEIRO TRIPLO CELULAR DE CONCRETO - FUNDAÇÃO DIRETA - ARMAÇÃO TÍPICA



NOTAS:

- 1) Cobrimento das armaduras de 4 cm
- 2) Concreto FCK = 25 MPa
- 3) Aço CA-50
- 4) Estrutura projetada para o TB-45 da NBR-7188
- 5) Em alguns bueiros, não são utilizadas as posições de armadura N9, N10, N16 e N17

ASSINATURA DAS AUTORIDADES

Engª Selma Schwab
Coordenadora da DP/GNT

Engº Mário S.Bortone
Gerente da DP/GPE

Engº Roger Gama Veloso
Diretor de Projetos

Engº Nelson A. Reis
Vice - Diretor Geral



DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS

CADERNO DE BUEIROS CELULARES
Bueiro Triplo Celular de Concreto - Fundação Direta -
Armação Típica

BTCC 150 x 150		TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)					
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,28 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,34 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,52 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	6,3	16,7	160	c.24	1	8,0	18,2	160	c.22	1	6,3	20,0	160	c.20	1	6,3	20,0	165	c.20	1	8,0	18,2	165	c.22
2	6,3	8,3	285	c.24	2	8,0	9,1	285	c.22	2	6,3	10,0	285	c.20	2	6,3	10,0	290	c.20	2	8,0	9,1	290	c.22
3	8,0	33,3	157	c.12	3	8,0	30,8	157	c.13	3	8,0	36,4	157	c.11	3	8,0	40,0	162	c.10	3	10,0	33,3	162	c.12
4	6,3	8,0	157	c.25	4	6,3	12,5	157	c.16	4	6,3	8,0	157	c.25	4	6,3	8,0	162	c.25	4	6,3	10,5	162	c.19
5	6,3	8,0	190	c.50	5	6,3	8,0	190	c.50	5	6,3	8,0	190	c.50	5	6,3	8,0	195	c.50	5	6,3	8,0	195	c.50
6	6,3	8,0	160	c.50	6	6,3	8,0	160	c.50	6	6,3	8,0	160	c.50	6	6,3	8,0	165	c.50	6	6,3	8,0	165	c.50
7	6,3	8,0	172	c.25	7	6,3	8,0	172	c.25	7	6,3	8,0	172	c.25	7	6,3	8,0	182	c.25	7	6,3	9,5	192	c.21
8	6,3	16,0	172	c.25	8	6,3	16,0	172	c.25	8	6,3	16,0	172	c.25	8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25
9	6,3	20,0	141	c.20	9	----	----	----	----	9	6,3	20,0	141	c.20	9	6,3	20,0	162	c.20	9	6,3	20,0	162	c.20
10	6,3	20,0	196	c.20	10	----	----	----	----	10	6,3	20,0	196	c.20	10	6,3	20,0	232	c.20	10	6,3	20,0	232	c.20
11	6,3	48,0	Corr.	c.20	11	6,3	48,0	Corr.	c.20	11	6,3	48,0	Corr.	c.20	11	6,3	48,0	Corr.	c.20	11	6,3	48,0	Corr.	c.20
12	6,3	48,0	Corr.	c.20	12	6,3	48,0	Corr.	c.20	12	6,3	48,0	Corr.	c.20	12	6,3	48,0	Corr.	c.20	12	8,0	48,0	Corr.	c.20
13	6,3	12,0	Corr.	c.20	13	6,3	16,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20
14	6,3	36,0	Corr.	c.20	14	6,3	48,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20
15	12,5	24,0	Corr.	----	15	12,5	24,0	Corr.	----	15	12,5	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----
17	----	----	----	----	17	----	----	----	----	17	----	----	----	----	17	6,3	168,4	Var.	c.19	17	6,3	266,7	Var.	c.12

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																					
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			Ø (mm)
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	
6,3	0,252	86,6	6,3	0,252	62,7	6,3	0,252	89,1	6,3	0,252	93,3	6,3	0,252	98,8	6,3	0,252	99,5	6,3	0,252	112,3	6,3
8,0	0,393	20,5	8,0	0,393	40,6	8,0	0,393	22,5	8,0	0,393	25,5	8,0	0,393	22,2	8,0	0,393	18,9	8,0	0,393	43,8	8,0
10,0	0,624	----	10,0	0,624	----	10,0	0,624	----	10,0	0,624	----	10,0	0,624	33,7	10,0	0,624	72,6	10,0	0,624	41,7	10,0
12,5	0,988	23,7	12,5	0,988	23,7	12,5	0,988	23,7	12,5	0,988	23,7	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5
16,0	1,570	----	16,0	1,570	----	16,0	1,570	----	16,0	1,570	----	16,0	1,570	33,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0
TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			304,8

DIMENSÕES DAS DOBRAS DAS ARMADURAS																			
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)
1	60	40	1	60	40	1	60	40	1	60	45	1	60	45	1	60	45	1	60
2	60	----	2	60	----	2	60	----	2	60	----	2	60	----	2	60	----	2	60
3	150	7	3	150	7	3	150	7	3	150	12	3	150	12	3	150	12	3	150
4	150	7	4	150	7	4	150	7	4	150	12	4	150	12	4	150	12	4	150
5	60	130	5	60	130	5	60	130	5	60	135	5	60	135	5	60	135	5	60
6	100	60	6	100	60	6	100	60	6	105	60	6	105	60	6	105	60	6	110
7	172	----	7	172	----	7	172	----	7	182	----	7	182	----	7	182	----	7	192
8	172	----	8	172	----	8	172	----	8	182	----	8	182	----	8	182	----	8	192
9	101	----	9	----	----	9	101	----	9	122	----	9	122	----	9	122	----	9	143
10	98	----	10	----	----	10	98	----	10	116	----	10	116	----	10	116	----	10	135

ASSINATURA DAS AUTORIDADES										 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS									
										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear) DES. 82									
Eng ^o Selma Schwab Coordenadora da DP/GNT					Eng ^o Mário S.Bortone Gerente da DP/GPE					Eng ^o Roger Gama Veloso Diretor de Projetos					Eng ^o Nelson A. Reis Vice - Diretor Geral				

BTCC 200 x 150					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																		
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000				
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,29 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,53 MPa				
LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	6,3	16,7	210	c.24	1	10,0	16,7	210	c.24	1	10,0	15,4	210	c.26	1	12,5	15,4	210	c.26	1	12,5	15,4	215	c.26	1	12,5	20,0	215	c.20	1	12,5	18,2	220	c.22	1	16,0	16,7	220	c.24
2	6,3	8,3	380	c.24	2	10,0	8,3	380	c.24	2	10,0	7,7	380	c.26	2	12,5	7,7	380	c.26	2	12,5	7,7	385	c.26	2	12,5	10,0	385	c.20	2	12,5	9,1	390	c.22	2	16,0	8,3	390	c.24
3	6,3	40,0	212	c.10	3	8,0	36,4	212	c.11	3	10,0	33,3	212	c.12	3	12,5	28,6	212	c.14	3	12,5	33,3	217	c.12	3	12,5	36,4	217	c.11	3	12,5	40,0	222	c.10	3	16,0	30,8	222	c.13
4	6,3	8,0	212	c.25	4	6,3	15,4	212	c.13	4	6,3	12,5	212	c.16	4	6,3	18,2	212	c.11	4	6,3	18,2	217	c.11	4	8,0	15,4	217	c.13	4	8,0	15,4	222	c.13	4	8,0	20,0	222	c.10
5	6,3	8,0	215	c.50	5	6,3	8,0	215	c.50	5	6,3	8,0	215	c.50	5	6,3	8,0	215	c.50	5	6,3	8,0	220	c.50	5	6,3	8,0	220	c.50	5	6,3	8,0	225	c.50	5	6,3	8,0	225	c.50
6	6,3	8,0	190	c.50	6	6,3	8,0	190	c.50	6	6,3	8,0	190	c.50	6	6,3	8,0	190	c.50	6	6,3	8,0	195	c.50	6	6,3	8,0	195	c.50	6	6,3	8,0	200	c.50	6	6,3	8,0	200	c.50
7	6,3	8,0	182	c.25	7	6,3	8,0	182	c.25	7	6,3	8,0	182	c.25	7	6,3	8,0	182	c.25	7	6,3	9,5	192	c.21	7	6,3	9,5	192	c.21	7	6,3	11,8	202	c.17	7	6,3	11,8	202	c.17
8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25	8	6,3	19,0	192	c.21	8	6,3	19,0	192	c.21	8	6,3	23,5	202	c.17	8	6,3	23,5	202	c.17
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	6,3	20,0	204	c.20
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	6,3	20,0	306	c.20
11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20
12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	10,0	60,0	Corr.	c.20	12	10,0	60,0	Corr.	c.20
13	6,3	12,0	Corr.	c.20	13	6,3	16,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20
14	6,3	36,0	Corr.	c.20	14	6,3	48,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	84,2	Var.	c.19	16	6,3	100,0	Var.	c.16	16	8,0	106,7	Var.	c.15
17	----	----	----	----	17	----	----	----	----	17	----	----	----	----	17	6,3	246,2	Var.	c.13	17	6,3	228,6	Var.	c.14	17	8,0	213,3	Var.	c.15	17	8,0	246,2	Var.	c.13	17	10,0	246,2	Var.	c.13
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)				
6,3	0,252	123,8	6,3	0,252	73,8	6,3	0,252	88,0	6,3	0,252	106,4	6,3	0,252	113,3	6,3	0,252	83,7	6,3	0,252	94,3	6,3	0,252	64,3	6,3	0,252	65,8	6,3	0,252	128,1	6,3	0,252	64,3	6,3	0,252	64,3				
8,0	0,393	----	8,0	0,393	30,3	8,0	0,393	----	8,0	0,393	23,6	8,0	0,393	23,6	8,0	0,393	82,0	8,0	0,393	70,5	8,0	0,393	65,8	8,0	0,393	65,8	8,0	0,393	128,1	8,0	0,393	65,8	8,0	0,393	65,8				
10,0	0,624	----	10,0	0,624	41,6	10,0	0,624	82,5	10,0	0,624	----	10,0	0,624	----	10,0	0,624	37,4	10,0	0,624	37,4	10,0	0,624	128,1	10,0	0,624	128,1	10,0	0,624	128,1	10,0	0,624	128,1	10,0	0,624	128,1				
12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	120,8	12,5	0,988	133,4	12,5	0,988	158,6	12,5	0,988	162,4	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----				
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	253,5	16,0	1,570	253,5	16,0	1,570	253,5	16,0	1,570	253,5	16,0	1,570	253,5				
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----				
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----				
TOTAL (Kg/m)		161,5	TOTAL (Kg/m)		183,4	TOTAL (Kg/m)		208,2	TOTAL (Kg/m)		288,5	TOTAL (Kg/m)		308,0	TOTAL (Kg/m)		362,0	TOTAL (Kg/m)		402,3	TOTAL (Kg/m)		511,7	TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)						
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																							
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)				
1	80	50	1	80	50	1	80	50	1	80	50	1	80	55	1	80	55	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60				
2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----				
3	200	12	3	200	12	3	200	12	3	200	12	3	200	17	3	200	17	3	200	17	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22				
4	200	12	4	200	12	4	200	12	4	200	12	4	200	17	4	200	17	4	200	17	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22				
5	80	135	5	80	135	5	80	135	5	80	135	5	80	140	5	80	140	5	80	140	5	80	145	5	80	145	5	80	145	5	80	145	5	80	145				
6	130	60	6	130	60	6	130	60	6	130	60	6	130	60	6	135	60	6	135	60	6	140	60	6	140	60	6	140	60	6	140	60	6	140	60				
7	182	----	7	182	----	7	182	----	7	182	----	7	182	----	7	192	----	7	192	----	7	202	----	7	202	----	7	202	----	7	202	----	7	202	----				
8	182	----	8	182	----	8	182	----	8	182	----	8	182	----	8	192	----	8	192	----	8	202	----	8	202	----	8	202	----	8	202	----	8	202	----				
9	122	----	9	----	----	9	122	----	9	122	----	9	122	----	9	143	----	9	143	----	9	164	----	9	164	----	9	164	----	9	164	----	9</						

BTCC 200 x 200		TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																										
ALTURA DO ATERRO (cm) 0 < h < 100		ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000												
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa		TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,24 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,30 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,36 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,42 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,54 MPa												
LISTA DE FERROS (POR METRO LINEAR)																																												
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS									
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	6,3	16,7	210	c.24	1	10,0	16,7	210	c.24	1	8,0	20,0	210	c.20	1	10,0	20,0	210	c.20	1	10,0	20,0	215	c.20	1	12,5	18,2	215	c.22	1	12,5	18,2	220	c.22	1	16,0	16,7	220	c.24					
2	6,3	8,3	380	c.24	2	10,0	8,3	380	c.24	2	8,0	10,0	380	c.20	2	10,0	10,0	380	c.20	2	10,0	10,0	385	c.20	2	12,5	9,1	385	c.22	2	12,5	9,1	390	c.22	2	16,0	8,3	390	c.24					
3	6,3	40,0	212	c.10	3	8,0	40,0	212	c.10	3	10,0	33,3	212	c.12	3	12,5	28,6	212	c.14	3	12,5	30,8	217	c.13	3	12,5	33,3	217	c.12	3	12,5	36,4	222	c.11	3	16,0	28,6	222	c.14					
4	6,3	8,0	212	c.25	4	6,3	15,4	212	c.13	4	6,3	12,5	212	c.16	4	6,3	18,2	212	c.11	4	6,3	18,2	217	c.11	4	8,0	15,4	217	c.13	4	6,3	20,0	222	c.10	4	8,0	20,0	222	c.10					
5	6,3	8,0	240	c.50	5	6,3	10,5	240	c.38	5	6,3	8,0	240	c.50	5	6,3	8,0	240	c.50	5	6,3	8,0	245	c.50	5	6,3	8,0	245	c.50	5	6,3	8,0	250	c.50	5	6,3	8,0	250	c.50					
6	6,3	8,0	210	c.50	6	6,3	10,5	210	c.38	6	6,3	8,0	210	c.50	6	6,3	8,0	210	c.50	6	6,3	8,0	215	c.50	6	6,3	8,0	215	c.50	6	6,3	8,0	220	c.50	6	6,3	8,0	220	c.50					
7	6,3	8,0	232	c.25	7	6,3	8,0	232	c.25	7	6,3	8,0	232	c.25	7	6,3	8,0	232	c.25	7	6,3	9,5	242	c.21	7	6,3	9,5	242	c.21	7	6,3	11,8	252	c.17	7	6,3	11,8	252	c.17					
8	6,3	16,0	232	c.25	8	6,3	16,0	232	c.25	8	6,3	16,0	232	c.25	8	6,3	16,0	232	c.25	8	6,3	19,0	242	c.21	8	6,3	19,0	242	c.21	8	6,3	23,5	252	c.17	8	6,3	23,5	252	c.17					
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	6,3	20,0	204	c.20					
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	6,3	20,0	306	c.20					
11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20					
12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	10,0	60,0	Corr.	c.20					
13	6,3	18,0	Corr.	c.20	13	6,3	20,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20					
14	6,3	54,0	Corr.	c.20	14	6,3	60,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20					
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----					
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	88,9	Var.	c.18	16	6,3	106,7	Var.	c.15	16	8,0	114,3	Var.	c.14					
17	----	----	----	----	17	----	----	----	----	17	----	----	----	----	17	6,3	246,2	Var.	c.13	17	6,3	228,6	Var.	c.14	17	8,0	213,3	Var.	c.15	17	8,0	246,2	Var.	c.13	17	10,0	228,6	Var.	c.14					
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																												
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50								
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	133,8	6,3	0,252	84,6	6,3	0,252	98,0	6,3	0,252	116,3	6,3	0,252	123,8	6,3	0,252	94,8	6,3	0,252	117,9	6,3	0,252	75,7	8,0	0,393	----	8,0	0,393	33,3	8,0	0,393	31,4	8,0	0,393	23,6	8,0	0,393	82,0	8,0	0,393	80,7	8,0	0,393	67,5
10,0	0,624	----	10,0	0,624	41,6	10,0	0,624	44,1	10,0	0,624	49,9	10,0	0,624	50,9	10,0	0,624	----	10,0	0,624	----	10,0	0,624	121,6	12,5	0,988	----	12,5	0,988	----	12,5	0,988	59,9	12,5	0,988	66,0	12,5	0,988	144,7	12,5	0,988	154,5	12,5	0,988	----
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	245,9	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		171,5	TOTAL (Kg/m)		197,2	TOTAL (Kg/m)		211,2	TOTAL (Kg/m)		287,4	TOTAL (Kg/m)		302,0	TOTAL (Kg/m)		359,2	TOTAL (Kg/m)		390,8	TOTAL (Kg/m)		510,7	TOTAL (Kg/m)		TOTAL (Kg/m)		TOTAL (Kg/m)		TOTAL (Kg/m)		TOTAL (Kg/m)		TOTAL (Kg/m)		TOTAL (Kg/m)								
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																												
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	80	50	1	80	50	1	80	50	1	80	50	1	80	55	1	80	55	1	80	55	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60						
2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----			
3	200	12	3	200	12	3	200	12	3	200	12	3	200	17	3	200	17	3	200	17	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22			
4	200	12	4	200	12	4	200	12	4	200	12	4	200	17	4	200	17	4	200	17	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22			
5	80	160	5	80	160	5	80	160	5	80	160	5	80	165	5	80	165	5	80	165	5	80	170	5	80	170	5	80	170	5	80	170	5	80	170	5	80	170	5	80	170			
6	130	80	6	130	80	6	130	80	6	130	80	6	130	80	6	135	80	6	135	80	6	135	80	6	140	80	6	140	80	6	140	80	6	140	80	6	140	80	6	140	80			
7	232	----	7	232	----	7	232	----	7	232	----	7	242	----	7	242	----	7	242	----	7	242	----	7	252	----	7	252	----	7	252	----	7	252	----	7	252	----	7	252	----			
8	232	----	8	232	----	8	232	----	8	232	----	8	242	----	8	242	----	8	242	----	8	242	----	8	252	----	8	252	----	8	252	----	8	252	----	8	252	----	8	252	----			
9	122	----	9	----	----	9	122	----	9	122	----	9	143	----	9	143	----	9	143	----	9	143	----	9	164	----	9	164	----	9	164	----	9	164	----	9	164	----	9	164	----			
10	116	----	10	----	----	10	116	----	10	116	----	10	135	----	10	135	----	10	135	----	10	135	----	10	153	----	10	153	----	10	153	----	10	153	----	10	153	----	10	153	----			
ASSINATURA DAS AUTORIDADES															 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																													
Engª Selma Schwab Coordenadora da DP/GNT															Engº Mário S.Bortone Gerente da DP/GPE					Engº Roger Gama Veloso Diretor de Projetos					Engº Nelson A. Reis Vice - Diretor Geral					CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear) DES. 86														

BTCC 200 x 250					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																										
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000												
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,19 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,25 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,31 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,37 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,43 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,55 MPa												
LISTA DE FERROS (POR METRO LINEAR)																																															
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS							
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)								
1	6,3	16,7	210	c.24	1	10,0	16,7	210	c.24	1	8,0	20,0	210	c.20	1	10,0	20,0	210	c.20	1	10,0	20,0	215	c.20	1	12,5	18,2	215	c.22	1	12,5	16,7	220	c.24	1	16,0	15,4	220	c.26								
2	6,3	8,3	380	c.24	2	10,0	8,3	380	c.10	2	8,0	10,0	380	c.20	2	10,0	10,0	380	c.20	2	10,0	10,0	385	c.20	2	12,5	9,1	385	c.22	2	12,5	8,3	390	c.24	2	16,0	7,7	390	c.26								
3	8,0	30,8	212	c.13	3	8,0	40,0	212	c.12	3	10,0	33,3	212	c.10	3	10,0	40,0	212	c.10	3	10,0	40,0	217	c.10	3	12,5	33,3	217	c.12	3	12,5	36,4	222	c.11	3	12,5	40,0	222	c.10								
4	6,3	8,7	212	c.23	4	6,3	16,7	212	c.12	4	6,3	13,3	212	c.15	4	6,3	18,2	212	c.11	4	6,3	18,2	217	c.11	4	8,0	15,4	217	c.13	4	8,0	15,4	222	c.13	4	8,0	20,0	222	c.10								
5	6,3	8,0	265	c.50	5	6,3	16,7	265	c.24	5	6,3	11,8	265	c.34	5	6,3	16,7	265	c.24	5	6,3	10,0	270	c.40	5	6,3	13,3	270	c.30	5	6,3	8,0	275	c.50	5	6,3	8,7	275	c.46								
6	6,3	8,0	230	c.50	6	6,3	16,7	230	c.24	6	6,3	11,8	230	c.34	6	6,3	16,7	230	c.24	6	6,3	10,0	235	c.40	6	6,3	13,3	235	c.30	6	6,3	8,0	240	c.50	6	6,3	8,7	240	c.46								
7	6,3	8,0	282	c.25	7	6,3	8,0	282	c.25	7	6,3	8,0	282	c.25	7	6,3	8,0	282	c.25	7	6,3	9,5	292	c.21	7	6,3	9,5	292	c.21	7	6,3	11,8	302	c.17	7	6,3	11,8	302	c.17								
8	6,3	16,0	282	c.25	8	6,3	16,0	282	c.25	8	6,3	16,0	282	c.25	8	6,3	16,0	282	c.25	8	6,3	19,0	292	c.21	8	6,3	19,0	292	c.21	8	6,3	23,5	302	c.17	8	6,3	23,5	302	c.17								
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	6,3	20,0	204	c.20								
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	6,3	20,0	306	c.20								
11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20								
12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	10,0	60,0	Corr.	c.20								
13	6,3	22,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20								
14	6,3	66,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20								
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----								
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	94,1	Var.	c.17	16	6,3	114,3	Var.	c.14	16	8,0	123,1	Var.	c.13								
17	----	----	----	----	17	----	----	----	----	17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	6,3	228,6	Var.	c.14	17	8,0	213,3	Var.	c.15	17	8,0	228,6	Var.	c.14	17	10,0	228,6	Var.	c.14								
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																															
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50								
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	120,7	6,3	0,252	103,3	6,3	0,252	111,2	6,3	0,252	133,0	6,3	0,252	134,9	6,3	0,252	110,8	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2	6,3	0,252	117,2						
8,0	0,393	25,7	8,0	0,393	33,3	8,0	0,393	31,4	8,0	0,393	23,6	8,0	0,393	23,6	8,0	0,393	82,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0	8,0	0,393	90,0						
10,0	0,624	----	10,0	0,624	41,6	10,0	0,624	44,1	10,0	0,624	102,8	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0	10,0	0,624	105,0						
12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7	12,5	0,988	144,7						
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7						
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----			
TOTAL (Kg/m)		184,1	TOTAL (Kg/m)		215,9	TOTAL (Kg/m)		224,4	TOTAL (Kg/m)		297,1	TOTAL (Kg/m)		301,2	TOTAL (Kg/m)		375,2	TOTAL (Kg/m)		393,0	TOTAL (Kg/m)		502,9																								
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																															
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	80	50	1	80	50	1	80	50	1	80	50	1	80	55	1	80	55	1	80	55	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60			
2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----
3	200	12	3	200	12	3	200	12	3	200	12	3	200	17	3	200	17	3	200	17	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22			
4	200	12	4	200	12	4	200	12	4	200	12	4	200	17	4	200	17	4	200	17	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22			
5	80	185	5	80	185	5	80	185	5	80	185	5	80	190	5	80	190	5	80	190	5	80	195	5	80	195	5	80	195	5	80	195	5	80	195	5	80	195	5	80	195	5	80	195			
6	130	100	6	130	100	6	130	100	6	130	100	6	130	100	6	135	100	6	135	100	6	135	100	6	140	100	6	140	100	6	140	100	6	140	100	6	140	100	6	140	100	6	140	100			

BTCC
200 x 300

ALTURA DO ATERRO (cm)
0 < h < 100

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,13 MPa

ALTURA DO ATERRO (cm)
100 < h < 250

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,15 MPa

ALTURA DO ATERRO (cm)
250 < h < 500

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,20 MPa

ALTURA DO ATERRO (cm)
500 < h < 750

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,26 MPa

ALTURA DO ATERRO (cm)
750 < h < 1000

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,32 MPa

ALTURA DO ATERRO (cm)
1000 < h < 1250

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,38 MPa

ALTURA DO ATERRO (cm)
1250 < h < 1500

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,44 MPa

ALTURA DO ATERRO (cm)
1500 < h < 2000

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,56 MPa

LISTA DE FERROS (POR METRO LINEAR)

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

6,3

16,7

210

c.24

2

6,3

8,3

380

c.24

3

8,0

30,8

212

c.13

4

6,3

9,5

212

c.21

5

6,3

13,3

290

c.30

6

6,3

13,3

250

c.30

7

6,3

8,0

332

c.25

8

6,3

16,0

332

c.25

9

6,3

20,0

162

c.20

10

6,3

20,0

232

c.20

11

6,3

60,0

Corr.

c.20

12

6,3

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

17

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

10,0

15,4

210

c.26

2

10,0

7,7

380

c.26

3

8,0

40,0

212

c.10

4

6,3

18,2

212

c.11

5

8,0

18,2

290

c.22

6

8,0

18,2

250

c.22

7

6,3

8,0

332

c.25

8

6,3

16,0

332

c.25

9

10

11

6,3

60,0

Corr.

c.20

12

6,3

60,0

Corr.

c.20

13

6,3

30,0

Corr.

c.20

14

6,3

90,0

Corr.

c.20

15

16,0

24,0

Corr.

16

17

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

8,0

20,0

210

c.20

2

8,0

10,0

380

c.20

3

10,0

30,8

212

c.13

4

6,3

15,4

212

c.13

5

8,0

15,4

290

c.26

6

8,0

15,4

250

c.26

7

6,3

8,0

332

c.25

8

6,3

16,0

332

c.25

9

6,3

20,0

162

c.20

10

6,3

20,0

232

c.20

11

6,3

60,0

Corr.

c.20

12

6,3

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

17

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

10,0

18,2

210

c.22

2

10,0

9,1

380

c.22

3

10,0

36,4

212

c.11

4

6,3

20,0

212

c.10

5

8,0

20,0

290

c.20

6

8,0

20,0

250

c.20

7

6,3

8,0

332

c.25

8

6,3

16,0

332

c.25

9

6,3

20,0

162

c.20

10

6,3

20,0

232

c.20

11

6,3

60,0

Corr.

c.20

12

8,0

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

17

6,3

200,0

Var.

c.16

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

10,0

18,2

215

c.22

2

10,0

9,1

385

c.22

3

10,0

40,0

217

c.10

4

6,3

20,0

217

c.10

5

8,0

15,4

295

c.26

6

8,0

15,4

255

c.26

7

6,3

9,5

342

c.21

8

6,3

19,0

342

c.21

9

6,3

20,0

183

c.20

10

6,3

20,0

270

c.20

11

6,3

60,0

Corr.

c.20

12

8,0

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

17

6,3

213,3

Var.

c.15

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

12,5

16,7

215

c.24

2

12,5

8,3

385

c.24

3

12,5

30,8

217

c.13

4

8,0

16,7

217

c.12

5

8,0

20,0

295

c.20

6

8,0

20,0

255

c.20

7

6,3

9,5

342

c.21

8

6,3

19,0

342

c.21

9

6,3

20,0

183

c.20

10

6,3

20,0

270

c.20

11

8,0

60,0

Corr.

c.20

12

8,0

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

6,3

106,7

Var.

c.15

17

6,3

320,0

Var.

c.10

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

12,5

20,0

215

c.20

2

12,5

10,0

385

c.20

3

12,5

33,3

217

c.12

4

8,0

20,0

217

c.10

5

10,0

16,7

295

c.24

6

10,0

16,7

255

c.24

7

6,3

9,5

342

c.21

8

6,3

19,0

342

c.21

9

6,3

20,0

183

c.20

10

6,3

20,0

270

c.20

11

8,0

60,0

Corr.

c.20

12

8,0

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

6,3

160,0

Var.

c.10

17

8,0

266,7

Var.

c.12

LISTA DE FERROS

N

Ø
(mm)

Quant.

Comp.
(cm)

Esp.
(cm)

1

12,5

20,0

220

c.20

2

12,5

10,0

390

c.20

3

12,5

36,4

222

c.11

4

10,0

15,4

222

c.13

5

8,0

20,0

300

c.20

6

8,0

20,0

260

c.20

7

6,3

11,8

352

c.17

8

6,3

23,5

352

c.17

9

6,3

20,0

204

c.20

10

6,3

20,0

306

c.20

11

8,0

60,0

Corr.

c.20

12

8,0

60,0

Corr.

c.20

13

6,3

28,0

Corr.

c.20

14

6,3

84,0

Corr.

c.20

15

16,0

24,0

Corr.

16

8,0

133,3

Var.

c.12

17

10,0

213,3

Var.

c.15

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

138,4

8,0

0,393

25,7

10,0

0,624

12,5

0,988

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

201,8

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

90,3

8,0

0,393

72,0

10,0

0,624

38,4

12,5

0,988

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

238,4

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

106,6

8,0

0,393

64,1

10,0

0,624

40,7

12,5

0,988

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

249,1

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

118,7

8,0

0,393

66,0

10,0

0,624

93,6

12,5

0,988

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

316,0

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

130,7

8,0

0,393

56,9

10,0

0,624

100,4

12,5

0,988

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

325,7

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

133,7

8,0

0,393

104,6

10,0

0,624

12,5

0,988

133,1

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

409,1

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

97,4

8,0

0,393

120,8

10,0

0,624

57,3

12,5

0,988

151,9

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

465,1

RESUMO - AÇO CA -50

Ø
(mm)

Kg/m

PESO (Kg)

6,3

0,252

85,2

8,0

0,393

122,1

10,0

0,624

99,9

12,5

0,988

161,8

16,0

1,570

37,7

20,0

2,480

25,0

3,980

TOTAL (Kg/m)

506,7

DIMENSÕES DAS DOBRAS DAS ARMADURAS

N

a (cm)

b (cm)

1

80

50

2

80

3

200

12

4

200

12

5

80

210

6

130

120

7

332

8

332

9

122

10

116

N

a (cm)

b (cm)

1

80

50

2

80

3

200

12

4

200

12

5

80

210

6

130

120

7

332

8

332

9

122

10

116

N

a (cm)

b (cm)

1

80

50

2

80

3

200

12

4

200

12

5

80

210

6

130

120

7

332

8

332

9

122

10

116

N

a (cm)

b (cm)

1

80

55

2

80

3

200

17

4

200

17

5

80

215

6

135

120

7

342

8

342

9

143

10

135

N

a (cm)

b (cm)

1

80

55

2

80

3

200

17

4

200

17

5

80

215

6

135

120

7

342

8

342

9

143

10

135

N

a (cm)

b (cm)

1

80

55

2

80

3

200

17

4

200

17

5

80

215

6

135

120

7

342

8

342

9

143

10

135

N

a (cm)

b (cm)

1

80

55

2

80

3

200

17

4

200

17

5

80

215

6

135

120

7

342

8

342

9

143

10

135

N

a (cm)

b (cm)

1

80

60

2

80

3

200

22

4

200

22

5

80

220

6

140

120

7

352

8

352

9

164

10

153

ASSINATURA DAS AUTORIDADES

Eng° Selma Schwab
Coordenadora da DP/GNT

Eng° Mário S.Bortone
Gerente da DP/GPE

Eng° Roger Gama Veloso
Diretor de Projetos

Eng° Nelson A. Reis
Vice - Diretor Geral

CADERNO DE BUEIROS CELULARES

Tabela de Armadura Para Bueiro Triplo Celular
Fundação Direta (Por Metro Linear)

DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

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DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

DESA

<

BTCC 200 x 350					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																		
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000				
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa				
LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	6,3	16,7	210	c.24	1	10,0	15,4	210	c.26	1	8,0	18,2	210	c.22	1	10,0	18,2	210	c.22	1	10,0	18,2	215	c.22	1	12,5	15,4	215	c.26	1	10,0	20,0	220	c.20	1	12,5	18,2	220	c.22
2	6,3	8,3	380	c.24	2	10,0	7,7	380	c.26	2	8,0	9,1	380	c.22	2	10,0	9,1	380	c.22	2	10,0	9,1	385	c.22	2	12,5	7,7	385	c.26	2	10,0	10,0	390	c.20	2	12,5	9,1	390	c.22
3	8,0	30,8	212	c.13	3	8,0	40,0	212	c.10	3	10,0	30,8	212	c.13	3	10,0	33,3	212	c.12	3	10,0	36,4	217	c.11	3	10,0	40,0	217	c.10	3	12,5	30,8	222	c.13	3	12,5	33,3	222	c.12
4	6,3	11,1	212	c.18	4	6,3	18,2	212	c.11	4	6,3	16,7	212	c.12	4	8,0	15,4	212	c.13	4	8,0	15,4	217	c.13	4	8,0	20,0	217	c.10	4	8,0	18,2	222	c.11	4	10,0	18,2	222	c.11
5	8,0	15,4	315	c.26	5	10,0	16,7	315	c.24	5	10,0	15,4	315	c.26	5	12,5	14,3	315	c.28	5	10,0	18,2	320	c.22	5	10,0	20,0	320	c.20	5	10,0	18,2	325	c.22	5	12,5	15,4	325	c.26
6	8,0	15,4	270	c.26	6	10,0	16,7	270	c.24	6	10,0	15,4	270	c.26	6	12,5	14,3	270	c.28	6	10,0	18,2	275	c.22	6	10,0	20,0	275	c.20	6	10,0	18,2	280	c.22	6	12,5	15,4	280	c.26
7	6,3	8,0	382	c.25	7	6,3	8,0	382	c.25	7	6,3	8,0	382	c.25	7	6,3	8,0	382	c.25	7	6,3	9,5	392	c.21	7	6,3	9,5	392	c.21	7	6,3	11,8	402	c.17	7	6,3	11,8	402	c.17
8	8,0	16,0	382	c.25	8	6,3	16,0	382	c.25	8	6,3	16,0	382	c.25	8	6,3	16,0	382	c.25	8	6,3	19,0	392	c.21	8	6,3	19,0	392	c.21	8	6,3	23,5	402	c.17	8	6,3	23,5	402	c.17
9	8,0	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	6,3	20,0	204	c.20
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	6,3	20,0	306	c.20
11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	6,3	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20	11	8,0	60,0	Corr.	c.20
12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	6,3	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20	12	8,0	60,0	Corr.	c.20
13	6,3	32,0	Corr.	c.20	13	6,3	36,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20
14	6,3	96,0	Corr.	c.20	14	6,3	108,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	123,1	Var.	c.13	16	6,3	133,3	Var.	c.12	16	8,0	145,5	Var.	c.11
17	----	----	----	----	17	----	----	----	----	17	----	----	----	----	17	6,3	177,8	Var.	c.18	17	6,3	200,0	Var.	c.16	17	6,3	290,9	Var.	c.11	17	8,0	213,3	Var.	c.15	17	10,0	213,3	Var.	c.15
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50															
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	
6,3	0,252	128,2	6,3	0,252	99,4	6,3	0,252	114,4	6,3	0,252	104,2	6,3	0,252	102,4	6,3	0,252	131,5	6,3	0,252	105,5	6,3	0,252	105,5	6,3	0,252	125,1	6,3	0,252	85,7										
8,0	0,393	61,1	8,0	0,393	33,3	8,0	0,393	28,6	8,0	0,393	49,0	8,0	0,393	72,9	8,0	0,393	76,8	8,0	0,393	125,1	8,0	0,393	125,1	8,0	0,393	93,5													
10,0	0,624	----	10,0	0,624	99,4	10,0	0,624	97,0	10,0	0,624	89,5	10,0	0,624	163,1	10,0	0,624	128,4	10,0	0,624	120,5	10,0	0,624	120,5	10,0	0,624	103,7													
12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	82,7	12,5	0,988	----	12,5	0,988	62,0	12,5	0,988	67,6	12,5	0,988	67,6	12,5	0,988	239,7													
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7													
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----										
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----										
TOTAL (Kg/m)		227,0	TOTAL (Kg/m)		269,8	TOTAL (Kg/m)		277,7	TOTAL (Kg/m)		363,1	TOTAL (Kg/m)		376,1	TOTAL (Kg/m)		436,4	TOTAL (Kg/m)		456,4	TOTAL (Kg/m)		560,3																
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																							
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)				
1	80	50	1	80	50	1	80	50	1	80	50	1	80	55	1	80	55	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60	1	80	60				
2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----	2	80	----				
3	200	12	3	200	12	3	200	12	3	200	12	3	200	17	3	200	17	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22	3	200	22				
4	200	12	4	200	12	4	200	12	4	200	12	4	200	17	4	200	17	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22	4	200	22				
5	80	235	5	80	235	5	80	235	5	80	235	5	80	240	5	80	240	5	80	245	5	80	245	5	80	245	5	80	245	5	80	245	5	80	245				
6	130	140	6	130	140	6	130	140	6	130	140	6	130	140	6	135	140	6	135	140	6	140	140	6	140	140	6	140	140	6	140	140	6	140	140				
7	382	----	7	382	----	7	382	----	7	382	----	7	382	----	7	392	----	7	392	----	7	402	----	7	402	----	7	402	----	7	402	----	7	402	----				
8	382	----	8	382	----	8	382	----	8	382	----	8	382	----	8	392	----	8	392	----	8	402	----	8	402	----	8	402	----	8	402	----	8	402	----				
9	122	----	9	----	----	9	122	----	9	122	----	9	122	----	9	143	----	9	143	----	9	164	----	9	164	----	9	164	----	9	164	----	9	164	----				
10	116	----	10	----	----	10	116	----	10	116	----	10	116	----	10	135	----	10	135	----	10	153	----	10	153	----	10	153	----	10	153	----	10	153	----				
ASSINATURA DAS AUTORIDADES															 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																								
Eng ^o Selma Schwab Coordenadora da DP/GNT															Eng ^o Mário S.Bortone Diretor da DP/GPE																								
Eng ^o Roger Gama Veloso Diretor de Projetos															Eng ^o Nelson A. Reis Vice - Diretor Geral																								
CADERNO DE BUEIROS CELULARES															Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)																								
															DES. 88																								

BTCC 250 x 150					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																							
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000									
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,30 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,53 MPa									
LISTA DE FERROS (POR METRO LINEAR)																																												
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS									
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	6,3	20,0	255	c.20	1	12,5	14,3	255	c.28	1	12,5	18,2	255	c.22	1	12,5	20,0	260	c.20	1	16,0	15,4	265	c.26	1	16,0	18,2	265	c.22	1	16,0	20,0	265	c.20	1	20,0	15,4	270	c.26					
2	6,3	10,0	470	c.20	2	12,5	7,1	470	c.28	2	12,5	9,1	470	c.22	2	12,5	10,0	475	c.20	2	16,0	7,7	480	c.26	2	16,0	9,1	480	c.22	2	16,0	10,0	480	c.20	2	20,0	7,7	485	c.26					
3	8,0	33,3	262	c.12	3	10,0	36,4	262	c.11	3	12,5	33,3	262	c.12	3	12,5	36,4	267	c.11	3	12,5	40,0	272	c.10	3	16,0	30,8	272	c.13	3	16,0	36,4	272	c.11	3	16,0	40,0	277	c.10					
4	6,3	10,5	262	c.19	4	8,0	16,7	262	c.12	4	8,0	18,2	262	c.11	4	8,0	20,0	267	c.10	4	8,0	20,0	272	c.10	4	10,0	18,2	272	c.11	4	12,5	15,4	272	c.13	4	12,5	16,7	277	c.12					
5	6,3	8,0	235	c.50	5	6,3	13,3	235	c.30	5	6,3	13,3	235	c.30	5	6,3	10,0	240	c.40	5	6,3	8,0	245	c.50	5	6,3	8,0	245	c.50	5	6,3	10,5	245	c.38	5	6,3	8,0	250	c.50					
6	6,3	8,0	215	c.50	6	6,3	13,3	215	c.30	6	6,3	13,3	215	c.30	6	6,3	10,0	220	c.40	6	6,3	8,0	225	c.50	6	6,3	8,0	225	c.50	6	6,3	10,5	225	c.38	6	6,3	8,0	230	c.50					
7	6,3	8,0	182	c.25	7	6,3	8,0	182	c.25	7	6,3	8,0	182	c.25	7	6,3	9,5	192	c.21	7	6,3	11,8	202	c.17	7	6,3	11,8	202	c.17	7	6,3	11,8	202	c.17	7	8,0	9,1	212	c.22					
8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25	8	6,3	16,0	182	c.25	8	6,3	19,0	192	c.21	8	6,3	23,5	202	c.17	8	6,3	23,5	202	c.17	8	6,3	23,5	202	c.17	8	8,0	18,2	212	c.22					
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20					
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20					
11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	10,0	78,0	Corr.	c.20					
12	6,3	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	12,5	78,0	Corr.	c.20	12	12,5	78,0	Corr.	c.20					
13	6,3	12,0	Corr.	c.20	13	6,3	16,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20	13	6,3	12,0	Corr.	c.20					
14	6,3	36,0	Corr.	c.20	14	6,3	48,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20	14	6,3	36,0	Corr.	c.20					
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----					
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	106,7	Var.	c.15	16	6,3	145,5	Var.	c.11	16	8,0	133,3	Var.	c.12					
17	----	----	----	----	17	6,3	188,2	Var.	c.17	17	6,3	246,2	Var.	c.13	17	6,3	266,7	Var.	c.12	17	6,3	290,9	Var.	c.11	17	8,0	266,7	Var.	c.12	17	10,0	213,3	Var.	c.15	17	10,0	266,7	Var.	c.12					
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																												
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50								
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	123,0	6,3	0,252	81,3	6,3	0,252	108,1	6,3	0,252	116,3	6,3	0,252	108,5	6,3	0,252	108,5	6,3	0,252	55,4	6,3	0,252	55,4	6,3	0,252	64,1	6,3	0,252	64,1	6,3	0,252	21,8	6,3	0,252	21,8	6,3	0,252	21,8						
8,0	0,393	34,3	8,0	0,393	47,8	8,0	0,393	49,4	8,0	0,393	51,6	8,0	0,393	52,0	8,0	0,393	52,0	8,0	0,393	132,6	8,0	0,393	132,6	8,0	0,393	70,7	8,0	0,393	70,7	8,0	0,393	101,0	8,0	0,393	101,0	8,0	0,393	101,0						
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	----	10,0	0,624	----	10,0	0,624	48,7	10,0	0,624	48,7	10,0	0,624	79,6	10,0	0,624	79,6	10,0	0,624	78,5	10,0	0,624	78,5	10,0	0,624	155,2	10,0	0,624	155,2	10,0	0,624	155,2						
12,5	0,988	----	12,5	0,988	69,0	12,5	0,988	174,3	12,5	0,988	194,3	12,5	0,988	107,5	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	----	12,5	0,988	118,4	12,5	0,988	118,4	12,5	0,988	122,8	12,5	0,988	122,8	12,5	0,988	122,8						
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	159,8	16,0	1,570	159,8	16,0	1,570	313,5	16,0	1,570	313,5	16,0	1,570	351,7	16,0	1,570	351,7	16,0	1,570	211,6	16,0	1,570	211,6	16,0	1,570	211,6						
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	195,7	20,0	2,480	195,7	20,0	2,480	195,7			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----			
TOTAL (Kg/m)		195,0	TOTAL (Kg/m)		295,3	TOTAL (Kg/m)		369,5	TOTAL (Kg/m)		399,9	TOTAL (Kg/m)		476,5	TOTAL (Kg/m)		581,1	TOTAL (Kg/m)		683,4	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1	TOTAL (Kg/m)		808,1			
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																												
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	100	55	1	100	55	1	100	55	1	100	60	1	100	65	1	100	65	1	100	65	1	100	65	1	100	65	1	100	65	1	100	65	1	100	70	1	100	70	1	100	70			
2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----
3	250	12	3	250	12	3	250	12	3	250	17	3	250	22	3	250	22	3	250	22	3	250	22	3	250	22	3	250	22	3	250	22	3	250	27	3	250	27	3	250	27			
4	250	12	4	250	12	4	250	12	4	250	17	4	250	22	4	250	22	4	250	22	4	250	22	4	250	22	4	250	22	4	250	22	4	250	27	4	250	27	4	250	27			
5	100	135	5	100	135	5	100	135	5	100	140	5	100	145	5	100	145	5	100	145	5	100	145	5	100	145	5	100	145	5	100	145	5	100	150	5	100	150	5	100	150			
6	155	60	6	155	60	6	155	60	6	155	60	6	160	60	6	165	60	6	165	60	6	165	60	6	165	60	6	165	60	6	165	60	6	170	60	6	170	60	6	170	60			
7	182	----	7	182	----	7	182	----	7	182	----	7	192																															

BTCC 250 x 200					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																						
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,24 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,30 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,36 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,43 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,55 MPa																								
LISTA DE FERROS (POR METRO LINEAR)																																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																				
1	6,3	20,0	255	c.20	1	12,5	14,3	255	c.28	1	12,5	18,2	255	c.22	1	12,5	20,0	260	c.20	1	16,0	15,4	265	c.26	1	16,0	18,2	265	c.22	1	16,0	18,2	270	c.22	1	16,0	20,0	275	c.22																				
2	6,3	10,0	470	c.20	2	12,5	7,1	470	c.28	2	12,5	9,1	470	c.22	2	12,5	10,0	475	c.20	2	16,0	7,7	480	c.26	2	16,0	9,1	480	c.22	2	16,0	9,1	485	c.22	2	16,0	10,0	490	c.22																				
3	8,0	36,4	262	c.11	3	10,0	36,4	262	c.11	3	12,5	30,8	262	c.13	3	12,5	36,4	267	c.11	3	12,5	40,0	272	c.10	3	16,0	30,8	272	c.13	3	16,0	33,3	277	c.12	3	16,0	36,4	282	c.11																				
4	6,3	11,1	262	c.18	4	8,0	16,7	262	c.12	4	8,0	18,2	262	c.11	4	8,0	18,2	267	c.11	4	8,0	20,0	272	c.10	4	10,0	16,7	272	c.12	4	10,0	16,7	277	c.12	4	10,0	20,0	282	c.10																				
5	6,3	8,0	260	c.50	5	6,3	15,4	260	c.26	5	6,3	13,3	260	c.30	5	6,3	9,5	265	c.42	5	6,3	8,0	270	c.50	5	6,3	8,0	270	c.50	5	6,3	8,0	275	c.50	5	6,3	8,0	280	c.50																				
6	6,3	8,0	235	c.50	6	6,3	15,4	235	c.26	6	6,3	13,3	235	c.30	6	6,3	9,5	240	c.42	6	6,3	8,0	245	c.50	6	6,3	8,0	245	c.50	6	6,3	8,0	250	c.50	6	6,3	8,0	255	c.50																				
7	6,3	8,0	232	c.25	7	6,3	8,0	232	c.25	7	6,3	8,0	232	c.25	7	6,3	9,5	242	c.21	7	6,3	11,8	252	c.17	7	6,3	11,8	252	c.17	7	8,0	9,1	262	c.22	7	8,0	10,5	272	c.19																				
8	6,3	16,0	232	c.25	8	6,3	16,0	232	c.25	8	6,3	16,0	232	c.25	8	6,3	19,0	242	c.21	8	6,3	23,5	252	c.17	8	6,3	23,5	252	c.17	8	8,0	18,2	262	c.22	8	8,0	21,1	272	c.19																				
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	162	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20																				
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	232	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20																				
11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20																				
12	6,3	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	12,5	78,0	Corr.	c.20																				
13	6,3	18,0	Corr.	c.20	13	6,3	20,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20																				
14	6,3	54,0	Corr.	c.20	14	6,3	60,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20																				
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----																				
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	106,7	Var.	c.15	16	6,3	133,3	Var.	c.12	16	8,0	114,3	Var.	c.19																				
17	----	----	----	----	17	6,3	200,0	Var.	16,0	17	6,3	246,2	Var.	c.13	17	6,3	266,7	Var.	c.12	17	6,3	290,9	Var.	c.11	17	8,0	266,7	Var.	c.12	17	8,0	290,9	Var.	c.11	17	10,0	246,2	Var.	c.13																				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																											
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																						
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																								
6,3	0,252	133,3	6,3	0,252	93,7	6,3	0,252	118,7	6,3	0,252	126,4	6,3	0,252	119,9	6,3	0,252	66,8	6,3	0,252	50,2	6,3	0,252	66,8	6,3	0,252	50,2	6,3	0,252	28,9																														
8,0	0,393	37,5	8,0	0,393	47,8	8,0	0,393	49,4	8,0	0,393	49,8	8,0	0,393	52,0	8,0	0,393	132,6	8,0	0,393	176,7	8,0	0,393	132,6	8,0	0,393	176,7	8,0	0,393	144,6																														
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	----	10,0	0,624	----	10,0	0,624	48,7	10,0	0,624	77,0	10,0	0,624	77,5	10,0	0,624	77,0	10,0	0,624	77,5	10,0	0,624	141,2																														
12,5	0,988	----	12,5	0,988	69,0	12,5	0,988	167,8	12,5	0,988	194,3	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	----	12,5	0,988	77,1																														
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	159,8	16,0	1,570	313,5	16,0	1,570	328,9	16,0	1,570	313,5	16,0	1,570	328,9	16,0	1,570	362,1																														
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----																														
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----																														
TOTAL (Kg/m)		208,5	TOTAL (Kg/m)		307,7	TOTAL (Kg/m)		373,6	TOTAL (Kg/m)		408,2	TOTAL (Kg/m)		487,9	TOTAL (Kg/m)		589,9	TOTAL (Kg/m)		633,3	TOTAL (Kg/m)		753,9																																				
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																											
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																								
1	100	55	1	100	55	1	100	55	1	100	60	1	100	65	1	100	65	1	100	65	1	100	65	1	100	65	1	100	70	1	100	75																											
2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----																											
3	250	12	3	250	12	3	250	12	3	250	17	3	250	22	3	250	22	3	250	22	3	250	22	3	250	22	3	250	27	3	250	32																											
4	250	12	4	250	12	4	250	12	4	250	17	4	250	22	4	250	22	4	250	22	4	250	22	4	250	22	4	250	27	4	250	32																											
5	100	160	5	100	160	5	100	160	5	100	165	5	100	170	5	100	170	5	100	170	5	100	170	5	100	170	5	100	175	5	100	180																											
6	155	80	6	155	80	6	155	80	6	160	80	6	165	80	6	165	80	6	165	80	6	170	80	6	170	80	6	170	80	6	175	80																											
7	232	----	7	232	----	7	232	----	7	242	----	7	252	----	7	252	----	7	252	----	7	262	----	7	262	----	7	262	----	7	272	----																											
8	232	----	8	232	----	8	232	----	8	242	----	8	252	----	8	252	----	8	252	----	8	262	----	8	262	----	8	262	----	8	272	----																											
9	122	----	9	----	----	9	122	----	9	143	----	9	164	----	9	164	----	9	164	----	9	185	----	9	185	----	9	185	----	9	206	----																											
10	116	----	10	----	----	10	116	----	10	135	----	10	153	----	10	153	----	10	153	----	10	172	----	10	172	----	10	172	----	10	190	----																											
ASSINATURA DAS AUTORIDADES															 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																												
Eng ^o Selma Schwab Coordenadora da DP/GNT															Eng ^o Mário S.Bortone Gerente da DP/GPE															Eng ^o Roger Gama Veloso Diretor de Projetos										Eng ^o Nelson A. Reis Vice - Diretor Geral					CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)										DES. 91				

BTCC 250 x 250	TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,19 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,25 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,31 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,37 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,44 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	6,3	20,0	255	c.20	1	12,5	14,3	255	c.28	1	10,0	20,0	260	c.20	1	12,5	20,0	265	c.20	1	16,0	18,2	270	c.22
2	6,3	10,0	470	c.20	2	12,5	7,1	470	c.28	2	10,0	10,0	475	c.20	2	12,5	10,0	480	c.20	2	16,0	9,1	485	c.22
3	8,0	36,4	262	c.11	3	10,0	36,4	262	c.11	3	10,0	40,0	267	c.10	3	12,5	33,3	267	c.12	3	16,0	33,3	277	c.12
4	6,3	11,8	262	c.17	4	8,0	16,7	262	c.12	4	6,3	20,0	267	c.10	4	8,0	20,0	272	c.10	4	10,0	16,7	277	c.12
5	6,3	8,7	285	c.46	5	6,3	20,0	285	c.20	5	6,3	10,0	290	c.40	5	6,3	14,3	290	c.28	5	6,3	11,8	295	c.34
6	6,3	8,7	255	c.46	6	6,3	20,0	255	c.20	6	6,3	10,0	260	c.40	6	6,3	14,3	260	c.28	6	6,3	11,8	265	c.34
7	6,3	8,0	282	c.25	7	6,3	8,0	282	c.25	7	6,3	9,5	292	c.21	7	6,3	9,5	292	c.21	7	6,3	11,8	302	c.17
8	6,3	16,0	282	c.25	8	6,3	16,0	282	c.25	8	6,3	19,0	292	c.21	8	6,3	19,0	292	c.21	8	6,3	23,5	302	c.17
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20
11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20
12	6,3	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20
13	6,3	22,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20
14	6,3	66,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	114,3	Var.	c.14
17	----	----	----	----	17	6,3	200,0	Var.	c.16	17	----	----	----	----	17	6,3	290,9	Var.	c.11	17	8,0	246,2	Var.	c.13

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																				
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	142,7	6,3	0,252	110,8	6,3	0,252	113,0	6,3	0,252	122,1	6,3	0,252	149,9	6,3	0,252	82,7	6,3	0,252	55,2
8,0	0,393	37,5	8,0	0,393	47,8	8,0	0,393	30,7	8,0	0,393	82,3	8,0	0,393	21,4	8,0	0,393	127,8	8,0	0,393	182,0
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	128,7	10,0	0,624	----	10,0	0,624	48,7	10,0	0,624	77,0	10,0	0,624	77,5
12,5	0,988	----	12,5	0,988	69,0	12,5	0,988	----	12,5	0,988	186,2	12,5	0,988	207,3	12,5	0,988	----	12,5	0,988	----
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	304,1	16,0	1,570	328,9
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)	217,9		TOTAL (Kg/m)	324,8		TOTAL (Kg/m)	310,1		TOTAL (Kg/m)	428,3		TOTAL (Kg/m)	465,0		TOTAL (Kg/m)	591,6		TOTAL (Kg/m)	643,6	

DIMENSÕES DAS DOBRAS DAS ARMADURAS																				
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)
1	100	55	1	100	55	1	100	60	1	100	60	1	100	65	1	100	65	1	100	75
2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----
3	250	12	3	250	12	3	250	17	3	250	17	3	250	22	3	250	22	3	250	32
4	250	12	4	250	12	4	250	17	4	250	17	4	250	22	4	250	22	4	250	32
5	100	185	5	100	185	5	100	190	5	100	190	5	100	195	5	100	195	5	100	205
6	155	100	6	155	100	6	160	100	6	160	100	6	165	100	6	165	100	6	170	100
7	282	----	7	282	----	7	292	----	7	292	----	7	302	----	7	302	----	7	312	----
8	282	----	8	282	----	8	292	----	8	292	----	8	302	----	8	302	----	8	312	----
9	122	----	9	----	----	9	143	----	9	143	----	9	164	----	9	164	----	9	185	----
10	116	----	10	----	----	10	135	----	10	135	----	10	153	----	10	153	----	10	172	----

ASSINATURA DAS AUTORIDADES										 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS									
										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear) DES. 92									
Engª Selma Schwab Coordenadora da DP/GNT					Engº Mário S.Bortone Gerente da DP/GPE					Engº Roger Gama Veloso Diretor de Projetos					Engº Nelson A. Reis Vice - Diretor Geral				

BTCC 250 x 300					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																										
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000												
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,15 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,38 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa												
LISTA DE FERROS (POR METRO LINEAR)																																															
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS												
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)								
1	8,0	14,3	255	c.28	1	12,5	14,3	255	c.28	1	10,0	20,0	260	c.20	1	12,5	20,0	260	c.20	1	12,5	20,0	265	c.20	1	16,0	16,7	265	c.24	1	16,0	16,7	270	c.24	1	16,0	18,2	275	c.22								
2	8,0	7,1	470	c.28	2	12,5	7,1	470	c.28	2	10,0	10,0	475	c.20	2	12,5	10,0	475	c.20	2	12,5	10,0	480	c.20	2	16,0	8,3	480	c.24	2	16,0	8,3	485	c.24	2	16,0	9,1	490	c.22								
3	8,0	40,0	262	c.10	3	10,0	36,4	262	c.11	3	10,0	40,0	267	c.10	3	12,5	33,3	267	c.12	3	12,5	36,4	272	c.11	3	12,5	40,0	272	c.10	3	16,0	30,8	277	c.13	3	16,0	36,4	282	c.11								
4	6,3	12,5	262	c.16	4	8,0	18,2	262	c.11	4	6,3	20,0	267	c.10	4	8,0	20,0	267	c.10	4	8,0	20,0	272	c.10	4	10,0	18,2	272	c.11	4	10,0	18,2	277	c.11	4	10,0	20,0	282	c.10								
5	6,3	13,3	310	c.30	5	8,0	20,0	310	c.20	5	6,3	16,7	315	c.24	5	8,0	15,4	315	c.26	5	6,3	16,7	320	c.24	5	8,0	15,4	320	c.26	5	6,3	13,3	325	c.30	5	6,3	8,0	330	c.50								
6	6,3	13,3	275	c.30	6	8,0	20,0	275	c.20	6	6,3	16,7	280	c.24	6	8,0	15,4	280	c.26	6	6,3	16,7	285	c.24	6	8,0	15,4	285	c.26	6	6,3	13,3	290	c.30	6	6,3	8,0	295	c.50								
7	6,3	8,0	332	c.25	7	6,3	8,0	332	c.25	7	6,3	9,5	342	c.21	7	6,3	9,5	342	c.21	7	6,3	11,8	352	c.17	7	6,3	11,8	352	c.17	7	8,0	9,1	362	c.22	7	8,0	10,5	372	c.19								
8	6,3	16,0	332	c.25	8	6,3	16,0	332	c.25	8	6,3	19,0	342	c.21	8	6,3	19,0	342	c.21	8	6,3	23,5	352	c.17	8	6,3	23,5	352	c.17	8	8,0	18,2	362	c.22	8	8,0	21,1	372	c.19								
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20								
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20								
11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20								
12	6,3	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	12,5	78,0	Corr.	c.20								
13	6,3	28,0	Corr.	c.20	13	6,3	30,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20								
14	6,3	84,0	Corr.	c.20	14	6,3	90,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20								
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----								
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	114,3	Var.	c.14	16	6,3	145,5	Var.	c.11	16	8,0	123,1	Var.	c.13								
17	----	----	----	----	17	6,3	188,2	Var.	c.17	17	----	----	----	----	17	6,3	246,2	Var.	c.13	17	6,3	266,7	Var.	c.12	17	8,0	246,2	Var.	c.13	17	8,0	266,7	Var.	c.12	17	10,0	228,6	Var.	c.14								
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																															
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50											
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)									
6,3	0,252	135,3	6,3	0,252	89,4	6,3	0,252	133,8	6,3	0,252	109,1	6,3	0,252	150,4	6,3	0,252	76,5	6,3	0,252	72,3	6,3	0,252	40,8	6,3	0,252	72,3	6,3	0,252	40,8	6,3	0,252	40,8	6,3	0,252	40,8	6,3	0,252	40,8									
8,0	0,393	68,6	8,0	0,393	95,4	8,0	0,393	30,7	8,0	0,393	118,3	8,0	0,393	82,7	8,0	0,393	164,4	8,0	0,393	181,3	8,0	0,393	159,4	8,0	0,393	181,3	8,0	0,393	159,4	8,0	0,393	159,4	8,0	0,393	159,4	8,0	0,393	159,4									
10,0	0,624	----	10,0	0,624	59,5	10,0	0,624	128,7	10,0	0,624	----	10,0	0,624	----	10,0	0,624	79,6	10,0	0,624	80,1	10,0	0,624	133,6	10,0	0,624	80,1	10,0	0,624	133,6	10,0	0,624	133,6	10,0	0,624	133,6	10,0	0,624	133,6									
12,5	0,988	----	12,5	0,988	69,0	12,5	0,988	----	12,5	0,988	186,2	12,5	0,988	197,6	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	77,1	12,5	0,988	----	12,5	0,988	77,1	12,5	0,988	77,1	12,5	0,988	77,1	12,5	0,988	77,1									
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	169,7	16,0	1,570	305,6	16,0	1,570	347,4	16,0	1,570	305,6	16,0	1,570	347,4	16,0	1,570	347,4	16,0	1,570	347,4	16,0	1,570	347,4									
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----									
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----									
TOTAL (Kg/m)			241,6			TOTAL (Kg/m)			351,0			TOTAL (Kg/m)			330,9			TOTAL (Kg/m)			451,3			TOTAL (Kg/m)			468,4			TOTAL (Kg/m)			597,7			TOTAL (Kg/m)			639,3			TOTAL (Kg/m)			758,3		
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																															
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)									
1	100	55	1	100	55	1	100	60	1	100	60	1	100	65	1	100	65	1	100	65	1	100	70	1	100	70	1	100	75	1	100	75	1	100	75	1	100	75									
2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----									
3	250	12	3	250	12	3	250	17	3	250	17	3	250	22	3	250	22	3	250	22	3	250	27	3	250	27	3	250	32	3	250	32	3	250	32	3	250	32									
4	250	12	4	250	12	4	250	17	4	250	17	4	250	22	4	250	22	4	250	22	4	250	27	4	250	27	4	250	32	4	250	32	4	250	32	4	250	32									
5	100	210	5	100	210	5	100	215	5	100	215	5	100	220	5	100	220	5	100	220	5	100	225	5	100	225	5	100	230	5	100	230	5	100	230	5	100	230									
6	155	120	6	155	120	6	160	120	6	160	120	6	165	120	6																																

BTCC 250 x 350					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																				
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000						
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,27 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa						
LISTA DE FERROS (POR METRO LINEAR)																																									
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS						
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)		
1	8,0	15,4	255	c.26	1	12,5	14,3	255	c.28	1	10,0	18,2	260	c.22	1	12,5	18,2	260	c.22	1	12,5	20,0	265	c.20	1	16,0	16,7	265	c.24	1	16,0	15,4	270	c.26	1	16,0	20,0	270	c.20		
2	8,0	7,7	470	c.26	2	12,5	7,1	470	c.28	2	10,0	9,1	475	c.22	2	12,5	9,1	475	c.22	2	12,5	10,0	480	c.20	2	16,0	8,3	480	c.24	2	16,0	7,7	485	c.26	2	16,0	10,0	485	c.20		
3	10,0	30,8	262	c.13	3	10,0	36,4	262	c.11	3	10,0	40,0	267	c.10	3	12,5	33,3	267	c.12	3	12,5	36,4	272	c.11	3	12,5	40,0	272	c.10	3	16,0	30,8	277	c.13	3	16,0	33,3	277	c.12		
4	6,3	13,3	262	c.15	4	8,0	18,2	262	c.11	4	8,0	15,4	267	c.13	4	8,0	20,0	267	c.10	4	10,0	15,4	272	c.13	4	10,0	18,2	272	c.11	4	10,0	18,2	277	c.11	4	12,5	18,2	277	c.11		
5	6,3	20,0	335	c.20	5	10,0	16,7	335	c.24	5	8,0	18,2	340	c.22	5	10,0	16,7	340	c.24	5	8,0	20,0	345	c.20	5	10,0	18,2	345	c.22	5	8,0	20,0	350	c.20	5	10,0	20,0	350	c.20		
6	6,3	20,0	295	c.20	6	10,0	16,7	295	c.24	6	8,0	18,2	300	c.22	6	10,0	16,7	300	c.24	6	8,0	20,0	305	c.20	6	10,0	18,2	305	c.22	6	8,0	20,0	310	c.20	6	10,0	20,0	310	c.20		
7	6,3	8,0	382	c.25	7	6,3	8,0	382	c.25	7	6,3	9,5	392	c.21	7	6,3	9,5	392	c.21	7	6,3	11,8	402	c.17	7	6,3	11,8	402	c.17	7	8,0	9,1	412	c.22	7	8,0	9,1	412	c.22		
8	6,3	16,0	382	c.25	8	6,3	16,0	382	c.25	8	6,3	19,0	392	c.21	8	6,3	19,0	392	c.21	8	6,3	23,5	402	c.17	8	6,3	23,5	402	c.17	8	8,0	18,2	412	c.22	8	8,0	18,2	412	c.22		
9	6,3	20,0	162	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	6,3	20,0	183	c.20	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	225	c.20		
10	6,3	20,0	232	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	6,3	20,0	270	c.20	10	6,3	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	344	c.20		
11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	6,3	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	10,0	78,0	Corr.	c.20	11	8,0	78,0	Corr.	c.20	11	10,0	78,0	Corr.	c.20		
12	6,3	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	8,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20	12	10,0	78,0	Corr.	c.20		
13	6,3	32,0	Corr.	c.20	13	6,3	36,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	8,0	32,0	Corr.	c.20		
14	6,3	96,0	Corr.	c.20	14	6,3	108,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20		
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----		
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	76,2	Var.	c.21	16	6,3	76,2	Var.	c.21	16	6,3	123,1	Var.	c.13	16	6,3	145,5	Var.	c.11	16	8,0	145,5	Var.	c.11		
17	----	----	----	----	17	6,3	188,2	Var.	c.17	17	----	----	----	----	17	6,3	246,2	Var.	c.13	17	6,3	266,7	Var.	c.12	17	8,0	246,2	Var.	c.13	17	8,0	266,7	Var.	c.12	17	10,0	266,7	Var.	c.12		
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																									
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50					
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)			
6,3	0,252	155,1	6,3	0,252	78,8	6,3	0,252	102,9	6,3	0,252	116,7	6,3	0,252	144,7	6,3	0,252	78,3	6,3	0,252	55,7	6,3	0,252	24,2	6,3	0,252	55,7	6,3	0,252	24,2	6,3	0,252	55,7	6,3	0,252	24,2	6,3	0,252	24,2			
8,0	0,393	29,7	8,0	0,393	80,0	8,0	0,393	92,6	8,0	0,393	82,3	8,0	0,393	112,4	8,0	0,393	109,7	8,0	0,393	238,5	8,0	0,393	138,1	8,0	0,393	238,5	8,0	0,393	138,1	8,0	0,393	238,5	8,0	0,393	138,1	8,0	0,393	138,1			
10,0	0,624	50,4	10,0	0,624	125,2	10,0	0,624	123,1	10,0	0,624	66,7	10,0	0,624	26,1	10,0	0,624	202,1	10,0	0,624	80,1	10,0	0,624	286,2	10,0	0,624	80,1	10,0	0,624	286,2	10,0	0,624	80,1	10,0	0,624	286,2	10,0	0,624	286,2			
12,5	0,988	----	12,5	0,988	69,0	12,5	0,988	----	12,5	0,988	177,3	12,5	0,988	197,6	12,5	0,988	107,5	12,5	0,988	----	12,5	0,988	49,8	12,5	0,988	107,5	12,5	0,988	49,8	12,5	0,988	107,5	12,5	0,988	49,8	12,5	0,988	49,8			
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	169,7	16,0	1,570	295,5	16,0	1,570	343,4	16,0	1,570	295,5	16,0	1,570	343,4	16,0	1,570	295,5	16,0	1,570	343,4	16,0	1,570	343,4			
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		272,9	TOTAL (Kg/m)		390,7	TOTAL (Kg/m)		356,3	TOTAL (Kg/m)		480,7	TOTAL (Kg/m)		518,5	TOTAL (Kg/m)		667,3	TOTAL (Kg/m)		669,8	TOTAL (Kg/m)		841,7	TOTAL (Kg/m)		841,7	TOTAL (Kg/m)		841,7	TOTAL (Kg/m)		841,7	TOTAL (Kg/m)		841,7	TOTAL (Kg/m)		841,7			
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																									
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)			
1	100	55	1	100	55	1	100	60	1	100	60	1	100	65	1	100	65	1	100	70	1	100	70	1	100	75	1	100	80	1	100	85	1	100	90	1	100	95	1	100	100
2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----	2	100	----
3	250	12	3	250	12	3	250	17	3	250	17	3	250	22	3	250	22	3	250	27	3	250	27	3	250	32	3	250	37	3	250	42	3	250	47	3	250	52	3	250	57
4	250	12	4	250	12	4	250	17	4	250	17	4	250	22	4	250	22	4	250	27	4	250	27	4	250	32	4	250	37	4	250	42	4	250	47	4	250	52	4	250	57
5	100	235	5	100	235	5	100	240	5	100	240	5	100	245	5	100	245	5	100	250	5	100	250	5	100	255	5	100	260	5	100	265	5	100	270	5	100	275	5	100	280
6	155	140	6	155	140	6	160	140	6	160	140	6	165	140	6</																										

BTCC 300 x 200					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																				
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000						
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,19 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,25 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,31 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,37 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,43 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,55 MPa						
LISTA DE FERROS (POR METRO LINEAR)																																									
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS						
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)		
1	8,0	15,4	305	c.26	1	12,5	16,7	305	c.24	1	16,0	15,4	305	c.26	1	16,0	18,2	310	c.22	1	16,0	20,0	310	c.20	1	20,0	15,4	315	c.26	1	20,0	16,7	320	c.24	1	20,0	18,2	325	c.22		
2	8,0	7,7	565	c.26	2	12,5	8,3	565	c.24	2	16,0	7,7	565	c.26	2	16,0	9,1	570	c.22	2	16,0	10,0	570	c.20	2	20,0	7,7	575	c.26	2	20,0	8,3	580	c.24	2	20,0	9,1	585	c.22		
3	8,0	33,3	317	c.12	3	10,0	40,0	317	c.10	3	12,5	36,4	317	c.11	3	16,0	28,6	322	c.14	3	16,0	33,3	322	c.12	3	16,0	36,4	327	c.11	3	16,0	40,0	332	c.10	3	20,0	33,3	337	c.12		
4	6,3	11,8	317	c.17	4	8,0	18,2	317	c.11	4	10,0	15,4	317	c.13	4	10,0	18,2	322	c.11	4	12,5	16,7	322	c.12	4	12,5	16,7	327	c.12	4	12,5	18,2	332	c.11	4	12,5	20,0	337	c.10		
5	6,3	8,0	285	c.50	5	6,3	15,4	285	c.26	5	6,3	15,4	285	c.26	5	6,3	13,3	290	c.30	5	6,3	18,2	290	c.22	5	6,3	13,3	295	c.30	5	6,3	8,0	300	c.50	5	6,3	8,0	305	c.50		
6	6,3	8,0	265	c.50	6	6,3	15,4	265	c.26	6	6,3	15,4	265	c.26	6	6,3	13,3	270	c.30	6	6,3	18,2	270	c.22	6	6,3	13,3	275	c.30	6	6,3	8,0	280	c.50	6	6,3	8,0	285	c.50		
7	6,3	9,5	242	c.21	7	6,3	9,5	242	c.21	7	6,3	9,5	242	c.21	7	6,3	11,8	252	c.17	7	6,3	11,8	252	c.17	7	8,0	9,1	262	c.22	7	8,0	10,5	272	c.19	7	8,0	11,8	282	c.17		
8	6,3	19,0	242	c.21	8	6,3	19,0	242	c.21	8	6,3	19,0	242	c.21	8	6,3	23,5	252	c.17	8	6,3	23,5	252	c.17	8	8,0	18,2	262	c.22	8	8,0	21,1	272	c.19	8	8,0	23,5	282	c.17		
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	10,0	20,0	246	c.20	9	10,0	20,0	267	c.20		
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	10,0	20,0	380	c.20	10	10,0	20,0	416	c.20		
11	6,3	90,0	Corr.	c.20	11	6,3	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20		
12	6,3	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20		
13	6,3	18,0	Corr.	c.20	13	6,3	20,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20		
14	6,3	54,0	Corr.	c.20	14	6,3	60,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20		
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----		
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	94,1	Var.	c.17	16	6,3	123,1	Var.	c.13	16	6,3	145,5	Var.	c.11	16	8,0	133,3	Var.	c.12		
17	----	----	----	----	17	6,3	160,0	Var.	c.20	17	6,3	228,6	Var.	c.14	17	6,3	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	10,0	200,0	Var.	c.16	17	10,0	266,7	Var.	c.12		
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																									
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																	
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)			
6,3	0,252	124,2	6,3	0,252	100,1	6,3	0,252	110,8	6,3	0,252	102,6	6,3	0,252	80,2	6,3	0,252	57,1	6,3	0,252	55,1	6,3	0,252	30,0	6,3	0,252	57,1	6,3	0,252	55,1	6,3	0,252	30,0	6,3	0,252	30,0	6,3	0,252	30,0			
8,0	0,393	77,0	8,0	0,393	58,0	8,0	0,393	70,7	8,0	0,393	75,5	8,0	0,393	107,5	8,0	0,393	146,0	8,0	0,393	33,8	8,0	0,393	77,9	8,0	0,393	146,0	8,0	0,393	33,8	8,0	0,393	77,9	8,0	0,393	77,9	8,0	0,393	77,9			
10,0	0,624	----	10,0	0,624	79,1	10,0	0,624	30,5	10,0	0,624	92,7	10,0	0,624	112,3	10,0	0,624	56,2	10,0	0,624	220,4	10,0	0,624	264,5	10,0	0,624	112,3	10,0	0,624	56,2	10,0	0,624	220,4	10,0	0,624	264,5	10,0	0,624	264,5			
12,5	0,988	----	12,5	0,988	96,7	12,5	0,988	114,0	12,5	0,988	----	12,5	0,988	53,1	12,5	0,988	142,9	12,5	0,988	148,6	12,5	0,988	155,5	12,5	0,988	142,9	12,5	0,988	148,6	12,5	0,988	148,6	12,5	0,988	155,5	12,5	0,988	155,5			
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	179,7	16,0	1,570	352,3	16,0	1,570	392,9	16,0	1,570	224,6	16,0	1,570	246,2	16,0	1,570	37,7	16,0	1,570	392,9	16,0	1,570	224,6	16,0	1,570	246,2	16,0	1,570	37,7	16,0	1,570	37,7			
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	230,1	20,0	2,480	251,9	20,0	2,480	557,0	20,0	2,480	230,1	20,0	2,480	251,9	20,0	2,480	251,9	20,0	2,480	557,0	20,0	2,480	557,0			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		238,9	TOTAL (Kg/m)		371,6	TOTAL (Kg/m)		505,7	TOTAL (Kg/m)		623,1	TOTAL (Kg/m)		746,0	TOTAL (Kg/m)		856,9	TOTAL (Kg/m)		956,0	TOTAL (Kg/m)		1122,6	TOTAL (Kg/m)		856,9	TOTAL (Kg/m)		956,0	TOTAL (Kg/m)		1122,6	TOTAL (Kg/m)		1122,6	TOTAL (Kg/m)		1122,6			
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																									
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)			
1	120	65	1	120	65	1	120	65	1	120	70	1	120	70	1	120	75	1	120	80	1	120	85	1	120	75	1	120	80	1	120	85	1	120	85	1	120	85			
2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----
3	300	17	3	300	17	3	300	17	3	300	22	3	300	22	3	300	27	3	300	32	3	300	37	3	300	27	3	300	32	3	300	37	3	300	37	3	300	37			
4	300	17	4	300	17	4	300	17	4	300	22	4	300	22	4	300	27	4	300	32	4	300	37	4	300	27	4	300	32	4	300	37	4	300	37	4	300	37			
5	120	165	5	120	165	5	120	165	5	120	170	5	120	170	5	120	175	5	120	180	5	120	185	5	120	175	5	120	180	5	120	185	5	120	185	5	120	185			
6	185	80	6	185	80	6	185	80	6	190	80	6	190	80	6	195	80	6	195	80	6	200	80	6	205	80	6	200	80	6	205	80	6	205	80	6	205	80			
7	242	----	7	242	----	7	242	----	7	252	----	7	252	----	7	262	----	7	262	----	7	272	----	7	282	----	7	272	----	7	282	----	7	282	----	7	282	----	7	282	----
8	242	----	8	242	----	8	242	----	8	252	----	8	252	----	8	262	----	8	262	----	8	272	----	8	282	----	8	272	----	8	282	----	8	282	----	8	282	----	8	282	----
9	143	----	9	----	----	9	143	----	9	164	----	9	164	----	9	185	----	9	185	----	9	206	----	9	227	----	9	206	----	9	227	----	9	227	----	9	227	----			

BTCC 300 x 250					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																						
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,38 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,44 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa								
LISTA DE FERROS (POR METRO LINEAR)																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)				
1	8,0	15,4	305	c.26	1	12,5	16,7	305	c.24	1	16,0	15,4	305	c.26	1	16,0	18,2	310	c.22	1	16,0	20,0	310	c.20	1	20,0	15,4	315	c.26	1	20,0	16,7	320	c.24	1	20,0	18,2	325	c.22				
2	8,0	7,7	565	c.26	2	12,5	8,3	565	c.24	2	16,0	7,7	565	c.26	2	16,0	9,1	570	c.22	2	16,0	10,0	570	c.20	2	20,0	7,7	575	c.26	2	20,0	8,3	580	c.24	2	20,0	9,1	585	c.22				
3	8,0	36,4	317	c.11	3	12,5	30,8	317	c.13	3	12,5	36,4	317	c.11	3	12,5	40,0	322	c.10	3	16,0	33,3	322	c.12	3	16,0	36,4	327	c.11	3	16,0	40,0	332	c.10	3	20,0	33,3	337	c.12				
4	6,3	12,5	317	c.16	4	8,0	18,2	317	c.11	4	10,0	16,7	317	c.12	4	10,0	18,2	322	c.11	4	12,5	16,7	322	c.12	4	12,5	16,7	327	c.12	4	12,5	18,2	332	c.11	4	12,5	20,0	337	c.10				
5	6,3	8,0	310	c.50	5	6,3	18,2	310	c.22	5	6,3	16,7	310	c.24	5	6,3	14,3	315	c.28	5	6,3	20,0	315	c.20	5	6,3	14,3	320	c.28	5	6,3	8,3	325	c.48	5	6,3	8,0	330	c.50				
6	6,3	8,0	285	c.50	6	6,3	18,2	285	c.22	6	6,3	16,7	285	c.24	6	6,3	14,3	290	c.28	6	6,3	20,0	290	c.20	6	6,3	14,3	295	c.28	6	6,3	8,3	300	c.48	6	6,3	8,0	305	c.50				
7	6,3	9,5	292	c.21	7	6,3	9,5	292	c.21	7	6,3	9,5	292	c.21	7	6,3	11,8	302	c.17	7	6,3	11,8	302	c.17	7	8,0	9,1	312	c.22	7	8,0	10,5	322	c.19	7	8,0	11,8	332	c.17				
8	6,3	19,0	292	c.21	8	6,3	19,0	292	c.21	8	6,3	19,0	292	c.21	8	6,3	23,5	302	c.17	8	6,3	23,5	302	c.17	8	8,0	18,2	312	c.22	8	8,0	21,1	322	c.19	8	8,0	23,5	332	c.17				
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	10,0	20,0	246	c.20	9	10,0	20,0	267	c.20				
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	10,0	20,0	380	c.20	10	10,0	20,0	416	c.20				
11	6,3	90,0	Corr.	c.20	11	6,3	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20				
12	6,3	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20				
13	6,3	22,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20				
14	6,3	66,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20				
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----				
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	100,0	Var.	c.16	16	6,3	123,1	Var.	c.13	16	6,3	145,5	Var.	c.11	16	8,0	133,3	Var.	c.12				
17	----	----	----	----	17	6,3	160,0	Var.	c.20	17	6,3	228,6	Var.	c.14	17	6,3	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	10,0	200,0	Var.	c.16	17	10,0	266,7	Var.	c.12				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																											
RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50				RESUMO - AÇO CA -50							
Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)		Ø (mm)	Kg/m	PESO (Kg)					
6,3	0,252	133,3		6,3	0,252	115,7		6,3	0,252	122,1		6,3	0,252	114,1		6,3	0,252	94,4		6,3	0,252	64,2		6,3	0,252	60,5		6,3	0,252	60,5		6,3	0,252	60,5		6,3	0,252	35,0					
8,0	0,393	80,9		8,0	0,393	58,0		8,0	0,393	70,7		8,0	0,393	75,5		8,0	0,393	107,5		8,0	0,393	151,4		8,0	0,393	40,0		8,0	0,393	40,0		8,0	0,393	40,0		8,0	0,393	84,8					
10,0	0,624	----		10,0	0,624	----		10,0	0,624	33,0		10,0	0,624	92,7		10,0	0,624	112,3		10,0	0,624	56,2		10,0	0,624	220,4		10,0	0,624	220,4		10,0	0,624	220,4		10,0	0,624	264,5					
12,5	0,988	----		12,5	0,988	193,1		12,5	0,988	114,0		12,5	0,988	127,3		12,5	0,988	53,1		12,5	0,988	142,9		12,5	0,988	148,6		12,5	0,988	148,6		12,5	0,988	148,6		12,5	0,988	155,5					
16,0	1,570	37,7		16,0	1,570	37,7		16,0	1,570	179,7		16,0	1,570	207,7		16,0	1,570	392,9		16,0	1,570	224,6		16,0	1,570	246,2		16,0	1,570	246,2		16,0	1,570	246,2		16,0	1,570	37,7					
20,0	2,480	----		20,0	2,480	----		20,0	2,480	----		20,0	2,480	----		20,0	2,480	----		20,0	2,480	230,1		20,0	2,480	251,9		20,0	2,480	251,9		20,0	2,480	251,9		20,0	2,480	557,0					
25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----		25,0	3,980	----					
TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)				TOTAL (Kg/m)			
251,9				404,5				519,5				617,3				760,2				869,4				TOTAL (Kg/m)				967,6				TOTAL (Kg/m)				1134,5							
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																											
N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)		N	a (cm)	b (cm)					
1	120	65		1	120	65		1	120	65		1	120	70		1	120	70		1	120	75		1	120	80		1	120	85		1	120	85		1	120	85					
2	120	----		2	120	----		2	120	----		2	120	----		2	120	----		2	120	----		2	120	----		2	120	----		2	120	----		2	120	----					
3	300	17		3	300	17		3	300	17		3	300	22		3	300	22		3	300	27		3	300	32		3	300	37		3	300	37		3	300	37					
4	300	17		4	300	17		4	300	17		4	300	22		4	300	22		4	300	27		4	300	32		4	300	37		4	300	37		4	300	37					
5	120	190		5	120	190		5	120	190		5	120	195		5	120	195		5	120	200		5	120	205		5	120	210		5	120	210		5	120	210					
6	185	100		6	185	100		6	185	100		6	190	100		6	190	100		6	190	100		6	195	100		6	200	100		6	200	100		6	205	100					
7	292	----		7	292	----		7	292	----		7	302	----		7	302	----		7	302	----		7	312	----		7	322	----		7	332	----		7	332	----					
8	292	----		8	292	----		8	292	----		8	302	----		8	302	----		8	302	----		8	312	----		8	322	----		8	332	----		8	332	----					
9	143	----		9	----	----		9	143	----		9	164	----		9	164	----		9	164	----		9	185	----		9	20														

BTCC 300 x 300					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																				
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000						
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,15 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,27 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa						
LISTA DE FERROS (POR METRO LINEAR)																																									
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS						
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)		
1	8,0	16,7	305	c.24	1	12,5	16,7	305	c.24	1	16,0	15,4	305	c.26	1	16,0	16,7	310	c.24	1	16,0	18,2	315	c.22	1	16,0	20,0	315	c.20	1	20,0	15,4	320	c.26	1	20,0	18,2	325	c.22		
2	8,0	8,3	565	c.24	2	12,5	8,3	565	c.24	2	16,0	7,7	565	c.26	2	16,0	8,3	570	c.24	2	16,0	9,1	575	c.22	2	16,0	10,0	575	c.20	2	20,0	7,7	580	c.26	2	20,0	9,1	585	c.22		
3	8,0	40,0	317	c.10	3	12,5	30,8	317	c.13	3	12,5	36,4	317	c.11	3	12,5	40,0	322	c.10	3	16,0	33,3	327	c.12	3	16,0	36,4	327	c.11	3	16,0	40,0	332	c.10	3	20,0	30,8	337	c.13		
4	6,3	13,3	317	c.15	4	8,0	20,0	317	c.10	4	10,0	16,7	317	c.12	4	10,0	18,2	322	c.11	4	10,0	20,0	327	c.10	4	12,5	16,7	327	c.12	4	12,5	18,2	332	c.11	4	12,5	20,0	337	c.10		
5	6,3	10,5	335	c.38	5	8,0	16,7	335	c.24	5	8,0	15,4	335	c.26	5	6,3	20,0	340	c.20	5	6,3	15,4	345	c.26	5	6,3	20,0	345	c.20	5	6,3	13,3	350	c.30	5	6,3	9,1	355	c.44		
6	6,3	10,5	305	c.38	6	8,0	16,7	305	c.24	6	8,0	15,4	305	c.26	6	6,3	20,0	310	c.20	6	6,3	15,4	315	c.26	6	6,3	20,0	315	c.20	6	6,3	13,3	320	c.30	6	6,3	9,1	325	c.44		
7	6,3	9,5	342	c.21	7	6,3	9,5	342	c.21	7	6,3	9,5	342	c.21	7	6,3	11,8	352	c.17	7	8,0	9,1	362	c.22	7	8,0	9,1	362	c.22	7	8,0	10,5	372	c.19	7	8,0	11,8	382	c.17		
8	6,3	19,0	342	c.21	8	6,3	19,0	342	c.21	8	6,3	19,0	342	c.21	8	8,0	18,2	362	c.22	8	8,0	18,2	362	c.22	8	8,0	18,2	362	c.22	8	8,0	21,1	372	c.19	8	8,0	23,5	382	c.17		
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	225	c.20	9	10,0	20,0	246	c.20	9	10,0	20,0	267	c.20		
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	344	c.20	10	10,0	20,0	380	c.20	10	10,0	20,0	416	c.20		
11	6,3	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20		
12	6,3	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20		
13	6,3	28,0	Corr.	c.20	13	6,3	30,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20		
14	6,3	84,0	Corr.	c.20	14	6,3	90,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20		
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----		
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	80,0	Var.	c.20	16	6,3	133,3	Var.	c.12	16	6,3	160,0	Var.	c.10	16	8,0	133,3	Var.	c.12		
17	----	----	----	----	17	6,3	160,0	Var.	c.20	17	6,3	228,6	Var.	c.14	17	6,3	290,9	Var.	c.11	17	8,0	200,0	Var.	c.16	17	8,0	290,9	Var.	c.11	17	10,0	320,0	Var.	c.10	17	10,0	266,7	Var.	c.12		
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																									
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																	
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)			
6,3	0,252	148,5	6,3	0,252	73,3	6,3	0,252	106,7	6,3	0,252	135,5	6,3	0,252	66,7	6,3	0,252	83,0	6,3	0,252	78,5	6,3	0,252	43,8	6,3	0,252	133,0	6,3	0,252	78,5	6,3	0,252	133,0	6,3	0,252	43,8	6,3	0,252	133,0			
8,0	0,393	88,3	8,0	0,393	137,7	8,0	0,393	109,5	8,0	0,393	75,5	8,0	0,393	169,2	8,0	0,393	156,7	8,0	0,393	133,0	8,0	0,393	91,8	8,0	0,393	133,0	8,0	0,393	133,0	8,0	0,393	133,0	8,0	0,393	133,0	8,0	0,393	133,0			
10,0	0,624	----	10,0	0,624	----	10,0	0,624	33,0	10,0	0,624	92,7	10,0	0,624	97,0	10,0	0,624	56,2	10,0	0,624	134,3	10,0	0,624	264,5	10,0	0,624	134,3	10,0	0,624	134,3	10,0	0,624	134,3	10,0	0,624	134,3	10,0	0,624	134,3			
12,5	0,988	----	12,5	0,988	193,1	12,5	0,988	114,0	12,5	0,988	127,3	12,5	0,988	----	12,5	0,988	142,9	12,5	0,988	148,6	12,5	0,988	155,5	12,5	0,988	142,9	12,5	0,988	142,9	12,5	0,988	142,9	12,5	0,988	142,9	12,5	0,988	142,9			
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	179,7	16,0	1,570	193,2	16,0	1,570	380,8	16,0	1,570	413,7	16,0	1,570	246,2	16,0	1,570	37,7	16,0	1,570	246,2	16,0	1,570	246,2	16,0	1,570	246,2	16,0	1,570	246,2	16,0	1,570	246,2			
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	233,0	20,0	2,480	536,1	20,0	2,480	233,0	20,0	2,480	233,0	20,0	2,480	233,0	20,0	2,480	233,0	20,0	2,480	233,0			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		274,5	TOTAL (Kg/m)		441,8	TOTAL (Kg/m)		542,9	TOTAL (Kg/m)		624,2	TOTAL (Kg/m)		713,7	TOTAL (Kg/m)		852,5	TOTAL (Kg/m)		973,6	TOTAL (Kg/m)		1129,4	TOTAL (Kg/m)		1129,4	TOTAL (Kg/m)		1129,4	TOTAL (Kg/m)		1129,4	TOTAL (Kg/m)		1129,4	TOTAL (Kg/m)		1129,4			
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																									
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)			
1	120	65	1	120	65	1	120	65	1	120	70	1	120	75	1	120	75	1	120	75	1	120	80	1	120	85	1	120	85	1	120	85	1	120	85	1	120	85			
2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----
3	300	17	3	300	17	3	300	17	3	300	22	3	300	27	3	300	27	3	300	27	3	300	32	3	300	37	3	300	37	3	300	37	3	300	37	3	300	37	3	300	37
4	300	17	4	300	17	4	300	17	4	300	22	4	300	27	4	300	27	4	300	27	4	300	32	4	300	37	4	300	37	4	300	37	4	300	37	4	300	37	4	300	37
5	120	215	5	120	215	5	120	215	5	120	220	5	120	225	5	120	225	5	120	225	5	120	230	5	120	235	5	120	235	5	120	235	5	120	235	5	120	235	5	120	235
6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120	6	185	120
7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----	7	342	----
8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----	8	342	----
9	143	----	9	----	----	9	143	----	9	143	----	9																													

BTCC 300 x 350	TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,22 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,28 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,34 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,40 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,46 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,58 MPa

LISTA DE FERROS (POR METRO LINEAR)																								
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	16,7	305	c.24	1	12,5	16,7	305	c.24	1	16,0	14,3	305	c.28	1	16,0	16,7	310	c.24	1	16,0	18,2	315	c.22
2	8,0	8,3	565	c.24	2	12,5	8,3	565	c.24	2	16,0	7,1	565	c.28	2	16,0	8,3	570	c.24	2	16,0	9,1	575	c.22
3	10,0	30,8	317	c.13	3	12,5	30,8	317	c.13	3	12,5	36,4	317	c.11	3	12,5	40,0	322	c.10	3	16,0	30,8	327	c.13
4	6,3	14,3	317	c.14	4	8,0	20,0	317	c.10	4	10,0	16,7	317	c.12	4	10,0	18,2	322	c.11	4	10,0	20,0	327	c.10
5	6,3	14,3	360	c.28	5	10,0	15,4	360	c.26	5	10,0	15,4	360	c.26	5	8,0	20,0	365	c.20	5	8,0	16,7	370	c.24
6	6,3	14,3	325	c.28	6	10,0	15,4	325	c.26	6	10,0	15,4	325	c.26	6	8,0	20,0	330	c.20	6	8,0	16,7	335	c.24
7	6,3	9,5	392	c.21	7	6,3	9,5	392	c.21	7	6,3	9,5	392	c.21	7	6,3	11,8	402	c.17	7	8,0	9,1	412	c.22
8	6,3	19,0	392	c.21	8	6,3	19,0	392	c.21	8	6,3	19,0	392	c.21	8	6,3	23,5	402	c.17	8	8,0	18,2	412	c.22
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20
11	6,3	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20
12	6,3	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20
13	6,3	32,0	Corr.	c.20	13	6,3	36,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20
14	6,3	96,0	Corr.	c.20	14	6,3	108,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	84,2	Var.	c.19	16	6,3	133,3	Var.	c.12
17	----	----	----	----	17	6,3	160,0	Var.	c.20	17	6,3	213,3	Var.	c.15	17	6,3	290,9	Var.	c.11	17	8,0	200,0	Var.	c.16

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50					
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)
6,3	0,252	164,7	6,3	0,252	83,0	6,3	0,252	112,3	6,3	0,252	111,3	6,3	0,252	45,8	6,3	0,252	53,8	6,3	0,252	32,3	6,3	0,252	68,8
8,0	0,393	38,4	8,0	0,393	95,7	8,0	0,393	70,7	8,0	0,393	130,1	8,0	0,393	220,9	8,0	0,393	162,1	8,0	0,393	213,2	8,0	0,393	102,2
10,0	0,624	60,9	10,0	0,624	65,8	10,0	0,624	98,9	10,0	0,624	92,7	10,0	0,624	97,0	10,0	0,624	180,1	10,0	0,624	134,3	10,0	0,624	264,5
12,5	0,988	----	12,5	0,988	193,1	12,5	0,988	114,0	12,5	0,988	127,3	12,5	0,988	----	12,5	0,988	58,8	12,5	0,988	148,6	12,5	0,988	155,5
16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	169,1	16,0	1,570	193,2	16,0	1,570	368,0	16,0	1,570	397,8	16,0	1,570	227,4	16,0	1,570	37,7
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	233,0	20,0	2,480	512,4
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----
TOTAL (Kg/m)		301,7	TOTAL (Kg/m)		475,3	TOTAL (Kg/m)		565,0	TOTAL (Kg/m)		654,6	TOTAL (Kg/m)		731,7	TOTAL (Kg/m)		852,6	TOTAL (Kg/m)		988,8	TOTAL (Kg/m)		1141,1

DIMENSÕES DAS DOBRAS DAS ARMADURAS																				
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)
1	120	65	1	120	65	1	120	65	1	120	70	1	120	75	1	120	75	1	120	80
2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----	2	120	----
3	300	17	3	300	17	3	300	17	3	300	22	3	300	27	3	300	27	3	300	32
4	300	17	4	300	17	4	300	17	4	300	22	4	300	27	4	300	27	4	300	32
5	120	240	5	120	240	5	120	240	5	120	245	5	120	250	5	120	250	5	120	255
6	185	140	6	185	140	6	185	140	6	190	140	6	195	140	6	195	140	6	200	140
7	392	----	7	392	----	7	392	----	7	402	----	7	412	----	7	412	----	7	422	----
8	392	----	8	392	----	8	392	----	8	402	----	8	412	----	8	412	----	8	422	----
9	143	----	9	----	----	9	143	----	9	164	----	9	185	----	9	185	----	9	206	----
10	135	----	10	----	----	10	135	----	10	153	----	10	172	----	10	172	----	10	190	----

ASSINATURA DAS AUTORIDADES Eng^o Selma Schwab Coordenadora da DP/GNT Eng^o Mário S.Bortone Gerente da DP/GPE Eng^o Roger Gama Veloso Diretor de Projetos Eng^o Nelson A. Reis Vice - Diretor Geral										 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS
										CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear) DES. 98

BTCC
300 x 400

ALTURA DO ATERRO (cm)
0 < h < 100

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,13 MPa

ALTURA DO ATERRO (cm)
100 < h < 250

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,17 MPa

ALTURA DO ATERRO (cm)
250 < h < 500

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,23 MPa

ALTURA DO ATERRO (cm)
500 < h < 750

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,29 MPa

ALTURA DO ATERRO (cm)
750 < h < 1000

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,35 MPa

ALTURA DO ATERRO (cm)
1000 < h < 1250

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,41 MPa

ALTURA DO ATERRO (cm)
1250 < h < 1500

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,47 MPa

ALTURA DO ATERRO (cm)
1500 < h < 2000

TENSÃO ADMISSÍVEL NO SOLO
DE FUNDAÇÃO = 0,59 MPa

LISTA DE FERROS (POR METRO LINEAR)

LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	16,7	305	c.24	1	12,5	16,7	305	c.24	1	12,5	20,0	305	c.20	1	16,0	16,7	310	c.24	1	16,0	18,2	310	c.22	1	16,0	20,0	315	c.20	1	20,0	15,4	320	c.26	1	20,0	16,7	325	c.24
2	8,0	8,3	565	c.24	2	12,5	8,3	565	c.24	2	12,5	10,0	565	c.20	2	16,0	8,3	570	c.24	2	16,0	9,1	570	c.22	2	16,0	10,0	575	c.20	2	20,0	7,7	580	c.26	2	20,0	8,3	585	c.24
3	10,0	33,3	317	c.12	3	12,5	30,8	317	c.13	3	12,5	36,4	317	c.11	3	12,5	40,0	322	c.10	3	16,0	30,8	322	c.13	3	16,0	33,3	327	c.12	3	16,0	36,4	332	c.11	3	16,0	40,0	337	c.10
4	6,3	15,4	317	c.13	4	8,0	20,0	317	c.10	4	10,0	16,7	317	c.12	4	10,0	20,0	322	c.10	4	12,5	16,7	322	c.12	4	12,5	18,2	327	c.11	4	12,5	18,2	332	c.11	4	16,0	14,3	337	c.14
5	6,3	20,0	385	c.20	5	10,0	20,0	385	c.20	5	10,0	20,0	385	c.20	5	10,0	20,0	390	c.20	5	12,5	16,7	390	c.24	5	12,5	16,7	395	c.24	5	10,0	18,2	400	c.22	5	10,0	18,2	405	c.22
6	6,3	20,0	345	c.20	6	10,0	20,0	345	c.20	6	10,0	20,0	345	c.20	6	10,0	20,0	350	c.20	6	12,5	16,7	350	c.24	6	12,5	16,7	355	c.24	6	10,0	18,2	360	c.22	6	10,0	18,2	365	c.22
7	6,3	9,5	442	c.21	7	6,3	9,5	442	c.21	7	6,3	9,5	442	c.21	7	6,3	11,8	452	c.17	7	6,3	11,8	452	c.17	7	8,0	9,1	462	c.22	7	8,0	10,5	472	c.19	7	8,0	11,8	482	c.17
8	6,3	19,0	442	c.21	8	6,3	19,0	442	c.21	8	6,3	19,0	442	c.21	8	6,3	23,5	452	c.17	8	6,3	23,5	452	c.17	8	8,0	18,2	462	c.22	8	8,0	21,1	472	c.19	8	8,0	23,5	482	c.17
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	10,0	20,0	246	c.20	9	10,0	20,0	267	c.20
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	10,0	20,0	380	c.20	10	10,0	20,0	416	c.20
11	6,3	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	8,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20	11	10,0	90,0	Corr.	c.20
12	6,3	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	8,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	10,0	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20	12	12,5	90,0	Corr.	c.20
13	6,3	38,0	Corr.	c.20	13	8,0	40,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20
14	6,3	114,0	Corr.	c.20	14	6,3	120,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	114,3	Var.	c.14	16	6,3	145,5	Var.	c.11	16	8,0	106,7	Var.	c.15	16	8,0	106,7	Var.	c.15	16	8,0	145,5	Var.	c.11
17	----	----	----	----	17	6,3	160,0	Var.	c.20	17	6,3	213,3	Var.	c.15	17	6,3	266,7	Var.	c.12	17	8,0	266,7	Var.	c.12	17	8,0	266,7	Var.	c.12	17	8,0	320,0	Var.	c.10	17	10,0	266,7	Var.	c.12

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)

RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
6,3	0,252	187,3	6,3	0,252	80,5	6,3	0,252	112,3	6,3	0,252	108,6	6,3	0,252	85,9	6,3	0,252	52,2	6,3	0,252	28,7	6,3	0,252	28,7	8,0	0,393	38,4	8,0	0,393	111,4	8,0	0,393	85,7	8,0	0,393	55,0	8,0	0,393	116,9	8,0	0,393	176,3	8,0	0,393	189,3	8,0	0,393	124,1	10,0	0,624	65,9	10,0	0,624	91,1	10,0	0,624	124,1	10,0	0,624	244,9	10,0	0,624	112,3	10,0	0,624	220,6	10,0	0,624	352,0	12,5	0,988	----	12,5	0,988	193,1	12,5	0,988	230,1	12,5	0,988	127,3	12,5	0,988	175,2	12,5	0,988	182,5	12,5	0,988	148,6	12,5	0,988	88,9	16,0	1,570	37,7	16,0	1,570	37,7	16,0	1,570	193,2	16,0	1,570	363,4	16,0	1,570	397,8	16,0	1,570	227,4	16,0	1,570	325,0	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	233,0	20,0	2,480	255,0	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----

BTCC 350 x 200					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																							
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000									
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 9,48 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,14 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,26 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,32 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,38 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,44 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,56 MPa									
LISTA DE FERROS (POR METRO LINEAR)																																												
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS									
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	8,0	20,0	350	c.20	1	16,0	14,3	350	c.28	1	16,0	18,2	350	c.22	1	16,0	20,0	355	c.20	1	20,0	16,7	360	c.24	1	20,0	18,2	365	c.22	1	20,0	20,0	365	c.20	1	25,0	15,4	375	c.26					
2	8,0	10,0	655	c.20	2	16,0	7,1	655	c.28	2	16,0	9,1	655	c.22	2	16,0	10,0	660	c.20	2	20,0	8,3	665	c.24	2	20,0	9,1	670	c.22	2	20,0	10,0	670	c.20	2	25,0	7,7	680	c.26					
3	8,0	40,0	367	c.10	3	12,5	33,3	367	c.12	3	16,0	30,8	367	c.13	3	16,0	36,4	372	c.11	3	16,0	40,0	377	c.10	3	20,0	30,8	382	c.13	3	20,0	33,3	382	c.12	3	20,0	40,0	392	c.10					
4	6,3	15,4	367	c.13	4	10,0	18,2	367	c.11	4	12,5	15,4	367	c.13	4	12,5	18,2	372	c.11	4	12,5	20,0	377	c.10	4	16,0	15,4	382	c.13	4	16,0	16,7	382	c.12	4	16,0	18,2	392	c.11					
5	6,3	9,5	305	c.42	5	8,0	15,4	305	c.26	5	8,0	18,2	305	c.22	5	8,0	18,2	310	c.22	5	8,0	18,2	315	c.22	5	8,0	15,4	320	c.26	5	8,0	20,0	320	c.20	5	6,3	14,3	330	c.28					
6	6,3	9,5	290	c.42	6	8,0	15,4	290	c.26	6	8,0	18,2	290	c.22	6	8,0	18,2	295	c.22	6	8,0	18,2	300	c.22	6	8,0	15,4	305	c.26	6	8,0	20,0	305	c.20	6	6,3	14,3	315	c.28					
7	6,3	9,5	242	c.21	7	6,3	9,5	242	c.21	7	6,3	9,5	242	c.21	7	6,3	11,8	252	c.17	7	8,0	9,1	262	c.22	7	8,0	10,5	272	c.19	7	8,0	10,5	272	c.19	7	10,0	8,7	292	c.23					
8	6,3	19,0	242	c.21	8	6,3	19,0	242	c.21	8	6,3	19,0	242	c.21	8	6,3	23,5	252	c.17	8	8,0	18,2	262	c.22	8	8,0	21,1	272	c.19	8	8,0	21,1	272	c.19	8	10,0	17,4	292	c.23					
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	183	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20	9	10,0	20,0	246	c.20	9	10,0	20,0	288	c.20					
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	270	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20	10	10,0	20,0	380	c.20	10	10,0	20,0	454	c.20					
11	6,3	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	12,5	108,0	Corr.	c.20	11	12,5	108,0	Corr.	c.20					
12	6,3	108,0	Corr.	c.20	12	8,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20					
13	6,3	18,0	Corr.	c.20	13	6,3	20,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20	13	6,3	18,0	Corr.	c.20					
14	6,3	54,0	Corr.	c.20	14	6,3	60,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20	14	6,3	54,0	Corr.	c.20					
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----					
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	106,7	Var.	c.15	16	6,3	133,3	Var.	c.12	16	8,0	114,3	Var.	c.14	16	8,0	145,5	Var.	c.11					
17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	8,0	213,3	Var.	c.15	17	8,0	266,7	Var.	c.12	17	8,0	290,9	Var.	c.11	17	8,0	320,0	Var.	c.10	17	10,0	266,7	Var.	c.12	17	10,0	290,9	Var.	c.11					
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																												
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																				
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	141,3	6,3	0,252	64,0	6,3	0,252	58,4	6,3	0,252	40,6	6,3	0,252	35,4	6,3	0,252	41,3	6,3	0,252	18,1	6,3	0,252	41,4																					
8,0	0,393	110,9	8,0	0,393	120,9	8,0	0,393	87,8	8,0	0,393	145,2	8,0	0,393	190,0	8,0	0,393	207,6	8,0	0,393	113,9	8,0	0,393	45,2																					
10,0	0,624	----	10,0	0,624	41,7	10,0	0,624	134,8	10,0	0,624	67,4	10,0	0,624	67,4	10,0	0,624	67,4	10,0	0,624	193,0	10,0	0,624	283,6																					
12,5	0,988	----	12,5	0,988	120,7	12,5	0,988	55,8	12,5	0,988	173,6	12,5	0,988	181,2	12,5	0,988	106,7	12,5	0,988	213,4	12,5	0,988	106,7																					
16,0	1,570	37,7	16,0	1,570	189,3	16,0	1,570	408,7	16,0	1,570	465,4	16,0	1,570	274,4	16,0	1,570	130,0	16,0	1,570	217,8	16,0	1,570	319,3																					
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	286,0	20,0	2,480	607,7	20,0	2,480	662,7	20,0	2,480	388,9																					
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	438,2																					
TOTAL (Kg/m)		389,9	TOTAL (Kg/m)		536,6	TOTAL (Kg/m)		745,5	TOTAL (Kg/m)		892,2	TOTAL (Kg/m)		1034,4	TOTAL (Kg/m)		1160,7	TOTAL (Kg/m)		1338,9	TOTAL (Kg/m)		1623,3																					
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																												
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	140	70	1	140	70	1	140	70	1	140	75	1	140	80	1	140	85	1	140	85	1	140	95																					
2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----																					
3	350	17	3	350	17	3	350	17	3	350	22	3	350	27	3	350	32	3	350	32	3	350	42																					
4	350	17	4	350	17	4	350	17	4	350	22	4	350	27	4	350	32	4	350	32	4	350	42																					
5	140	165	5	140	165	5	140	165	5	140	170	5	140	175	5	140	180	5	140	180	5	140	190																					
6	210	80	6	210	80	6	210	80	6	215	80	6	220	80	6	225	80	6	225	80	6	225	80																					
7	242	----	7	242	----	7	242	----	7	252	----	7	262	----	7	272	----	7	272	----	7	272	----																					
8	242	----	8	242	----	8	242	----	8	252	----	8	262	----	8	272	----	8	272	----	8	272	----																					
9	143	----	9	----	----	9	143	----	9	164	----	9	185	----	9	206	----	9	206	----	9	206	----																					
10	135	----	10	----	----	10	135	----	10	153	----	10	172	----	10	190	----	10	190	----	10	190	----																					
ASSINATURA DAS AUTORIDADES															 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																													
															CADERNO DE BUEIROS CELULARES																													
Engª Selma Schwab Coordenadora da DP/GNT															Engº Mário S.Bortone Gerente da DP/GPE					Engº Roger Gama Veloso Diretor de Projetos					Engº Nelson A. Reis Vice - Diretor Geral					Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)														
																														DES. 100														

BTCC 350 x 250					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																							
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000									
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,10 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,15 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,20 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,27 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,33 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,39 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,45 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,57 MPa									
LISTA DE FERROS (POR METRO LINEAR)																																												
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS									
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	8,0	20,0	350	c.20	1	16,0	14,3	350	c.28	1	16,0	18,2	355	c.22	1	16,0	20,0	355	c.20	1	20,0	16,7	360	c.24	1	20,0	18,2	365	c.22	1	20,0	18,2	370	c.22	1	25,0	15,4	375	c.26					
2	8,0	10,0	655	c.20	2	16,0	7,1	655	c.28	2	16,0	9,1	660	c.22	2	16,0	10,0	660	c.20	2	20,0	8,3	665	c.24	2	20,0	9,1	670	c.22	2	20,0	9,1	675	c.22	2	25,0	7,7	680	c.26					
3	10,0	30,8	367	c.13	3	12,5	33,3	367	c.12	3	12,5	40,0	372	c.10	3	16,0	36,4	372	c.11	3	16,0	40,0	377	c.10	3	16,0	40,0	382	c.10	3	20,0	33,3	387	c.12	3	20,0	36,4	392	c.11					
4	6,3	16,7	367	c.12	4	10,0	18,2	367	c.11	4	10,0	18,2	372	c.11	4	12,5	18,2	372	c.11	4	12,5	20,0	377	c.10	4	16,0	15,4	382	c.13	4	16,0	15,4	387	c.13	4	16,0	18,2	392	c.11					
5	6,3	10,5	330	c.38	5	8,0	16,7	330	c.24	5	6,3	18,2	335	c.22	5	8,0	18,2	335	c.22	5	8,0	16,7	340	c.24	5	6,3	20,0	345	c.20	5	6,3	14,3	350	c.28	5	6,3	11,8	355	c.34					
6	6,3	10,5	310	c.38	6	8,0	16,7	310	c.24	6	6,3	18,2	315	c.22	6	8,0	18,2	315	c.22	6	8,0	16,7	320	c.24	6	6,3	20,0	325	c.20	6	6,3	14,3	330	c.28	6	6,3	11,8	335	c.34					
7	6,3	9,5	292	c.21	7	6,3	9,5	292	c.21	7	6,3	11,8	302	c.17	7	6,3	11,8	302	c.17	7	8,0	9,1	312	c.22	7	8,0	10,5	322	c.19	7	8,0	11,8	332	c.17	7	10,0	8,7	342	c.23					
8	6,3	19,0	292	c.21	8	6,3	19,0	292	c.21	8	6,3	23,5	302	c.17	8	6,3	23,5	302	c.17	8	8,0	18,2	312	c.22	8	8,0	21,1	322	c.19	8	8,0	23,5	332	c.17	8	10,0	17,4	342	c.23					
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	204	c.20	9	8,0	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20	9	10,0	20,0	267	c.20	9	10,0	20,0	288	c.20					
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	306	c.20	10	8,0	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20	10	10,0	20,0	416	c.20	10	10,0	20,0	454	c.20					
11	6,3	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	12,5	108,0	Corr.	c.20					
12	6,3	108,0	Corr.	c.20	12	8,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	16,0	108,0	Corr.	c.20					
13	6,3	22,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20	13	6,3	22,0	Corr.	c.20					
14	6,3	66,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20	14	6,3	66,0	Corr.	c.20					
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----					
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	106,7	Var.	c.15	16	6,3	145,5	Var.	c.11	16	8,0	106,7	Var.	c.15	16	8,0	145,5	Var.	c.11					
17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	6,3	213,3	Var.	c.15	17	8,0	266,7	Var.	c.12	17	8,0	290,9	Var.	c.11	17	8,0	320,0	Var.	c.10	17	10,0	228,6	Var.	c.14	17	10,0	290,9	Var.	c.11					
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																												
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																				
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	152,8	6,3	0,252	73,7	6,3	0,252	136,3	6,3	0,252	49,0	6,3	0,252	39,4	6,3	0,252	81,2	6,3	0,252	46,7	6,3	0,252	46,7	6,3	0,252	42,7	6,3	0,252	46,7	6,3	0,252	46,7	6,3	0,252	46,7	6,3	0,252	42,7						
8,0	0,393	53,3	8,0	0,393	126,9	8,0	0,393	42,4	8,0	0,393	148,4	8,0	0,393	194,7	8,0	0,393	176,0	8,0	0,393	77,1	8,0	0,393	77,1	8,0	0,393	45,2	8,0	0,393	77,1	8,0	0,393	77,1	8,0	0,393	77,1	8,0	0,393	45,2						
10,0	0,624	70,5	10,0	0,624	41,7	10,0	0,624	109,6	10,0	0,624	67,4	10,0	0,624	67,4	10,0	0,624	67,4	10,0	0,624	258,2	10,0	0,624	258,2	10,0	0,624	291,7	10,0	0,624	258,2	10,0	0,624	258,2	10,0	0,624	258,2	10,0	0,624	291,7						
12,5	0,988	----	12,5	0,988	120,7	12,5	0,988	147,0	12,5	0,988	173,6	12,5	0,988	181,2	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7						
16,0	1,570	37,7	16,0	1,570	189,3	16,0	1,570	233,4	16,0	1,570	465,4	16,0	1,570	274,4	16,0	1,570	369,9	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	319,3	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	319,3						
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	286,0	20,0	2,480	316,0	20,0	2,480	638,9	20,0	2,480	638,9	20,0	2,480	353,9	20,0	2,480	638,9	20,0	2,480	638,9	20,0	2,480	638,9	20,0	2,480	353,9						
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	438,2	25,0	3,980	----	25,0	3,980	----	25,0	3,980	438,2	25,0	3,980	438,2						
TOTAL (Kg/m)		314,3	TOTAL (Kg/m)		552,3	TOTAL (Kg/m)		668,7	TOTAL (Kg/m)		903,8	TOTAL (Kg/m)		1043,1	TOTAL (Kg/m)		1117,2	TOTAL (Kg/m)		1258,8	TOTAL (Kg/m)		1597,7																					
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																												
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	140	70	1	140	70	1	140	75	1	140	75	1	140	80	1	140	85	1	140	90	1	140	95	1	140	100	1	140	105	1	140	110	1	140	115	1	140	120	1	140	125			
2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----
3	350	17	3	350	17	3	350	22	3	350	22	3	350	27	3	350	32	3	350	37	3	350	42	3	350	47	3	350	52	3	350	57	3	350	62	3	350	67	3	350	72			
4	350	17	4	350	17	4	350	22	4	350	22	4	350	27	4	350	32	4	350	37	4	350	42	4	350	47	4	350	52	4	350	57	4	350	62	4	350	67	4	350	72			
5	140	190	5	140	190	5	140	195	5	140	195	5	140	200	5	140	205	5	140	210	5	140	215	5	140	220	5	140	225	5	140	230	5	140	235	5	140	240	5	140	245			
6	210	100	6	210	100	6	215	100	6	215	100	6	220	100	6	225	100	6	230	100	6	235	100	6	240	100	6	245	100	6	250	100	6	255	100	6	260	100	6	265	100			
7	292	----	7	292	----	7	302	----	7	302	----	7	312	----	7	322	----	7	332	----	7	342	----	7	352	----	7	362	----	7	372	----	7	382	----	7	392	----	7	402	----			
8	292	----	8	292	----	8	302	----	8	302	----	8	312	----	8	322	----	8	332	----	8	342	----	8	352	----	8	362	----	8	372	----	8	382	----	8	392	----	8	402	----			
9	143	----	9	----	----	9	164	----	9	164	----	9	185	----	9	206																												

BTCC 350 x 300					TABELA DE ARMADURA PARA BUEIRO TRIPLIO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																		
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000				
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,16 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,21 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,27 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,34 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,40 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,46 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,58 MPa				
LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)
1	8,0	20,0	350	c.20	1	16,0	14,3	350	c.28	1	16,0	18,2	355	c.22	1	16,0	20,0	360	c.20	1	20,0	16,7	360	c.24	1	20,0	18,2	365	c.22	1	20,0	18,2	370	c.22	1	20,0	20,0	375	c.20
2	8,0	10,0	655	c.20	2	16,0	7,1	655	c.28	2	16,0	9,1	660	c.22	2	16,0	10,0	665	c.20	2	20,0	8,3	665	c.24	2	20,0	9,1	670	c.22	2	20,0	9,1	675	c.22	2	20,0	10,0	680	c.20
3	10,0	33,3	367	c.12	3	12,5	33,3	367	c.12	3	12,5	40,0	372	c.10	3	16,0	33,3	377	c.12	3	16,0	40,0	377	c.10	3	16,0	40,0	382	c.10	3	20,0	33,3	387	c.12	3	20,0	36,4	392	c.11
4	6,3	16,7	367	c.12	4	10,0	18,2	367	c.11	4	10,0	20,0	372	c.10	4	12,5	15,4	377	c.13	4	12,5	20,0	377	c.10	4	16,0	15,4	382	c.13	4	16,0	15,4	387	c.13	4	16,0	18,2	392	c.11
5	6,3	12,5	355	c.32	5	8,0	20,0	355	c.20	5	8,0	14,3	360	c.28	5	6,3	20,0	365	c.20	5	8,0	18,2	365	c.22	5	8,0	15,4	370	c.26	5	6,3	16,7	375	c.24	5	6,3	13,3	380	c.30
6	6,3	12,5	330	c.32	6	8,0	20,0	330	c.20	6	8,0	14,3	335	c.28	6	6,3	20,0	340	c.20	6	8,0	18,2	340	c.22	6	8,0	15,4	345	c.26	6	6,3	16,7	350	c.24	6	6,3	13,3	355	c.30
7	6,3	9,5	342	c.21	7	6,3	9,5	342	c.21	7	6,3	11,8	352	c.17	7	8,0	9,1	362	c.22	7	8,0	9,1	362	c.22	7	8,0	10,5	372	c.19	7	8,0	11,8	382	c.17	7	10,0	8,7	392	c.23
8	6,3	19,0	342	c.21	8	6,3	19,0	342	c.21	8	6,3	23,5	352	c.17	8	8,0	18,2	362	c.22	8	8,0	18,2	362	c.22	8	8,0	21,1	372	c.19	8	8,0	23,5	382	c.17	8	10,0	17,4	392	c.23
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20	9	10,0	20,0	267	c.20	9	10,0	20,0	288	c.20
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20	10	10,0	20,0	416	c.20	10	10,0	20,0	454	c.20
11	6,3	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20
12	6,3	108,0	Corr.	c.20	12	8,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	16,0	108,0	Corr.	c.20
13	6,3	28,0	Corr.	c.20	13	6,3	30,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20	13	6,3	28,0	Corr.	c.20
14	6,3	84,0	Corr.	c.20	14	6,3	90,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20	14	6,3	84,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	106,7	Var.	c.15	16	6,3	106,7	Var.	c.15	16	6,3	145,5	Var.	c.11	16	8,0	106,7	Var.	c.15	16	8,0	145,5	Var.	c.11
17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	6,3	213,3	Var.	c.15	17	6,3	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	8,0	320,0	Var.	c.10	17	10,0	228,6	Var.	c.14	17	10,0	290,9	Var.	c.11
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																							
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																		
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)				
6,3	0,252	167,1	6,3	0,252	81,3	6,3	0,252	117,0	6,3	0,252	110,7	6,3	0,252	45,4	6,3	0,252	53,5	6,3	0,252	58,7	6,3	0,252	52,9																
8,0	0,393	53,3	8,0	0,393	138,7	8,0	0,393	81,5	8,0	0,393	83,6	8,0	0,393	207,2	8,0	0,393	225,4	8,0	0,393	84,0	8,0	0,393	45,2																
10,0	0,624	76,3	10,0	0,624	41,7	10,0	0,624	113,8	10,0	0,624	134,8	10,0	0,624	67,4	10,0	0,624	67,4	10,0	0,624	258,2	10,0	0,624	367,2																
12,5	0,988	----	12,5	0,988	120,7	12,5	0,988	147,0	12,5	0,988	57,4	12,5	0,988	181,2	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	----																
16,0	1,570	37,7	16,0	1,570	189,3	16,0	1,570	233,4	16,0	1,570	452,2	16,0	1,570	274,4	16,0	1,570	369,9	16,0	1,570	131,2	16,0	1,570	319,3																
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	286,0	20,0	2,480	316,0	20,0	2,480	638,9	20,0	2,480	708,5																
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----																
TOTAL (Kg/m)		334,4	TOTAL (Kg/m)		571,7	TOTAL (Kg/m)		692,7	TOTAL (Kg/m)		838,7	TOTAL (Kg/m)		1061,6	TOTAL (Kg/m)		1138,9	TOTAL (Kg/m)		1277,7	TOTAL (Kg/m)		1493,1																
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																							
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)				
1	140	70	1	140	70	1	140	75	1	140	80	1	140	80	1	140	85	1	140	90	1	140	95																
2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----																
3	350	17	3	350	17	3	350	22	3	350	27	3	350	27	3	350	32	3	350	37	3	350	42																
4	350	17	4	350	17	4	350	22	4	350	27	4	350	27	4	350	32	4	350	37	4	350	42																
5	140	215	5	140	215	5	140	220	5	140	225	5	140	225	5	140	230	5	140	235	5	140	240																
6	210	120	6	210	120	6	215	120	6	220	120	6	220	120	6	225	120	6	230	120	6	235	120																
7	342	----	7	342	----	7	352	----	7	362	----	7	362	----	7	372	----	7	382	----	7	392	----																
8	342	----	8	342	----	8	352	----	8	362	----	8	362	----	8	372	----	8	382	----	8	392	----																
9	143	----	9	----	----	9	164	----	9	185	----	9	185	----	9	206	----	9	227	----	9	248	----																
10	135	----	10	----	----	10	153	----	10	172	----	10	172	----	10	190	----	10	208	----	10	227	----																
ASSINATURA DAS AUTORIDADES									Engª Selma Schwab Coordenadora da DP/GNT Engº Mário S.Bortone Gerente da DP/GPE Engº Roger Gama Veloso Diretor de Projetos Engº Nelson A. Reis Vice - Diretor Geral									DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)																					
																		DES. 100																					

BTCC 350 x 350	TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)						
ALTURA DO ATERRO (cm) 0 < h < 100	ALTURA DO ATERRO (cm) 100 < h < 250	ALTURA DO ATERRO (cm) 250 < h < 500	ALTURA DO ATERRO (cm) 500 < h < 750	ALTURA DO ATERRO (cm) 750 < h < 1000	ALTURA DO ATERRO (cm) 1000 < h < 1250	ALTURA DO ATERRO (cm) 1250 < h < 1500	ALTURA DO ATERRO (cm) 1500 < h < 2000
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,22 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,28 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,47 MPa	TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,59 MPa

LISTA DE FERROS (POR METRO LINEAR)																																							
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS				
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	8,0	20,0	350	c.20	1	16,0	14,3	350	c.28	1	16,0	18,2	355	c.22	1	16,0	20,0	360	c.20	1	20,0	16,7	360	c.24	1	20,0	18,2	365	c.22	1	20,0	18,2	370	c.22	1	20,0	20,0	375	c.20
2	8,0	10,0	655	c.20	2	16,0	7,1	655	c.28	2	16,0	9,1	660	c.22	2	16,0	10,0	665	c.20	2	20,0	8,3	665	c.24	2	20,0	9,1	670	c.22	2	20,0	9,1	675	c.22	2	20,0	10,0	680	c.20
3	10,0	36,4	367	c.11	3	12,5	36,4	367	c.11	3	12,5	40,0	372	c.10	3	16,0	33,3	377	c.12	3	16,0	36,4	377	c.11	3	16,0	40,0	382	c.10	3	20,0	33,3	387	c.12	3	20,0	36,4	392	c.11
4	6,3	18,2	367	c.11	4	10,0	20,0	367	c.10	4	10,0	20,0	372	c.10	4	12,5	16,7	377	c.12	4	12,5	20,0	377	c.10	4	16,0	15,4	382	c.13	4	16,0	15,4	387	c.13	4	16,0	18,2	392	c.11
5	6,3	16,7	380	c.24	5	10,0	18,2	380	c.22	5	8,0	18,2	385	c.22	5	8,0	18,2	390	c.22	5	10,0	16,7	390	c.24	5	8,0	20,0	395	c.20	5	8,0	16,7	400	c.24	5	6,3	20,0	405	c.20
6	6,3	16,7	350	c.24	6	10,0	18,2	350	c.22	6	8,0	18,2	355	c.22	6	8,0	18,2	360	c.22	6	10,0	16,7	360	c.24	6	8,0	20,0	365	c.20	6	8,0	16,7	370	c.24	6	6,3	20,0	375	c.20
7	6,3	9,5	392	c.21	7	6,3	9,5	392	c.21	7	6,3	11,8	402	c.17	7	8,0	9,1	412	c.22	7	8,0	9,1	412	c.22	7	8,0	10,5	422	c.19	7	8,0	11,8	432	c.17	7	10,0	8,7	442	c.23
8	6,3	19,0	392	c.21	8	6,3	19,0	392	c.21	8	6,3	23,5	402	c.17	8	8,0	18,2	412	c.22	8	8,0	18,2	412	c.22	8	8,0	21,1	422	c.19	8	8,0	23,5	432	c.17	8	10,0	17,4	442	c.23
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20	9	10,0	20,0	267	c.20	9	10,0	20,0	288	c.20
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20	10	10,0	20,0	416	c.20	10	10,0	20,0	454	c.20
11	6,3	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20
12	8,0	108,0	Corr.	c.20	12	8,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	16,0	108,0	Corr.	c.20
13	6,3	32,0	Corr.	c.20	13	8,0	36,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20
14	6,3	96,0	Corr.	c.20	14	6,3	108,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	114,3	Var.	c.14	16	6,3	145,5	Var.	c.11	16	8,0	106,7	Var.	c.15	16	8,0	106,7	Var.	c.15	16	8,0	160,0	Var.	c.10
17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	6,3	213,3	Var.	c.15	17	6,3	290,9	Var.	c.11	17	8,0	320,0	Var.	c.10	17	8,0	320,0	Var.	c.10	17	10,0	228,6	Var.	c.14	17	10,0	290,9	Var.	c.11

QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																										
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50					
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)			
6,3	0,252	158,0	6,3	0,252	81,9	6,3	0,252	125,4	6,3	0,252	79,2	6,3	0,252	50,7	6,3	0,252	57,6	6,3	0,252	32,3	6,3	0,252	71,6			
8,0	0,393	95,7	8,0	0,393	99,0	8,0	0,393	52,9	8,0	0,393	142,6	8,0	0,393	162,1	8,0	0,393	248,1	8,0	0,393	141,5	8,0	0,393	49,7			
10,0	0,624	83,4	10,0	0,624	128,7	10,0	0,624	181,2	10,0	0,624	134,8	10,0	0,624	145,5	10,0	0,624	67,4	10,0	0,624	258,2	10,0	0,624	375,4			
12,5	0,988	----	12,5	0,988	132,0	12,5	0,988	147,0	12,5	0,988	62,2	12,5	0,988	181,2	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	----			
16,0	1,570	37,7	16,0	1,570	189,3	16,0	1,570	233,4	16,0	1,570	452,2	16,0	1,570	253,1	16,0	1,570	369,9	16,0	1,570	131,2	16,0	1,570	319,3			
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	286,0	20,0	2,480	316,0	20,0	2,480	638,9	20,0	2,480	708,5			
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----			
TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)			TOTAL (Kg/m)		
374,8			630,9			739,9			871,0			1078,6			1165,7			1308,8			1524,5					

DIMENSÕES DAS DOBRAS DAS ARMADURAS																								
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	
1	140	70	1	140	70	1	140	75	1	140	80	1	140	80	1	140	85	1	140	90	1	140	95	
2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	
3	350	17	3	350	17	3	350	22	3	350	27	3	350	27	3	350	32	3	350	37	3	350	42	
4	350	17	4	350	17	4	350	22	4	350	27	4	350	27	4	350	32	4	350	37	4	350	42	
5	140	240	5	140	240	5	140	245	5	140	250	5	140	250	5	140	255	5	140	260	5	140	265	
6	210	140	6	210	140	6	215	140	6	220	140	6	220	140	6	225	140	6	230	140	6	235	140	
7	392	----	7	392	----	7	402	----	7	412	----	7	412	----	7	422	----	7	432	----	7	442	----	
8	392	----	8	392	----	8	402	----	8	412	----	8	412	----	8	422	----	8	432	----	8	442	----	
9	143	----	9	----	----	9	164	----	9	185	----	9	185	----	9	206	----	9	227	----	9	248	----	
10	135	----	10	----	----	10	153	----	10	172	----	10	172	----	10	190	----	10	208	----	10	227	----	

BTCC 350 x 400					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																							
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000									
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,13 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,29 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,42 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,48 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,60 MPa									
LISTA DE FERROS (POR METRO LINEAR)																																												
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS									
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)					
1	8,0	20,0	350	c.20	1	16,0	14,3	350	c.28	1	16,0	18,2	355	c.22	1	16,0	20,0	360	c.20	1	20,0	16,7	360	c.24	1	20,0	16,7	365	c.24	1	20,0	18,2	370	c.22	1	20,0	20,0	375	c.20					
2	8,0	10,0	655	c.20	2	16,0	7,1	655	c.28	2	16,0	9,1	660	c.22	2	16,0	10,0	665	c.20	2	20,0	8,3	665	c.24	2	20,0	8,3	670	c.24	2	20,0	9,1	675	c.22	2	20,0	10,0	680	c.20					
3	10,0	40,0	367	c.10	3	12,5	36,4	367	c.11	3	12,5	40,0	372	c.10	3	16,0	33,3	377	c.12	3	16,0	36,4	377	c.11	3	16,0	40,0	382	c.10	3	20,0	30,8	387	c.13	3	20,0	36,4	392	c.11					
4	6,3	18,2	367	c.11	4	10,0	20,0	367	c.10	4	10,0	20,0	372	c.10	4	12,5	16,7	377	c.12	4	12,5	20,0	377	c.10	4	16,0	15,4	382	c.13	4	16,0	15,4	387	c.13	4	16,0	18,2	392	c.11					
5	8,0	15,4	405	c.26	5	10,0	20,0	405	c.20	5	10,0	18,2	410	c.22	5	10,0	16,7	415	c.24	5	12,5	16,7	415	c.24	5	10,0	20,0	420	c.20	5	10,0	16,7	425	c.24	5	10,0	16,7	430	c.24					
6	8,0	15,4	370	c.26	6	10,0	20,0	370	c.20	6	10,0	18,2	375	c.22	6	10,0	16,7	380	c.24	6	12,5	16,7	380	c.24	6	10,0	20,0	385	c.20	6	10,0	16,7	390	c.24	6	10,0	16,7	395	c.24					
7	6,3	9,5	442	c.21	7	6,3	9,5	442	c.21	7	6,3	11,8	452	c.17	7	8,0	9,1	462	c.22	7	8,0	9,1	462	c.22	7	8,0	10,5	472	c.19	7	8,0	11,8	482	c.17	7	10,0	8,7	492	c.23					
8	6,3	19,0	442	c.21	8	6,3	19,0	442	c.21	8	6,3	23,5	452	c.17	8	8,0	18,2	462	c.22	8	8,0	18,2	462	c.22	8	8,0	21,1	472	c.19	8	8,0	23,5	482	c.17	8	10,0	17,4	492	c.23					
9	6,3	20,0	183	c.20	9	----	----	----	----	9	6,3	20,0	204	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	225	c.20	9	8,0	20,0	246	c.20	9	10,0	20,0	267	c.20	9	10,0	20,0	288	c.20					
10	6,3	20,0	270	c.20	10	----	----	----	----	10	6,3	20,0	306	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	344	c.20	10	8,0	20,0	380	c.20	10	10,0	20,0	416	c.20	10	10,0	20,0	454	c.20					
11	6,3	108,0	Corr.	c.20	11	8,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	12,5	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	10,0	108,0	Corr.	c.20	11	12,5	108,0	Corr.	c.20					
12	8,0	108,0	Corr.	c.20	12	8,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	10,0	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	12,5	108,0	Corr.	c.20	12	16,0	108,0	Corr.	c.20					
13	6,3	38,0	Corr.	c.20	13	8,0	40,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	6,3	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	8,0	38,0	Corr.	c.20	13	6,3	38,0	Corr.	c.20	13	6,3	38,0	Corr.	c.20					
14	6,3	114,0	Corr.	c.20	14	6,3	120,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20	14	6,3	114,0	Corr.	c.20					
15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----	15	16,0	24,0	Corr.	----					
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	114,3	Var.	c.14	16	6,3	114,3	Var.	c.11	16	6,3	145,5	Var.	c.11	16	8,0	114,3	Var.	c.14	16	8,0	160,0	Var.	c.10					
17	----	----	----	----	17	6,3	228,6	Var.	c.14	17	6,3	213,3	Var.	c.15	17	6,3	290,9	Var.	c.11	17	8,0	290,9	Var.	c.11	17	8,0	320,0	Var.	c.10	17	10,0	213,3	Var.	c.15	17	10,0	290,9	Var.	c.11					
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																												
RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50					RESUMO - AÇO CA -50				
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)						
6,3	0,252	136,9	6,3	0,252	88,5	6,3	0,252	126,4	6,3	0,252	85,2	6,3	0,252	47,2	6,3	0,252	54,0	6,3	0,252	38,3	6,3	0,252	38,3	6,3	0,252	49,7	6,3	0,252	100,1	6,3	0,252	100,1	6,3	0,252	49,7	6,3	0,252	49,7						
8,0	0,393	142,6	8,0	0,393	100,6	8,0	0,393	14,9	8,0	0,393	94,3	8,0	0,393	182,4	8,0	0,393	209,5	8,0	0,393	100,1	8,0	0,393	100,1	8,0	0,393	100,1	8,0	0,393	100,1	8,0	0,393	100,1	8,0	0,393	100,1	8,0	0,393	100,1						
10,0	0,624	91,6	10,0	0,624	142,5	10,0	0,624	270,4	10,0	0,624	217,6	10,0	0,624	----	10,0	0,624	167,9	10,0	0,624	336,1	10,0	0,624	336,1	10,0	0,624	336,1	10,0	0,624	336,1	10,0	0,624	336,1	10,0	0,624	336,1	10,0	0,624	336,1						
12,5	0,988	----	12,5	0,988	132,0	12,5	0,988	147,0	12,5	0,988	62,2	12,5	0,988	419,1	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7	12,5	0,988	106,7						
16,0	1,570	37,7	16,0	1,570	189,3	16,0	1,570	233,4	16,0	1,570	452,2	16,0	1,570	253,1	16,0	1,570	369,9	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2	16,0	1,570	131,2						
20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	----	20,0	2,480	286,0	20,0	2,480	289,1	20,0	2,480	614,9	20,0	2,480	614,9	20,0	2,480	614,9	20,0	2,480	614,9	20,0	2,480	614,9	20,0	2,480	614,9	20,0	2,480	614,9						
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----			
TOTAL (Kg/m)		408,8	TOTAL (Kg/m)		652,9	TOTAL (Kg/m)		792,1	TOTAL (Kg/m)		911,5	TOTAL (Kg/m)		1187,8	TOTAL (Kg/m)		1197,1	TOTAL (Kg/m)		1327,3	TOTAL (Kg/m)		1624,6	TOTAL (Kg/m)		1624,6	TOTAL (Kg/m)		1624,6	TOTAL (Kg/m)		1624,6	TOTAL (Kg/m)		1624,6	TOTAL (Kg/m)		1624,6						
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																												
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)						
1	140	70	1	140	70	1	140	75	1	140	80	1	140	80	1	140	85	1	140	90	1	140	95	1	140	95	1	140	95	1	140	95	1	140	95	1	140	95						
2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----	2	140	----			
3	350	17	3	350	17	3	350	22	3	350	27	3	350	27	3	350	32	3	350	37	3	350	42	3	350	42	3	350	42	3	350	42	3	350	42	3	350	42						
4	350	17	4	350	17	4	350	22	4	350	27	4	350	27	4	350	32	4	350	37	4	350	42	4	350	42	4	350	42	4	350	42	4	350	42	4	350	42						
5	140	265	5	140	265	5	140	270	5	140	275	5	140	275	5	140	280	5	140	285	5	140	290	5	140	290	5	140	290	5	140	290	5	140	290	5	140	290						
6	210	160	6	210	160	6	215	160	6	220	160	6	220	160	6	225	160	6	230	160	6	235	160	6	240	160	6	245	160	6	250	160	6	255	160	6	260	160						
7	442	----	7	442	----	7	452	----	7	462	----	7	472	----	7	482	----	7	492	----	7	502	----	7	512	----	7	522	----	7	532	----	7	542	----	7	552	----						
8	442	----	8	442	----	8	452	----	8	462	----	8	472	----	8	482	----	8	492	----	8	502	----	8	512	----	8	522	----	8	532	----	8	542	----	8	552	----						
9	143	----	9	----	----	9	164	----	9	185	----	9	206																															

BTCC 400 x 300					TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																						
ALTURA DO ATERRO (cm) 0 < h < 100					ALTURA DO ATERRO (cm) 100 < h < 250					ALTURA DO ATERRO (cm) 250 < h < 500					ALTURA DO ATERRO (cm) 500 < h < 750					ALTURA DO ATERRO (cm) 750 < h < 1000					ALTURA DO ATERRO (cm) 1000 < h < 1250					ALTURA DO ATERRO (cm) 1250 < h < 1500					ALTURA DO ATERRO (cm) 1500 < h < 2000																								
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,11MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,17 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,22 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,28 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,34 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,40 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,46 MPa					TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,59 MPa																								
LISTA DE FERROS (POR METRO LINEAR)																																																											
LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS					LISTA DE FERROS																								
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																				
1	8,0	20,0	400	c.20	1	16,0	15,4	400	c.26	1	16,0	20,0	400	c.20	1	20,0	16,7	405	c.24	1	20,0	20,0	410	c.20	1	20,0	20,0	415	c.20	1	25,0	15,4	420	c.26	1	25,0	18,2	425	c.22																				
2	8,0	10,0	750	c.20	2	16,0	7,7	750	c.26	2	16,0	10,0	750	c.20	2	20,0	8,3	755	c.24	2	20,0	10,0	760	c.20	2	20,0	10,0	765	c.20	2	25,0	7,7	770	c.26	2	25,0	9,1	775	c.22																				
3	10,0	33,3	422	c.12	3	12,5	40,0	422	c.10	3	16,0	33,3	422	c.12	3	16,0	36,4	427	c.11	3	20,0	30,8	432	c.13	3	20,0	33,3	437	c.12	3	20,0	36,4	442	c.11	3	25,0	30,8	447	c.13																				
4	6,3	16,7	422	c.12	4	10,0	20,0	422	c.10	4	12,5	16,7	422	c.10	4	12,5	20,0	427	c.10	4	16,0	15,4	432	c.13	4	16,0	16,7	437	c.12	4	16,0	18,2	442	c.11	4	16,0	20,0	447	c.10																				
5	6,3	10,5	380	c.38	5	8,0	18,2	380	c.22	5	8,0	18,2	380	c.22	5	8,0	18,2	385	c.22	5	8,0	18,2	390	c.22	5	8,0	15,4	395	c.26	5	6,3	18,2	400	c.22	5	6,3	15,4	405	c.26																				
6	6,3	10,5	360	c.38	6	8,0	18,2	360	c.22	6	8,0	18,2	360	c.22	6	8,0	18,2	365	c.22	6	8,0	18,2	370	c.22	6	8,0	15,4	375	c.26	6	6,3	18,2	380	c.22	6	6,3	15,4	385	c.26																				
7	6,3	11,8	352	c.17	7	6,3	11,8	352	c.17	7	6,3	11,8	352	c.17	7	8,0	9,1	362	c.22	7	8,0	10,5	372	c.19	7	8,0	11,8	382	c.17	7	10,0	8,7	392	c.23	7	10,0	9,5	402	c.21																				
8	6,3	23,5	352	c.17	8	6,3	23,5	352	c.17	8	6,3	23,5	352	c.17	8	8,0	18,2	362	c.22	8	8,0	21,1	372	c.19	8	8,0	23,5	382	c.17	8	10,0	17,4	392	c.23	8	10,0	19,0	402	c.21																				
9	8,0	20,0	220	c.20	9	----	----	----	----	9	8,0	20,0	220	c.20	9	8,0	20,0	241	c.20	9	10,0	20,0	262	c.20	9	10,0	20,0	283	c.20	9	10,0	20,0	304	c.20	9	12,5	20,0	325	c.20																				
10	8,0	20,0	338	c.20	10	----	----	----	----	10	8,0	20,0	338	c.20	10	8,0	20,0	374	c.20	10	10,0	20,0	412	c.20	10	10,0	20,0	448	c.20	10	10,0	20,0	486	c.20	10	12,5	20,0	522	c.20																				
11	6,3	120,0	Corr.	c.20	11	8,0	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20																				
12	6,3	120,0	Corr.	c.20	12	10,0	120,0	Corr.	c.20	12	10,0	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	16,0	120,0	Corr.	c.20	12	16,0	120,0	Corr.	c.20																				
13	6,3	26,0	Corr.	c.20	13	6,3	30,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20	13	6,3	26,0	Corr.	c.20																				
14	6,3	78,0	Corr.	c.20	14	6,3	90,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20	14	6,3	78,0	Corr.	c.20																				
15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----																				
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	94,1	Var.	c.17	16	6,3	123,1	Var.	c.13	16	6,3	160,0	Var.	c.10	16	8,0	145,5	Var.	c.11																				
17	----	----	----	----	17	6,3	200,0	Var.	c.16	17	6,3	266,7	Var.	c.12	17	8,0	228,6	Var.	c.14	17	8,0	266,7	Var.	c.12	17	8,0	290,9	Var.	c.11	17	10,0	213,3	Var.	c.15	17	10,0	290,9	Var.	c.11																				
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																											
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																			
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																					
6,3	0,252	155,3	6,3	0,252	87,3	6,3	0,252	98,5	6,3	0,252	26,2	6,3	0,252	43,0	6,3	0,252	49,8	6,3	0,252	94,6	6,3	0,252	56,9	8,0	0,393	104,8	8,0	0,393	100,1	8,0	0,393	96,8	8,0	0,393	200,1	8,0	0,393	175,0	8,0	0,393	186,5	8,0	0,393	----	8,0	0,393	49,2												
10,0	0,624	87,7	10,0	0,624	127,5	10,0	0,624	149,8	10,0	0,624	74,9	10,0	0,624	84,1	10,0	0,624	166,1	10,0	0,624	270,2	10,0	0,624	227,6	12,5	0,988	----	12,5	0,988	166,8	12,5	0,988	69,6	12,5	0,988	202,9	12,5	0,988	237,1	12,5	0,988	118,6	12,5	0,988	118,6	12,5	0,988	285,9												
16,0	1,570	----	16,0	1,570	187,4	16,0	1,570	464,0	16,0	1,570	244,0	16,0	1,570	104,4	16,0	1,570	114,6	16,0	1,570	314,7	16,0	1,570	328,8	20,0	2,480	59,5	20,0	2,480	59,5	20,0	2,480	59,5	20,0	2,480	382,7	20,0	2,480	781,3	20,0	2,480	816,0	20,0	2,480	458,5	20,0	2,480	59,5												
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	493,4	25,0	3,980	1136,5	TOTAL (Kg/m)	407,3	TOTAL (Kg/m)	728,6	TOTAL (Kg/m)	938,2	TOTAL (Kg/m)	1130,8	TOTAL (Kg/m)	1424,9	TOTAL (Kg/m)	1451,6	TOTAL (Kg/m)	1750,0	TOTAL (Kg/m)	2144,4																				
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																											
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																					
1	160	80	1	160	80	1	160	80	1	160	85	1	160	90	1	160	95	1	160	100	1	160	105	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	3	400	47															
2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	3	400	47	4	400	27	3	400	32	3	400	37	3	400	42	3	400	47	4	400	47															
3	400	22	3	400	22	3	400	22	3	400	27	3	400	32	3	400	37	3	400	42	3	400	47	5	160	220	5	160	220	5	160	220	5	160	235	5	160	240	5	160	245	6	240	120															
4	400	22	4	400	22	4	400	22	4	400	27	4	400	32	4	400	37	4	400	42	4	400	47	6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	7	352	----															
5	160	220	5	160	220	5	160	220	5	160	225	5	160	230	5	160	235	5	160	240	5	160	245	7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	8	352	----															
6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	6	240	120	8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	9	180	----															
7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	7	352	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----												
8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	8	352	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----												
9	180	----	9	----	----	9	180	----	9	201	----	9	201	----	9	222	----	9	222	----	9	243	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----												
10	169	----	10	----	----	10	169	----	10	169	----	10	169	----	10	206	----	10	206	----	10	224	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----	10	169	----												
ASSINATURA DAS AUTORIDADES															DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																												
Engº Selma Schwab Coordenadora da DP/GNT															Engº Mário S.Bortone Gerente da DP/GPE															Engº Roger Gama Veloso Diretor de Projetos															Engº Nelson A. Reis Vice - Diretor Geral														
															CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)															DES. 105																													

BTCC 400 x 350				TABELA DE ARMADURA PARA BUEIRO TRIPLO CELULAR - FUNDAÇÃO DIRETA (POR METRO LINEAR)																																																											
ALTURA DO ATERRO (cm) 0 < h < 100				ALTURA DO ATERRO (cm) 100 < h < 250				ALTURA DO ATERRO (cm) 250 < h < 500				ALTURA DO ATERRO (cm) 500 < h < 750				ALTURA DO ATERRO (cm) 750 < h < 1000				ALTURA DO ATERRO (cm) 1000 < h < 1250				ALTURA DO ATERRO (cm) 1250 < h < 1500				ALTURA DO ATERRO (cm) 1500 < h < 2000																																			
TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,12 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,18 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,23 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,29 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,35 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,41 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,47 MPa				TENSÃO ADMISSÍVEL NO SOLO DE FUNDAÇÃO = 0,60 MPa																																			
LISTA DE FERROS (POR METRO LINEAR)																																																															
LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS				LISTA DE FERROS																																			
N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)	N	Ø (mm)	Quant.	Comp. (cm)	Esp. (cm)																													
1	10,0	15,4	400	c.26	1	16,0	15,4	400	c.26	1	16,0	20,0	400	c.20	1	20,0	16,7	405	c.24	1	20,0	18,2	410	c.22	1	20,0	20,0	415	c.20	1	25,0	15,4	420	c.26	1	25,0	18,2	425	c.22																								
2	10,0	7,7	750	c.26	2	16,0	7,7	750	c.26	2	16,0	10,0	750	c.20	2	20,0	8,3	755	c.24	2	20,0	9,1	760	c.22	2	20,0	10,0	765	c.20	2	25,0	7,7	770	c.26	2	25,0	9,1	775	c.22																								
3	10,0	36,4	422	c.11	3	12,5	40,0	422	c.10	3	16,0	30,8	422	c.13	3	16,0	36,4	427	c.11	3	20,0	30,8	432	c.13	3	20,0	33,3	437	c.12	3	20,0	36,4	442	c.11	3	20,0	40,0	447	c.10																								
4	6,3	18,2	422	c.11	4	10,0	20,0	422	c.10	4	12,5	16,7	422	c.12	4	12,5	20,0	427	c.10	4	16,0	15,4	432	c.13	4	16,0	16,7	437	c.12	4	16,0	18,2	442	c.11	4	16,0	20,0	447	c.10																								
5	6,3	13,3	405	c.30	5	10,0	15,4	405	c.26	5	10,0	15,4	405	c.26	5	10,0	15,4	410	c.26	5	8,0	20,0	415	c.20	5	8,0	18,2	420	c.22	5	6,3	20,0	425	c.20	5	6,3	18,2	430	c.22																								
6	6,3	13,3	380	c.30	6	10,0	15,4	380	c.26	6	10,0	15,4	380	c.26	6	10,0	15,4	385	c.26	6	8,0	20,0	390	c.20	6	8,0	18,2	395	c.22	6	6,3	20,0	400	c.20	6	6,3	18,2	405	c.22																								
7	6,3	11,8	402	c.17	7	6,3	11,8	402	c.17	7	6,3	11,8	402	c.17	7	8,0	9,1	412	c.22	7	8,0	10,5	422	c.19	7	8,0	11,8	432	c.17	7	10,0	8,7	442	c.23	7	10,0	9,5	452	c.21																								
8	6,3	23,5	402	c.17	8	6,3	23,5	402	c.17	8	6,3	23,5	402	c.17	8	8,0	18,2	412	c.22	8	8,0	21,1	422	c.19	8	8,0	23,5	432	c.17	8	10,0	17,4	442	c.23	8	10,0	19,0	452	c.21																								
9	8,0	20,0	220	c.20	9	----	----	----	----	9	8,0	20,0	220	c.20	9	8,0	20,0	241	c.20	9	10,0	20,0	262	c.20	9	10,0	20,0	283	c.20	9	10,0	20,0	304	c.20	9	12,5	20,0	325	c.20																								
10	8,0	20,0	338	c.20	10	----	----	----	----	10	8,0	20,0	338	c.20	10	8,0	20,0	374	c.20	10	10,0	20,0	412	c.20	10	10,0	20,0	448	c.20	10	10,0	20,0	486	c.20	10	12,5	20,0	522	c.20																								
11	6,3	120,0	Corr.	c.20	11	8,0	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	10,0	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20	11	12,5	120,0	Corr.	c.20																								
12	8,0	120,0	Corr.	c.20	12	10,0	120,0	Corr.	c.20	12	10,0	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	12,5	120,0	Corr.	c.20	12	16,0	120,0	Corr.	c.20	12	16,0	120,0	Corr.	c.20																								
13	6,3	32,0	Corr.	c.20	13	6,3	36,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20	13	6,3	32,0	Corr.	c.20																								
14	6,3	96,0	Corr.	c.20	14	6,3	108,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20	14	6,3	96,0	Corr.	c.20																								
15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----	15	20,0	24,0	Corr.	----																								
16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	----	----	----	----	16	6,3	94,1	Var.	c.17	16	6,3	133,3	Var.	c.12	16	6,3	160,0	Var.	c.10	16	8,0	145,5	Var.	c.11																								
17	----	----	----	----	17	6,3	200,0	Var.	c.16	17	6,3	266,7	Var.	c.12	17	8,0	228,6	Var.	c.14	17	8,0	266,7	Var.	c.12	17	8,0	290,9	Var.	c.11	17	10,0	213,3	Var.	c.15	17	10,0	290,9	Var.	c.11																								
QUADRO RESUMO DE ARMADURA (POR METRO LINEAR)																																																															
RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50			RESUMO - AÇO CA -50																																							
Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)	Ø (mm)	Kg/m	PESO (Kg)																												
6,3	0,252	143,9	6,3	0,252	97,8	6,3	0,252	109,0	6,3	0,252	32,3	6,3	0,252	49,1	6,3	0,252	57,8	6,3	0,252	106,5	6,3	0,252	70,6	8,0	0,393	91,0	8,0	0,393	47,2	8,0	0,393	151,8	8,0	0,393	190,1	8,0	0,393	205,1	8,0	0,393	----	8,0	0,393	49,2																			
10,0	0,624	170,3	10,0	0,624	203,0	10,0	0,624	225,2	10,0	0,624	151,3	10,0	0,624	159,0	10,0	0,624	91,2	10,0	0,624	278,4	10,0	0,624	236,5	12,5	0,988	----	12,5	0,988	166,8	12,5	0,988	69,6	12,5	0,988	202,9	12,5	0,988	118,6	12,5	0,988	237,1	12,5	0,988	118,6	12,5	0,988	285,9																
16,0	1,570	----	16,0	1,570	187,4	16,0	1,570	447,4	16,0	1,570	244,0	16,0	1,570	104,4	16,0	1,570	114,6	16,0	1,570	314,7	16,0	1,570	328,8	20,0	2,480	59,5	20,0	2,480	59,5	20,0	2,480	59,5	20,0	2,480	382,7	20,0	2,480	746,1	20,0	2,480	816,0	20,0	2,480	458,5	20,0	2,480	502,9																
25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	----	25,0	3,980	493,4	25,0	3,980	588,5	TOTAL (Kg/m)	464,7	TOTAL (Kg/m)	761,7	TOTAL (Kg/m)	954,6	TOTAL (Kg/m)	1165,0	TOTAL (Kg/m)	1367,3	TOTAL (Kg/m)	1521,8	TOTAL (Kg/m)	1770,1	TOTAL (Kg/m)	2062,4																								
DIMENSÕES DAS DOBRAS DAS ARMADURAS																																																															
N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)	N	a (cm)	b (cm)																									
1	160	80	1	160	80	1	160	80	1	160	85	1	160	90	1	160	95	1	160	100	1	160	105	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----	2	160	----																			
3	400	22	3	400	22	3	400	22	3	400	27	3	400	32	3	400	37	3	400	42	3	400	47	4	400	22	4	400	22	4	400	22	4	400	27	4	400	32	4	400	37	4	400	42	4	400	47																
5	160	245	5	160	245	5	160	245	5	160	250	5	160	255	5	160	260	5	160	265	5	160	270	6	240	140	6	240	140	6	240	140	6	240	140	6	240	140	6	240	140	6	240	140	6	240	140																
7	402	----	7	402	----	7	402	----	7	402	----	7	412	----	7	422	----	7	432	----	7	442	----	8	402	----	8	402	----	8	402	----	8	402	----	8	412	----	8	422	----	8	432	----	8	442	----																
9	180	----	9	----	----	9	180	----	9	201	----	9	222	----	9	243	----	9	264	----	9	285	----	10	169	----	10	----	----	10	169	----	10	187	----	10	206	----	10	224	----	10	243	----	10	261	----																
ASSINATURA DAS AUTORIDADES																 DEPARTAMENTO DE ESTRADAS DE RODAGEM DO ESTADO DE MINAS GERAIS																																															
Engº Selma Schwab Coordenadora da DP/GNT																Engº Mário S.Bortone Gerente da DP/GPE																Engº Roger Gama Veloso Diretor de Projetos																Engº Nelson A. Reis Vice - Diretor Geral															
CADERNO DE BUEIROS CELULARES Tabela de Armadura Para Bueiro Triplo Celular Fundação Direta (Por Metro Linear)																																																															
DES. 001																																																															